Optoelectronics GROUP @ ZJU

浙江大学 ZHEJIANG UNIVERSITY

光谱仪微型化

杨宗银

浙江大学信息与电子工程学院 教授

2025年4月6日

学习工作经历

1988.8 – 至今　喜欢拆电器，制作小发明，梦想从事电子技术研究

2005.9 – 2006.6　**浙江**大学，应用**生物**，本科

2006.9 – 2009.6　**浙江**大学，**机械**工程，本科

2009.9 – 2012.3　**浙江**大学，**光学**工程，硕士

- 导师：童利民 教授

2014.10 – 2019.1　**剑桥**大学，**电子**工程，博士

- 导师：Tawfique Hasan 教授 副系主任
- 剑桥大学全额奖学金，全球仅80人

2019.4 – 2020.8　**剑桥**大学，电子工程，博后

- 导师：Andrea Ferrari 皇家工程院院士

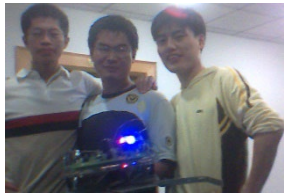
2019.7 – 2020.8　**剑桥**大学，国王学院，研究员

- Non-stipendiary Research Fellowship

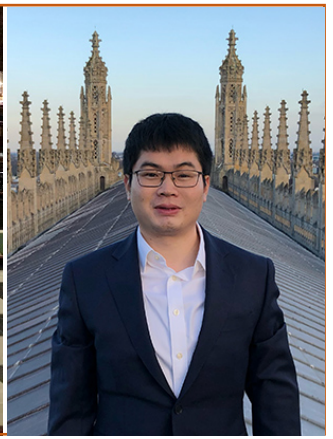
2020.9 – 至今　**浙江**大学，信息与电子工程学院，“百人计划”研究员

- 2024年评为长聘教授

全国机械设计大赛一等奖
浙大机器人大赛二等奖
浙江省机械设计大赛二等奖
浙大机械设计大赛二等奖
浙大“挑战杯” 一等奖
浙大电路焊接大赛特等奖...



剑桥墨水公司**创始董事**



❑ 发表论文、著作及专利

- 在Science, Nature Photonics, Nature Electronics, Nature Communications, Science Advances 等期刊发表**60**余篇论文
- **Science 3篇**（一作×2）**Nature Photon.**, **Science Advances, Light, Engineering 通讯**作者
- 中国专利**20**多项，申请英国专利**2**项，PCT专利**4**项
- 在**Springer**出版书本**1**部
- 担任INGEP2024国际会议**执行主席**，组织并担任OECC2023国际会议**Workshop主席**，受邀**大会报告3次**，**华为STW**会议1次，**华为“罗马论坛”**授课
- Nature Nano., Nature Photon., Nature Commun., Science Advances 等期刊的审稿人
- The Innovation (IF=32.1), 中国激光, FITEE 等杂志青年编委
- Journal of Physics D（IF=3.2）等期刊专题编辑

❑ 学术兼职与荣誉

- 国家优自奖“**特等奖**”，近年来**英国**理工科**首位**获得者
- 浙江大学启真优秀青年学者；青年创新奖
- 浙江大学校级**优秀班主任，创新创业先进个人**
- 第八届中国国际“互联网+”创新创业大赛 **全国金奖**
- 2024中国国际大学生创新大赛**全国金奖+银奖**
- 第十四届“挑战杯”创业竞赛**全国金奖**
- 浙江省国际大学生创新大赛**金奖×4项**
- 第七、八（**冠军**）、九届 浙江省“互联网+” 大赛 **金奖**
- 第十三、十四届浙江省“挑战杯” 创业计划竞赛 **金奖**
- 入选国家**海外优青**；浙江省**创新长期**项目（千人计划）
- 入选《麻省理工科技评论》 **35岁科技35人** （TR35）
- 阿里巴巴达摩院“**青橙奖**”
- 华为“**火花奖**” （解决通讯光源识别难题，最高价值奖）

Outline

- ❑ **光谱检测基础**
- ❑ 光谱仪微型化技术
- ❑ 成像光谱仪微型化技术
- ❑ 浙大光电子研究组研究工作

物质检测方法之“四大名谱”

质谱：将不同质量的离子按质荷比收集和记录，进行定性、定量分析及结构分析。

色谱：当流动相中所携带的混合物，按组分先后次序从固定相中分离流出，进行分析。

核磁谱：在外磁场中，原子核吸收一定频率的射频电磁波，自旋能级跃迁产生核磁共振信号。

光谱：每种物质都有特征的吸收、发射或散射光谱线，可以用来鉴别物质成分。

四大名谱的区别与联系

粒子的质量谱（质谱）

分子极性、大小、结合亲和力等（色谱）

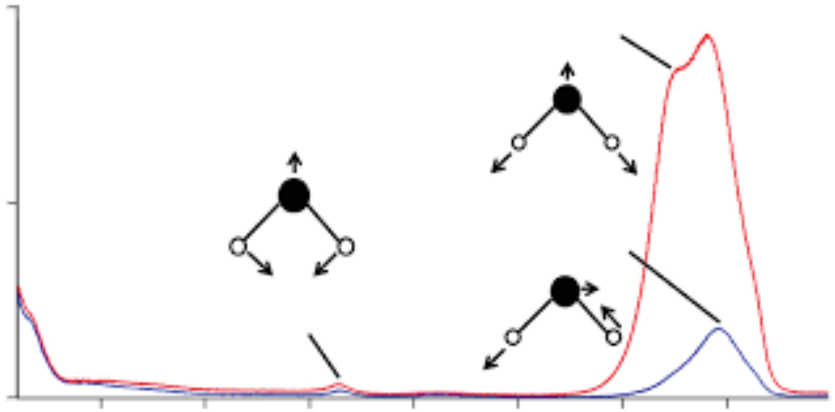
分子

原子

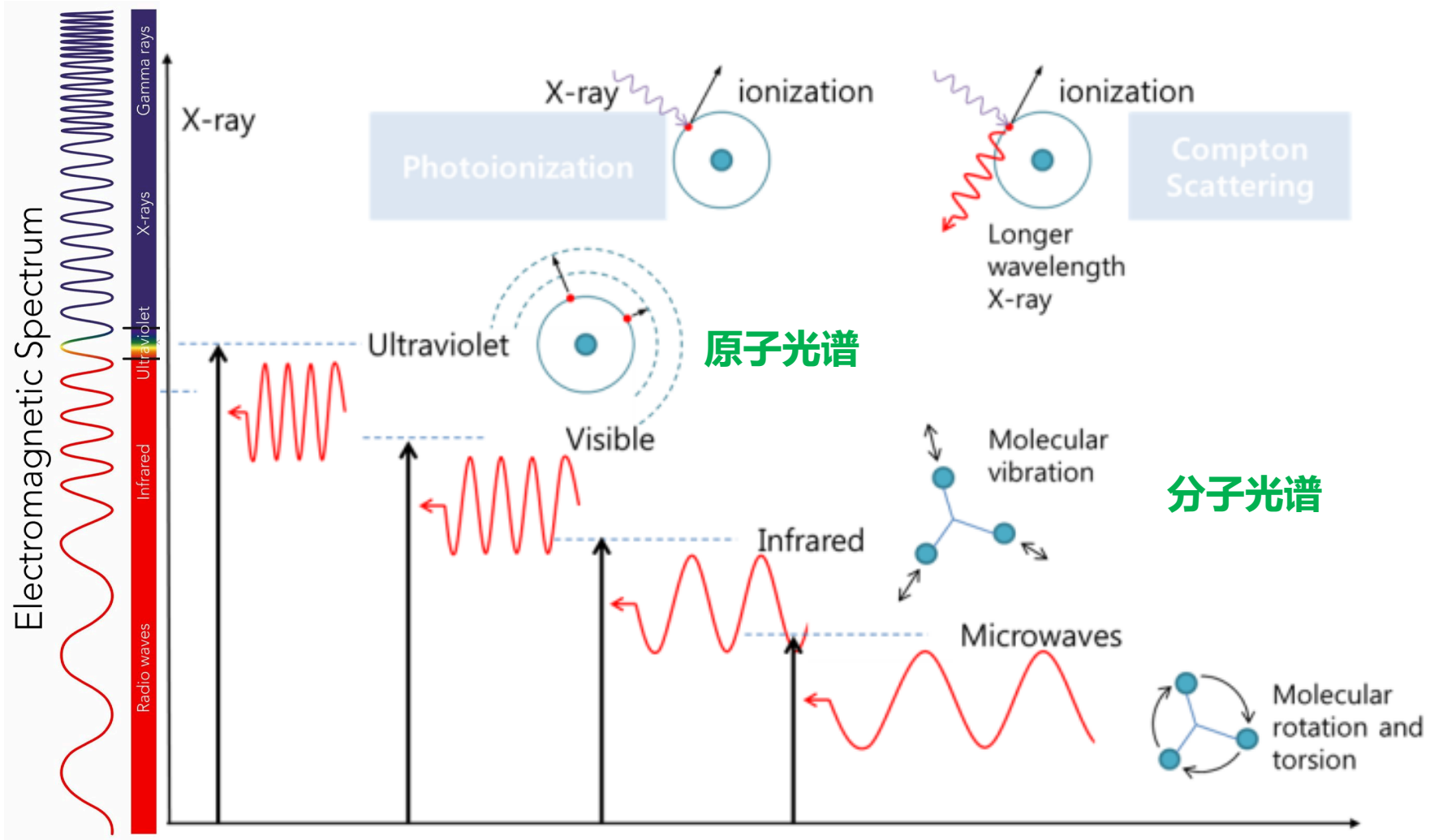
原子核外电子能级跃迁（紫外光谱）

原子核自旋能级跃迁（核磁共振）

分子跃迁（紫外-可见光谱），分子振动（红外光谱、拉曼光谱）



光谱分类


Electromagnetic Spectrum
Gamma rays
X-rays
Ultraviolet
Infrared
Radio waves
X-ray
X-ray
ionization
Photoionization
ionization
Compton Scattering
Longer wavelength X-ray
Ultraviolet
原子光谱
Visible
Molecular vibration
分子光谱
Infrared
Microwaves
Molecular rotation and torsion

光谱分析方法之间的区别

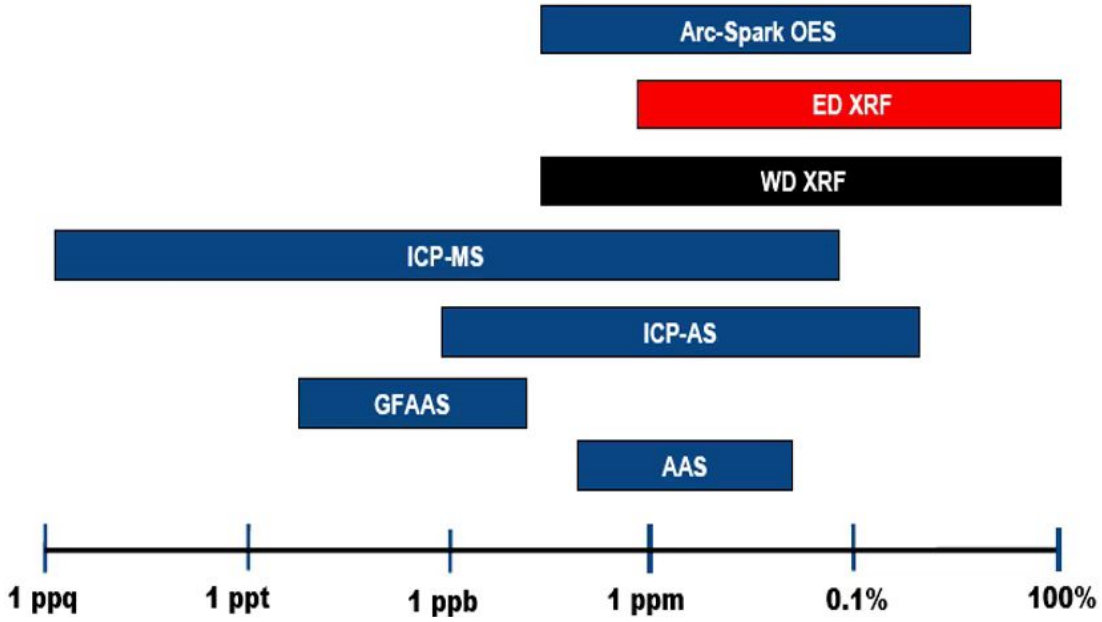
按波长区域分类：红外光谱、可见光谱和紫外光谱。

按检测的方式分类：发射、吸收、散射和荧光光谱。

按信号产生的机理分类：

分子光谱仪： 红外、紫外可见分光光度计、拉曼光谱仪

原子光谱仪： 原子吸收光谱仪（AAS）、X射线光谱仪（XRF、XRD和XPS）、火花/电弧/辉光放电光谱仪、等离子体光谱仪（LIBS和ICP-OES）



虽然都是“光谱”，但不同应用对光谱仪的性能要求差别很大，无法比较

物质检测方法之“四大名谱”

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光谱：每种物质都有特征的吸收、发射或散射光谱线，可以用来鉴别物质成分。

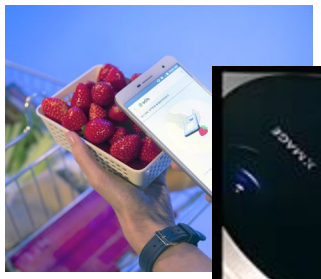
光谱：“四大名谱”中唯一可以小型化的系统

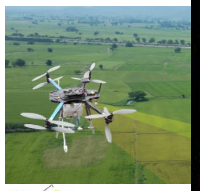
光谱仪微型化

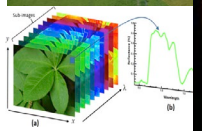
传统光谱仪体积大、价格昂贵，难以走出实验室

微型化

全新的应用场景

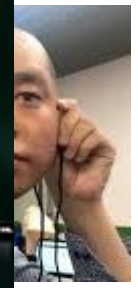






农业智能化

大健康，远程医疗传感器，航天应用

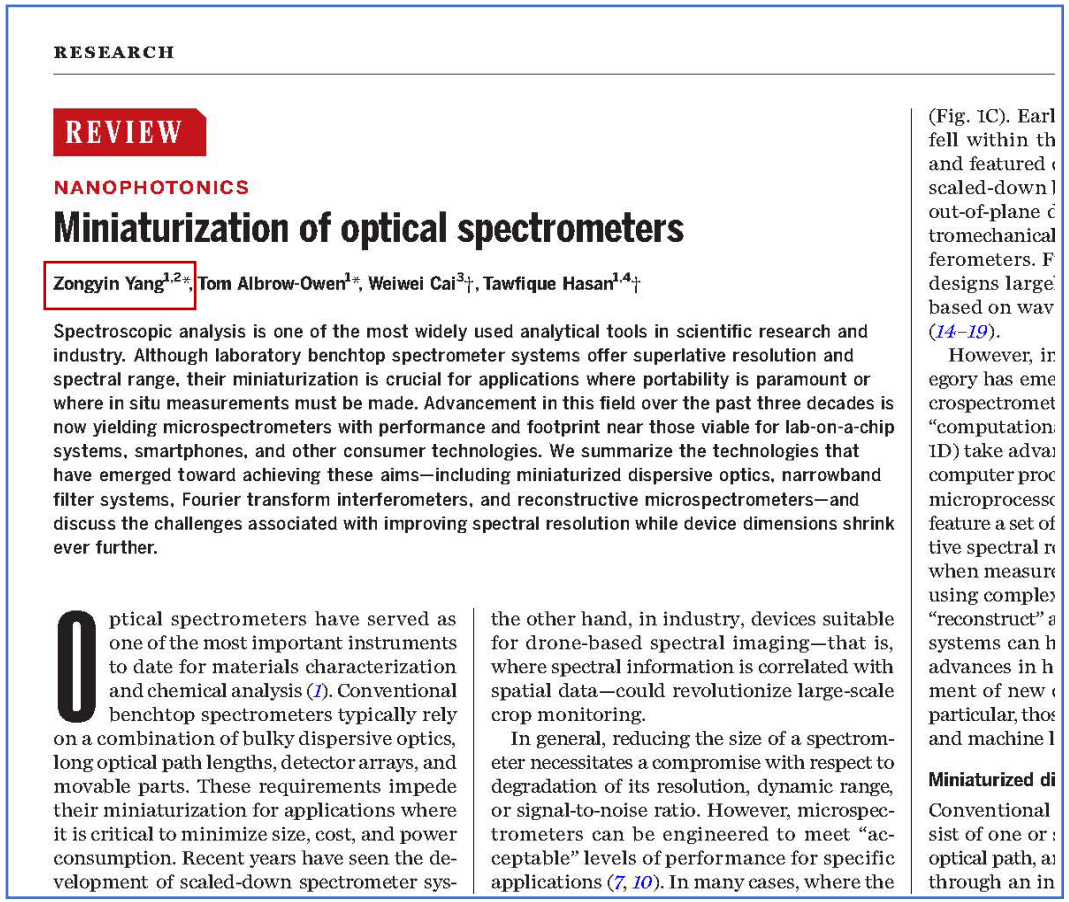


安防，安全识别

Outline

- 光谱检测基础
- **光谱仪微型化技术**
- 成像光谱仪微型化技术
- 浙大光电子研究组研究工作

提出光谱仪微型化理论与技术体系



RESEARCH

REVIEW

NANOPHOTONICS

Miniaturization of optical spectrometers

Zongyin Yang[1,2]*, Tom Albrow-Owen[1]*, Weiwei Cai[3]†, Tawfique Hasan[1,4]†

Spectroscopic analysis is one of the most widely used analytical tools in scientific research and industry. Although laboratory benchtop spectrometer systems offer superlative resolution and spectral range, their miniaturization is crucial for applications where portability is paramount or where in situ measurements must be made. Advancement in this field over the past three decades is now yielding microspectrometers with performance and footprint near those viable for lab-on-a-chip systems, smartphones, and other consumer technologies. We summarize the technologies that have emerged toward achieving these aims—including miniaturized dispersive optics, narrowband filter systems, Fourier transform interferometers, and reconstructive microspectrometers—and discuss the challenges associated with improving spectral resolution while device dimensions shrink ever further.

Optical spectrometers have served as one of the most important instruments to date for materials characterization and chemical analysis (*1*). Conventional benchtop spectrometers typically rely on a combination of bulky dispersive optics, long optical path lengths, detector arrays, and movable parts. These requirements impede their miniaturization for applications where it is critical to minimize size, cost, and power consumption. Recent years have seen the development of scaled-down spectrometer systems that fell within th and featured scaled-down out-of-plane tromechanical ferometers. F designs large based on wav (*14*–*19*).

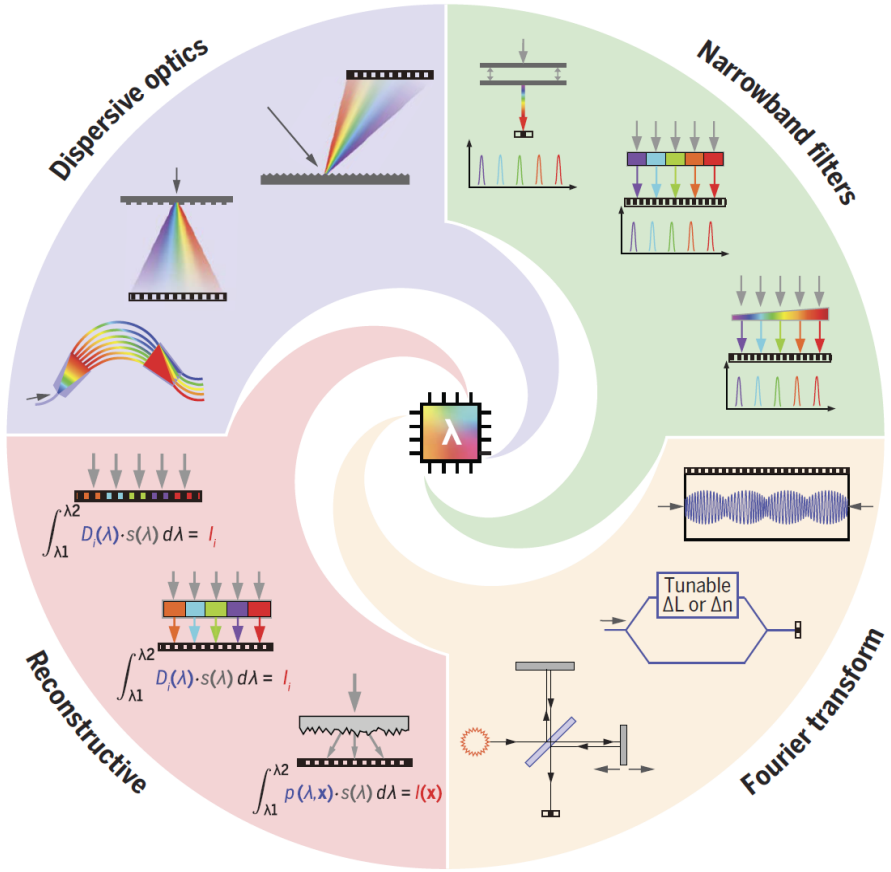
However, in egory has eme crospectromet "computation 1D) take adva computer proc microprocess feature a set of tive spectral r when measur using comple "reconstruct" a systems can h advances in h ment of new particular, thos and machine l

the other hand, in industry, devices suitable for drone-based spectral imaging—that is, where spectral information is correlated with spatial data—could revolutionize large-scale crop monitoring.

In general, reducing the size of a spectrometer necessitates a compromise with respect to degradation of its resolution, dynamic range, or signal-to-noise ratio. However, microspectrometers can be engineered to meet "acceptable" levels of performance for specific applications (*7*, *10*). In many cases, where the

Miniaturized di

Conventional sist of one or optical path, a through an in



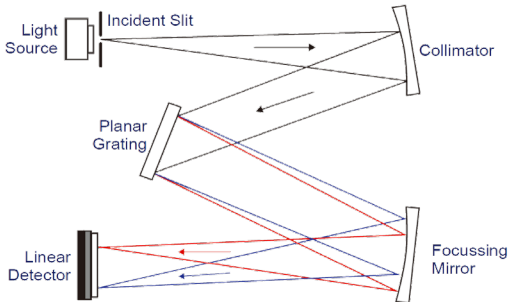
Science 综述：总结并提出了关于微型光谱仪的理论与技术体系

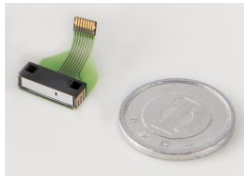
Z. Y. Yang *et al.*, ***Science*** 371, eabe0722 (2021)

W. W. Cai, **Z. Y. Yang*** *et al.*, ***Nature Photonics*** 18, 651 (2024)

Journal of Physics: Photonics 5, 042501 (2023)

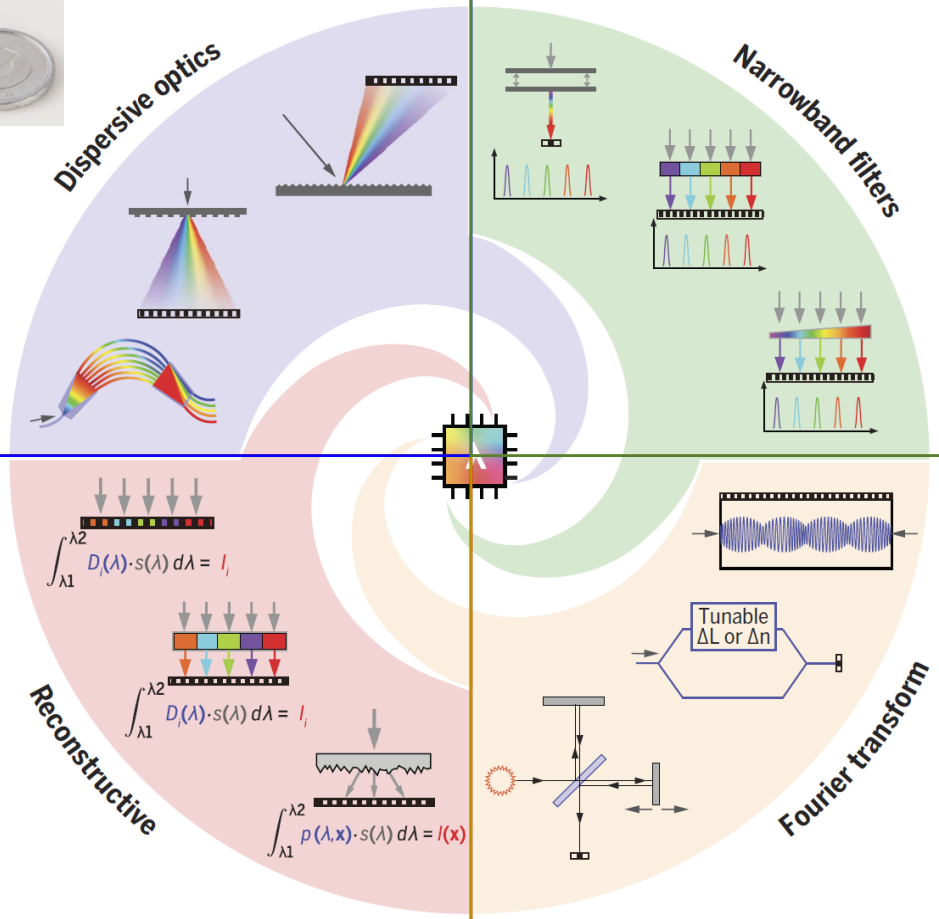
光谱仪微型化策略

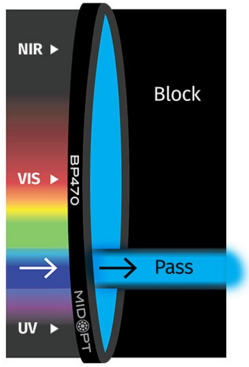




传统光谱仪体积庞大：光栅、探测器阵列、准直光路

缩小尺寸，**性能大幅下降**



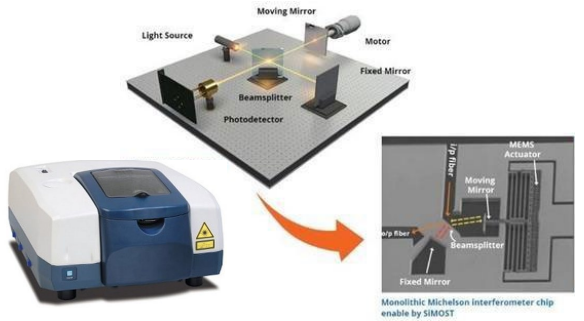


高分辨率与滤光片数量和成本矛盾

前三种路线在性能和成本方面有难以克服的挑战。

计算光谱用算力增强光学元件，能进一步减小器件的尺寸和成本。

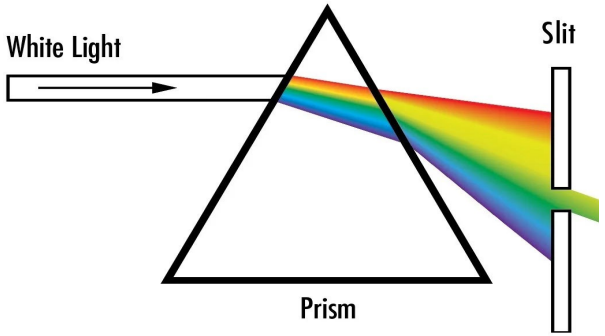
计算光谱⁺ 是大趋势



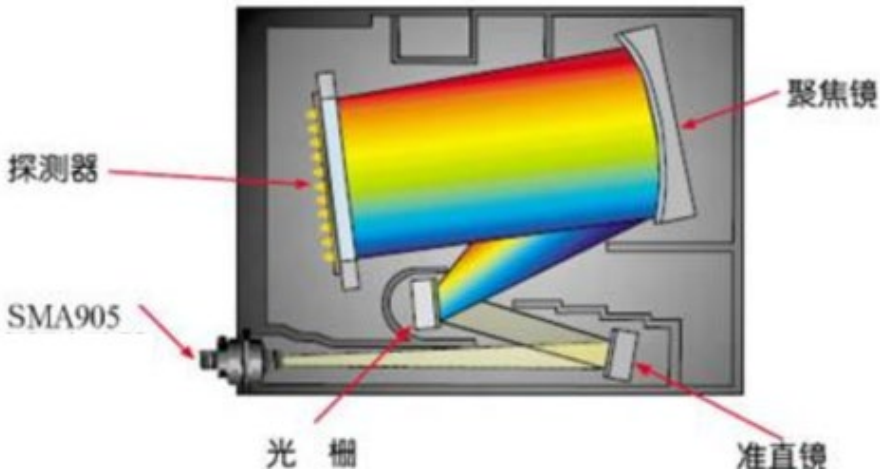
运动部件导致稳定性差，分辨率低；
波导通光量低，灵敏度低

Z. Y. Yang *et al.*, ***Science*** 371, eabe0722 (2021)

2.1 色散型光谱仪-基本原理

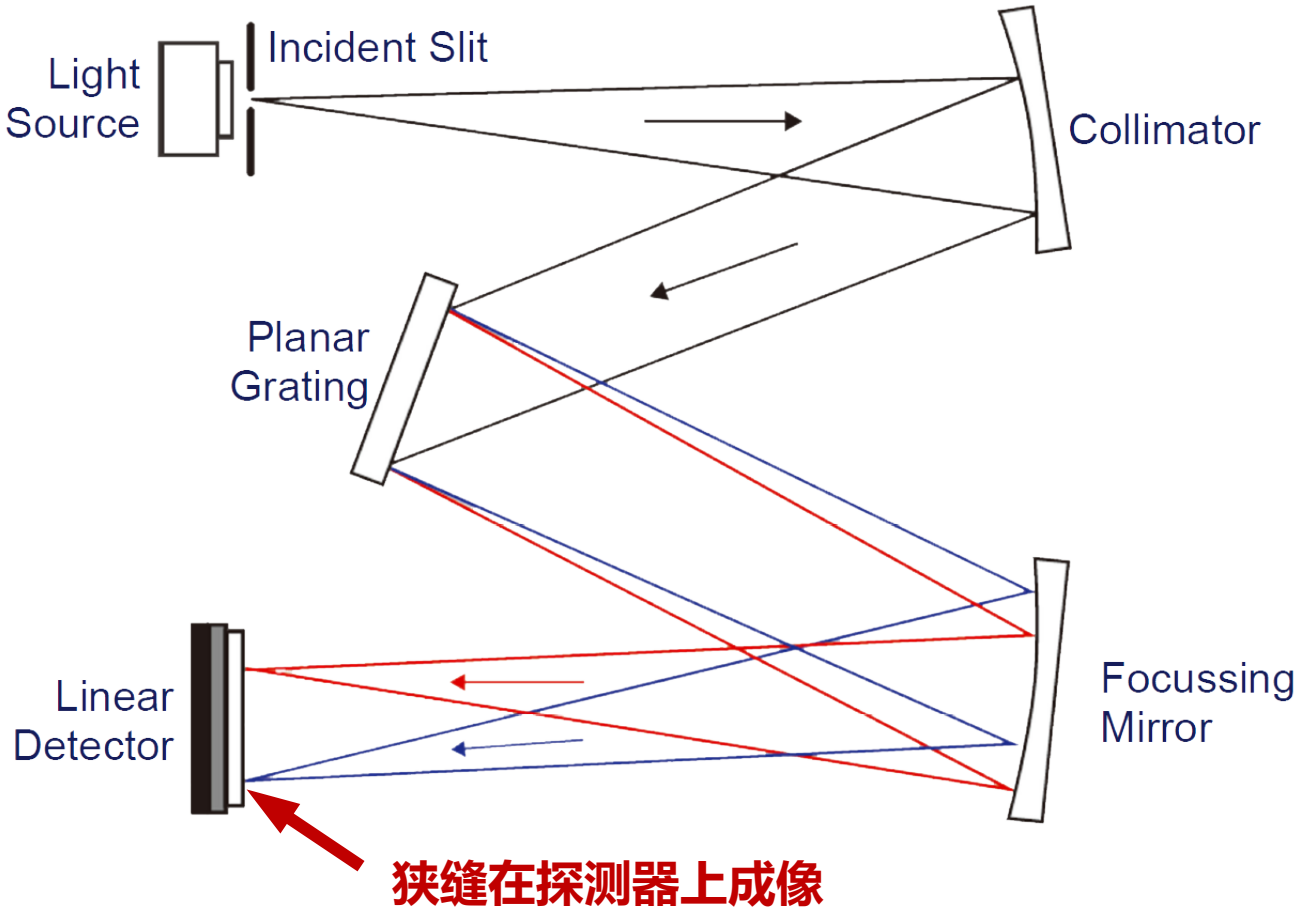


白光色散实验实际上光谱分辨率很低，从色散到光谱仪需要特殊光路。

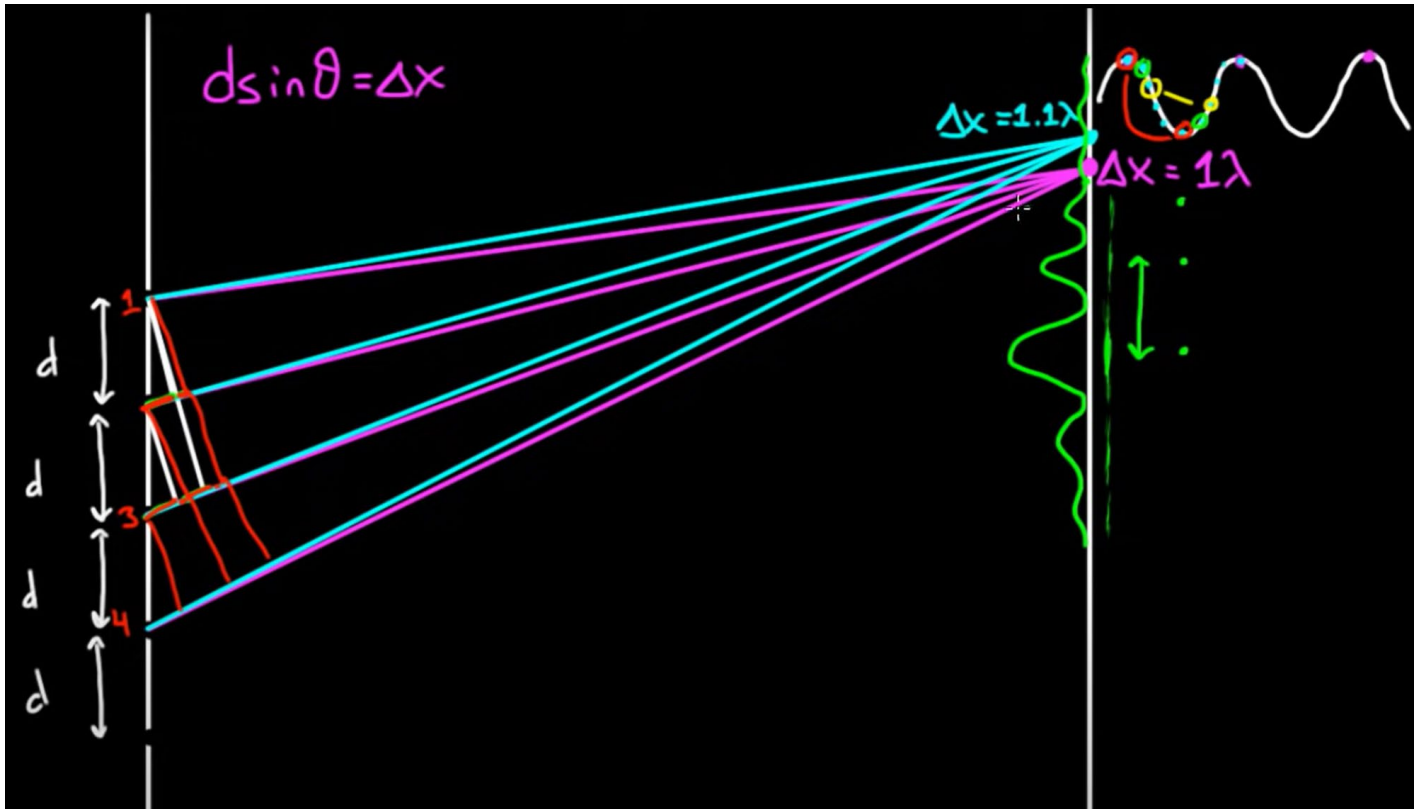


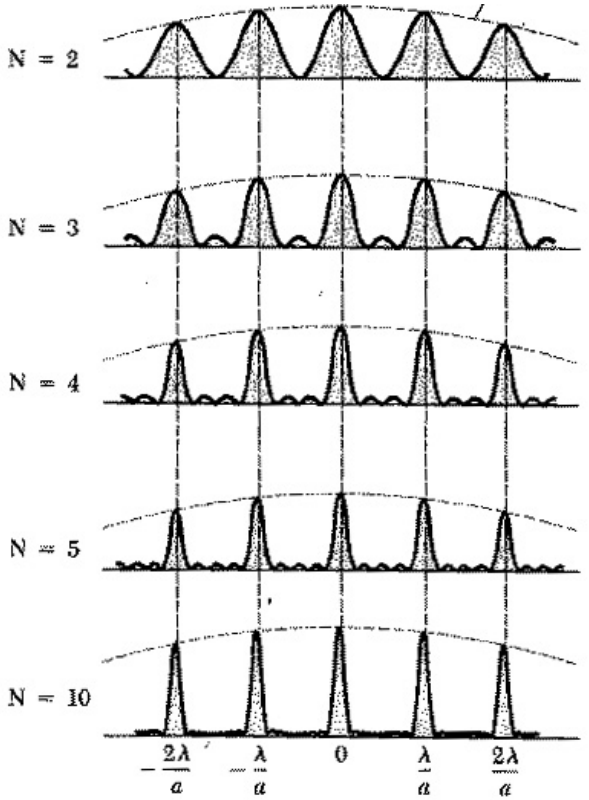
色散型光谱仪的基本结构：

经典Czerny-Turner （CT）型

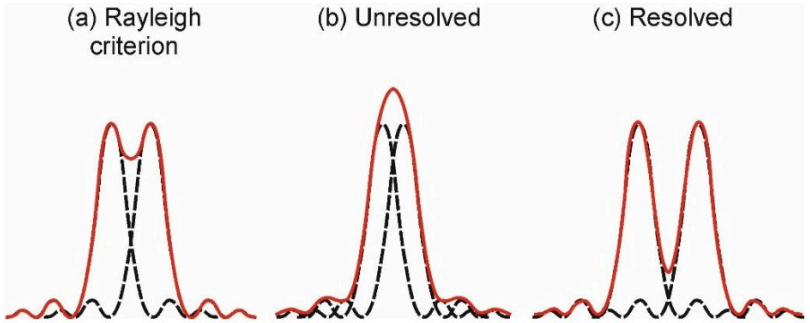


2.1 色散型光谱仪-光栅刻线与光谱分辨率



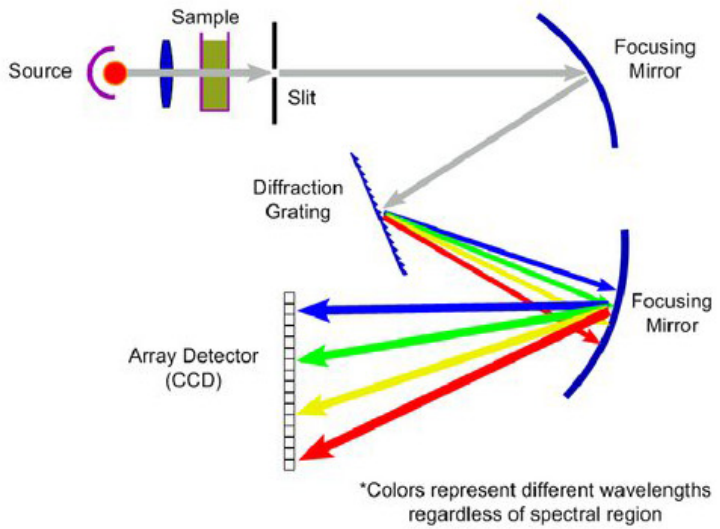


刻线数越多光谱分辨率越高

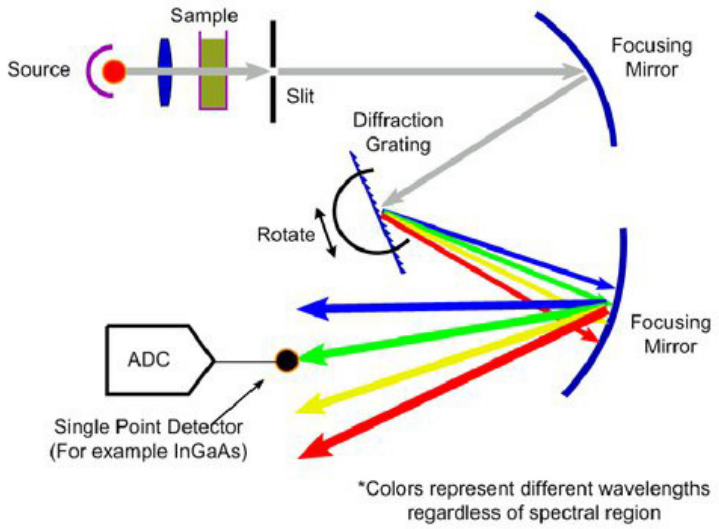


光谱分辨率判断标准-Rayleigh Criterion：两个峰间距等于其中一个峰的半高全宽，或者说其中一个峰的尾端刚好在另一个峰的中心位置。

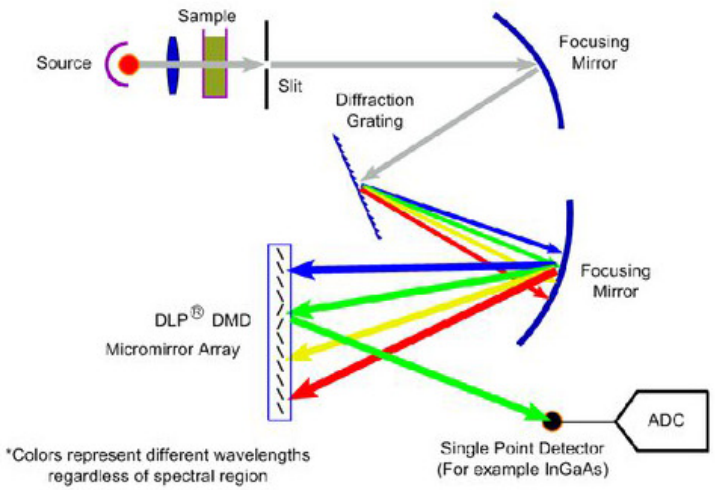
2.1 色散型光谱仪



Array detector spectrometer



Scanning grating spectrometer



DLP spectrometer

美国Ocean Optics海洋光学

日本Horiba光谱仪

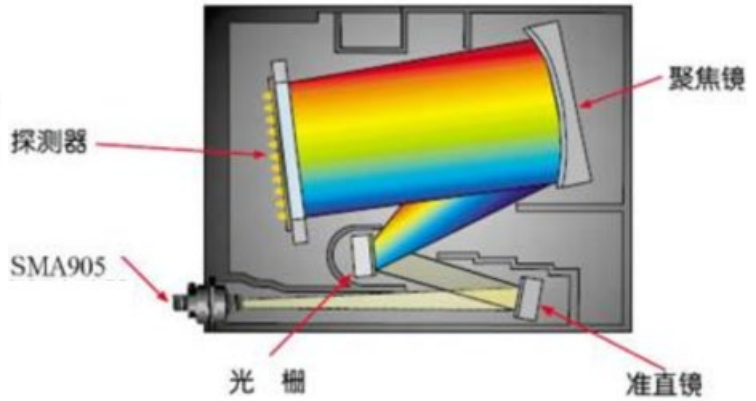
美国Ti DLP光谱仪

Czerny-Turner （CT）光谱仪结构及变种

2.1 色散型光谱仪

美国Ocean Optics海洋光学：交叉式CT设计，慧差校准好，交叉折叠使得结构更紧凑。

但是入射焦距和色散焦距比小于1，导致在探测器上的像大于狭缝，降低分辨率，并且交叉光路带来杂散光干扰。



荷兰Avantes：M型对称CT设计，入射焦距和色散焦距1:1，像差优化好，分辨率较交叉式的高，但是体积比交叉式的大。

基本型（M型）和交叉型CT结构的小型光纤光谱仪

2.1 色散型微光谱仪

海洋光学的STS系列超小体积可见光光谱仪，尺寸到40×42×24 mm

Ibsen的Pebble系列光谱仪尺寸只有20×15×8 mm

基于CT结构的分立元件色散型微型光谱仪的尺寸极限，分辨率大于5 nm

2.1 色散型微光谱仪

Bayspec的掌上型光谱仪

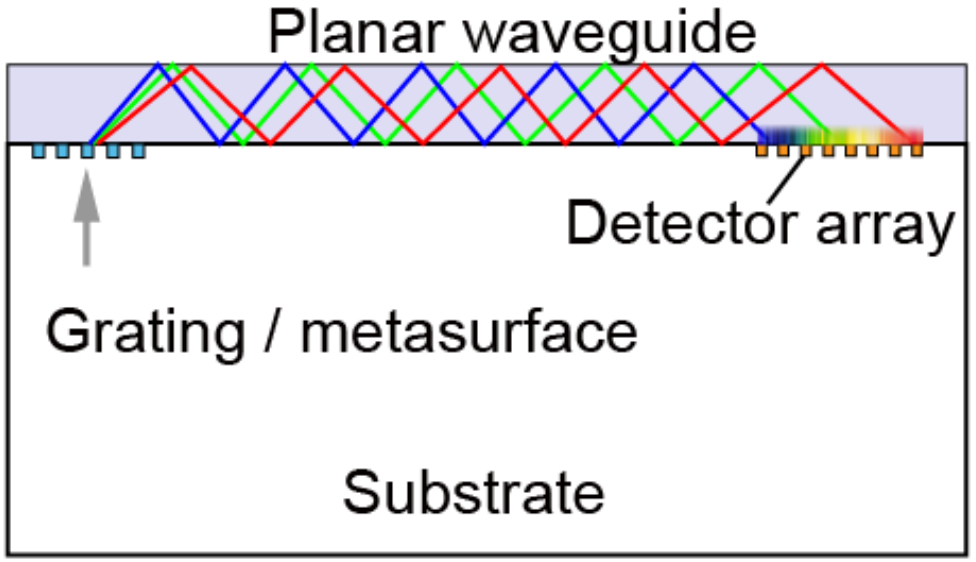
Goyalab的GoSpectro手持式光谱仪

LinkSquare手持食物分析仪

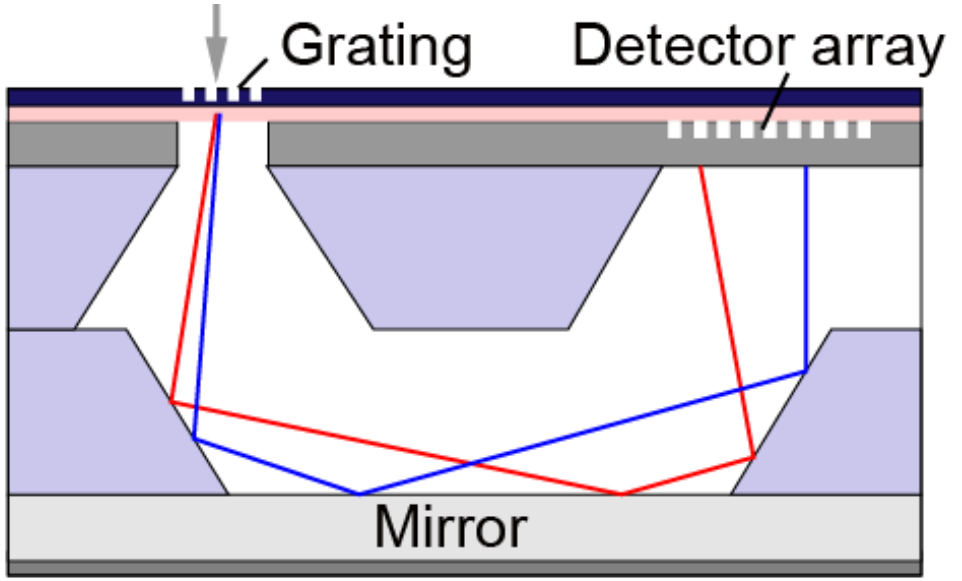
基于CT结构的分立元件色散型微型光谱仪的应用，分辨率大于10 nm

2.1 色散型微光谱仪

从分立元件到集成化设计，进一步减小体积



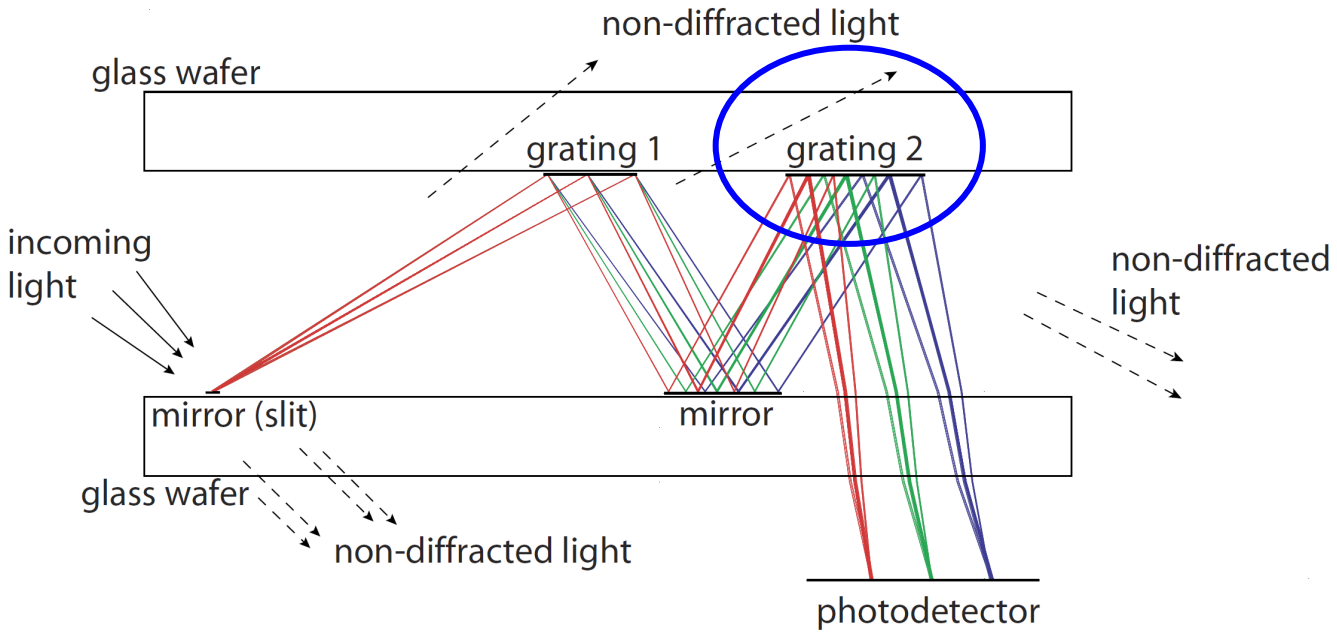
Don S. Goldman *et al*., ***Applied Optics*** 29, 4583 (1990)

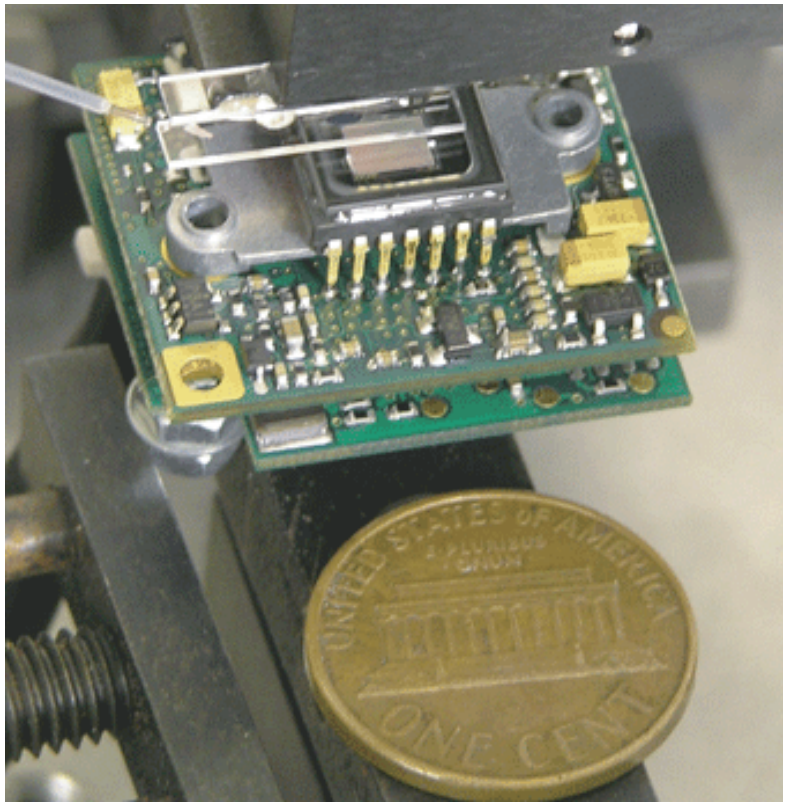


T. Kwa *et al*., ***Sensors and Actuators A*** 31, 259 (1992)

缺了准直光路！

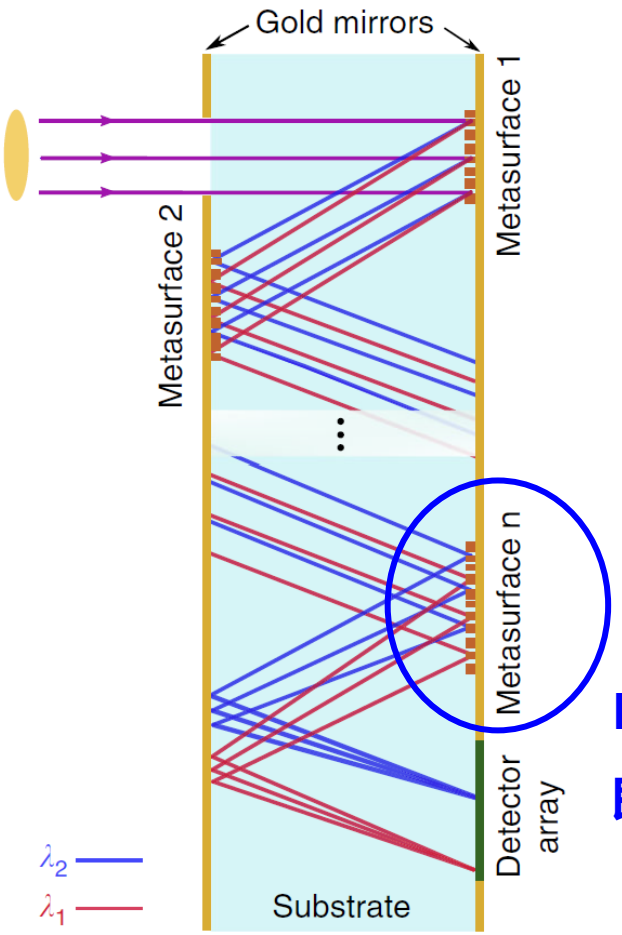
Grating 2：即分光又准直

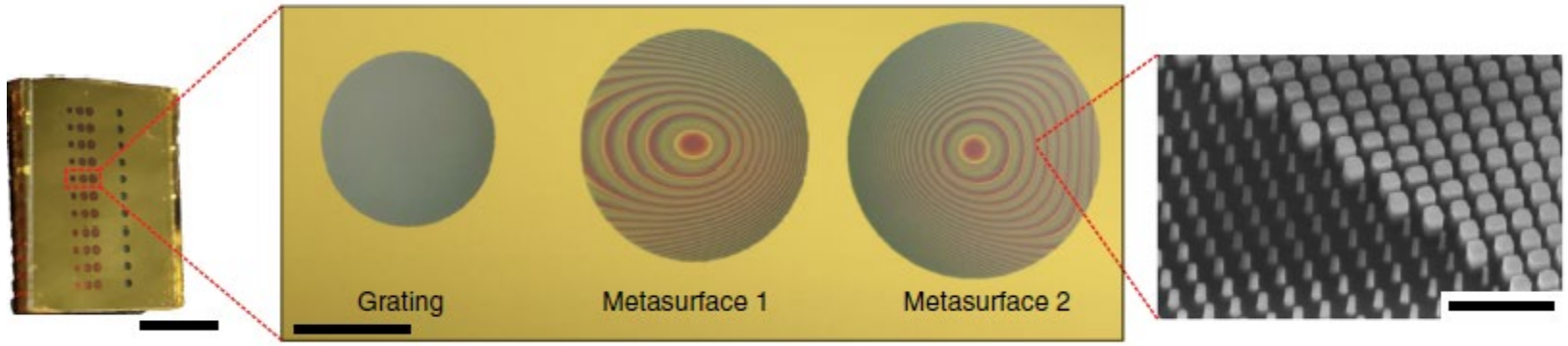




S. Grabarnik *et al.*, ***Opt. Express*** *15*, 3581 (2007)

2.1 色散型光谱仪



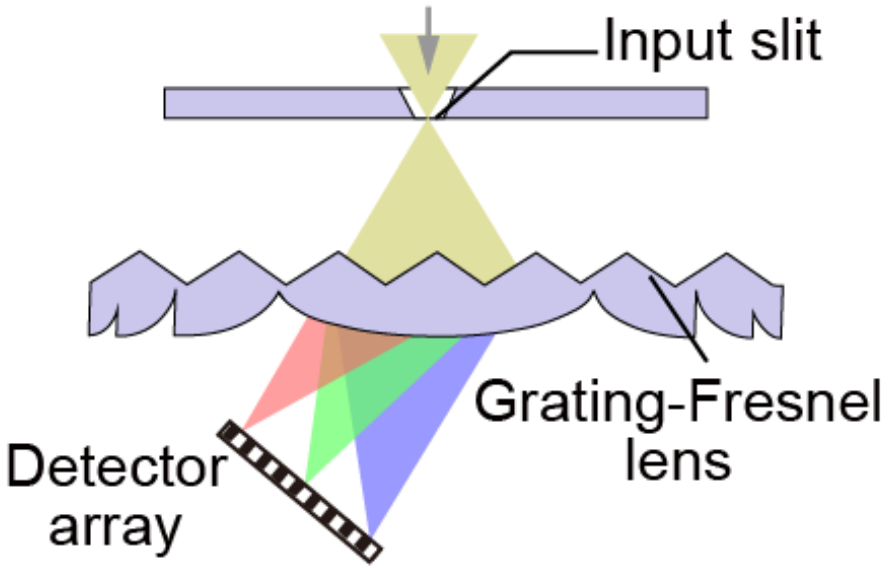


Metasurface:

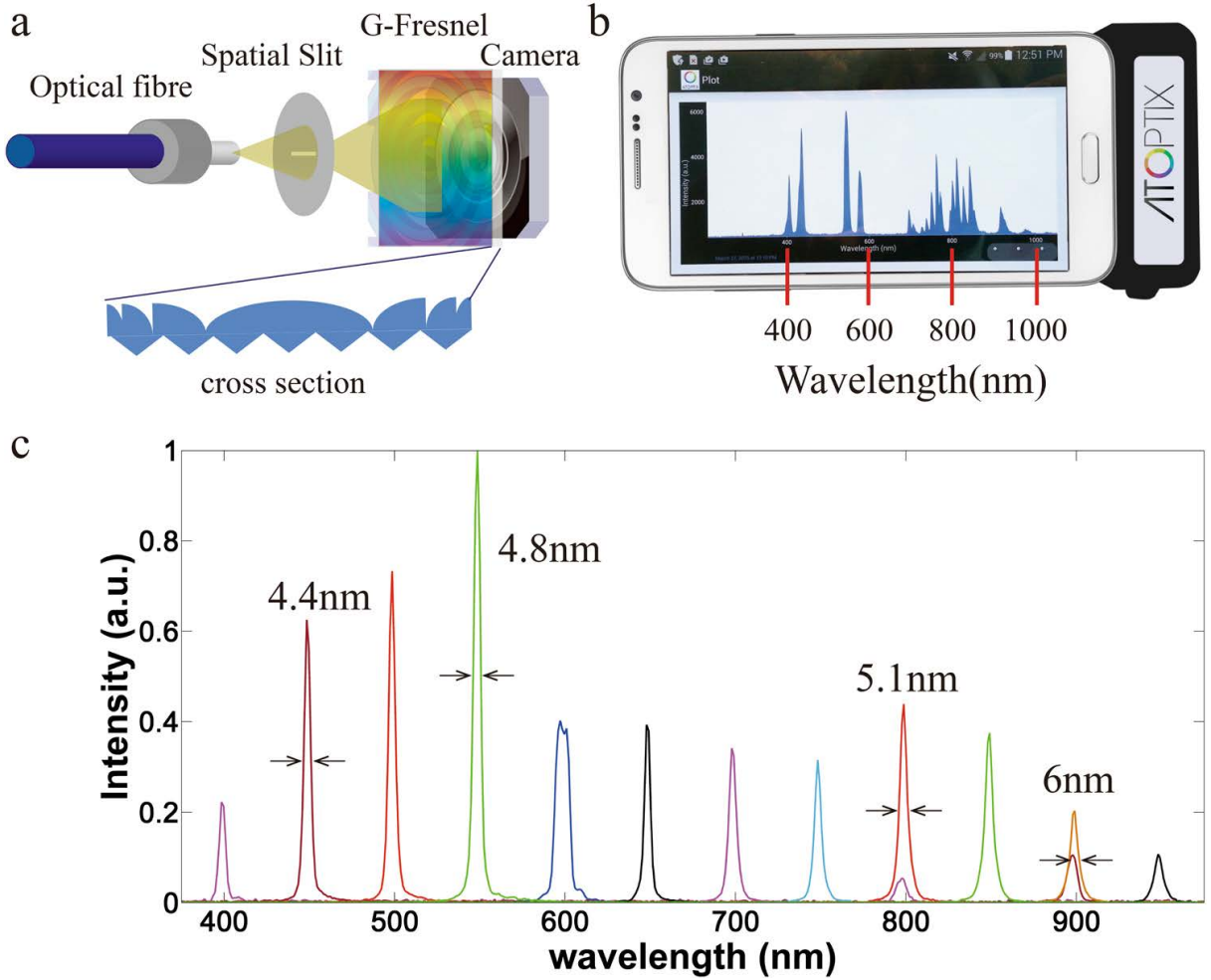
即分光又准直

M. Faraji-Dana *et al.*, ***Nature Communicatons*** *9*, 4196 (2018)

2.1 色散型微光谱仪

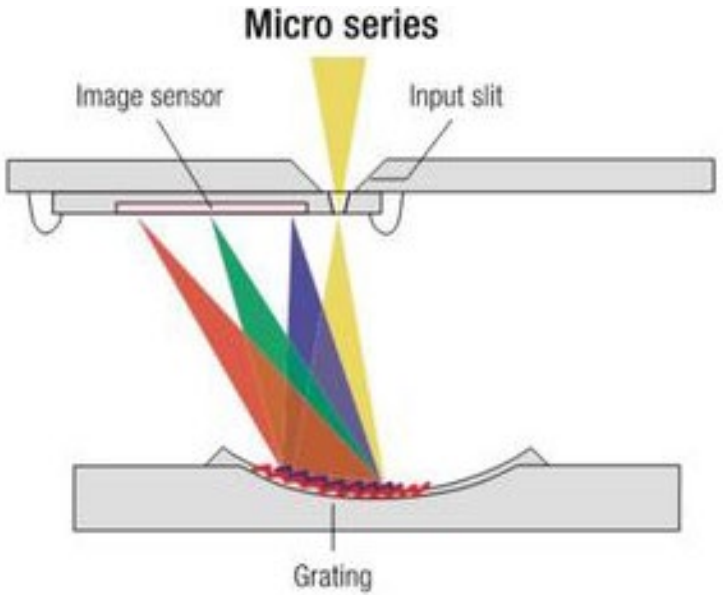


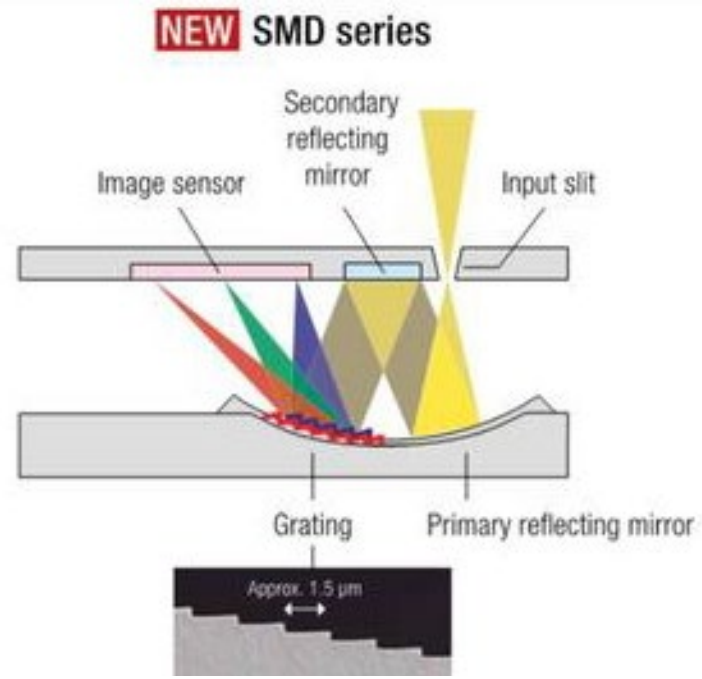
光栅与准直透镜（菲涅尔透镜）一体化设计

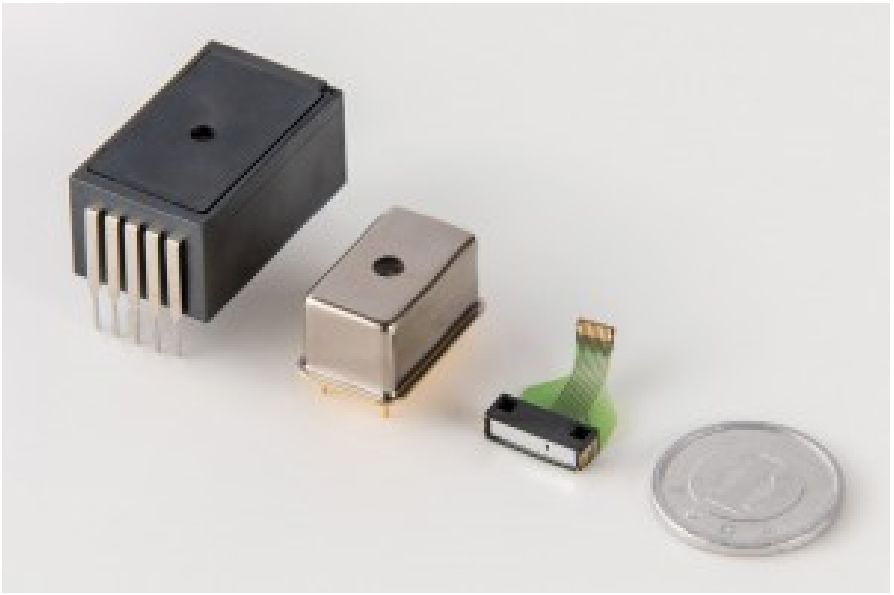


P. Edwards *et al*., ***Scientific Reports*** *7*, 12224 (2017)

2.1 色散型微光谱仪







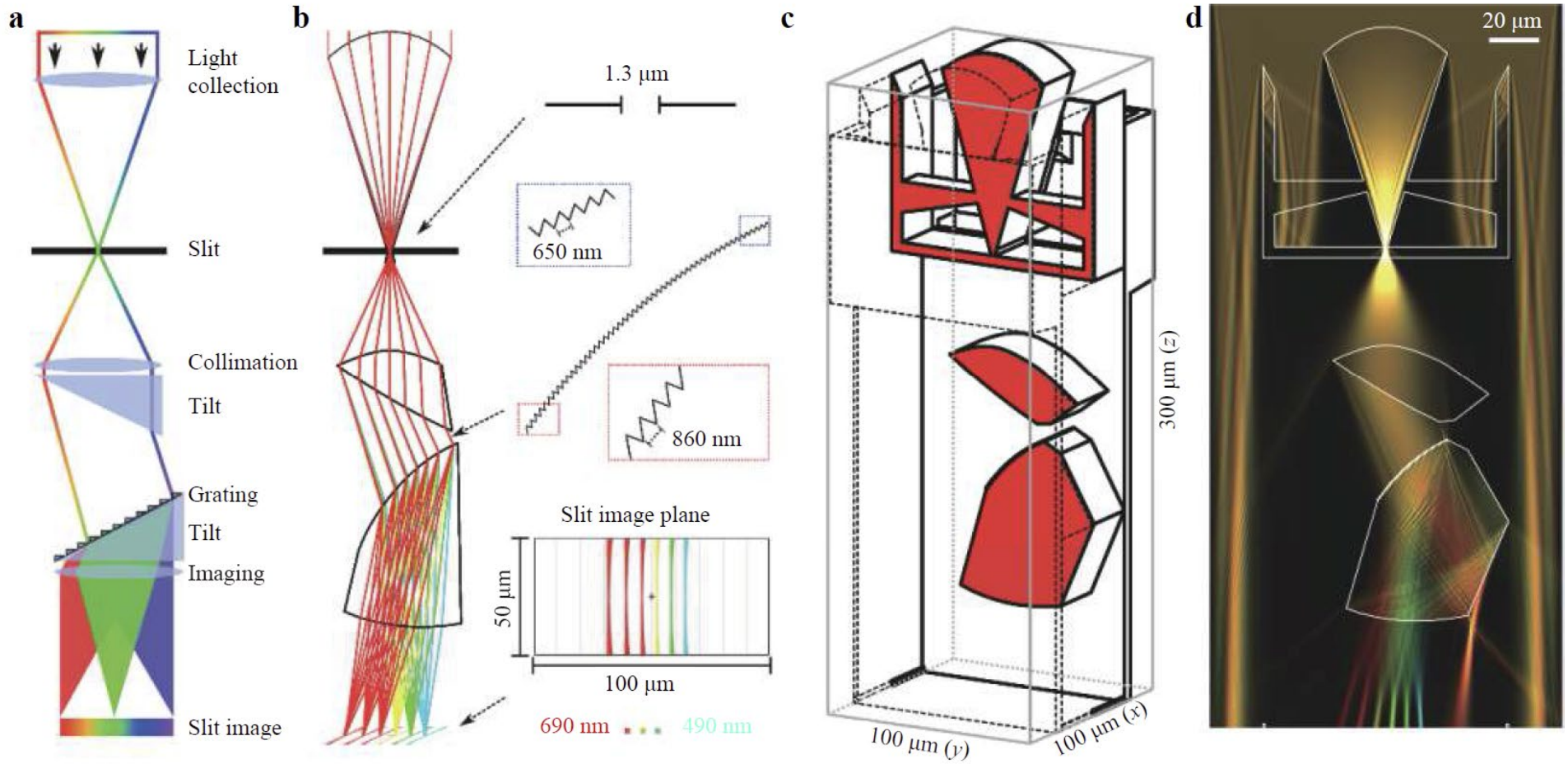
滨松（Hamamatsu）凹面光栅微型光谱仪

2.1 色散型微光谱仪

基于滨松微型光谱仪元件开发的小型检测仪

基于滨松微型光谱仪元件开发的照度仪

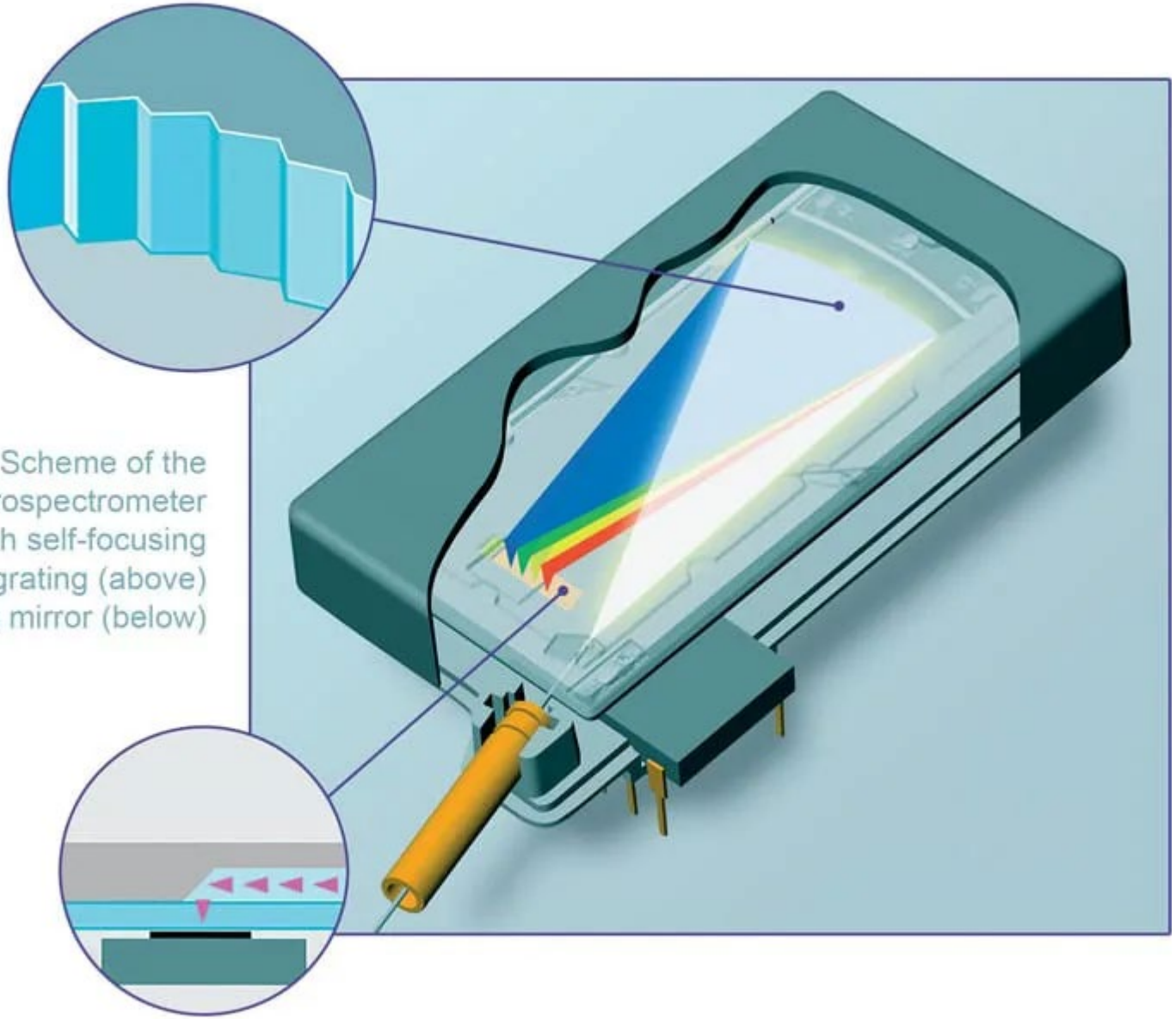
2.1 色散型微光谱仪



A. Toulouse *et al.*, ***Light: Advanced Manufacturing*** 1, 5 (2020)

3D打印色散型微光谱仪，仅100×100×300 μm³，分辨率10 nm（532 nm）

2.1 片上集成色散型微光谱仪

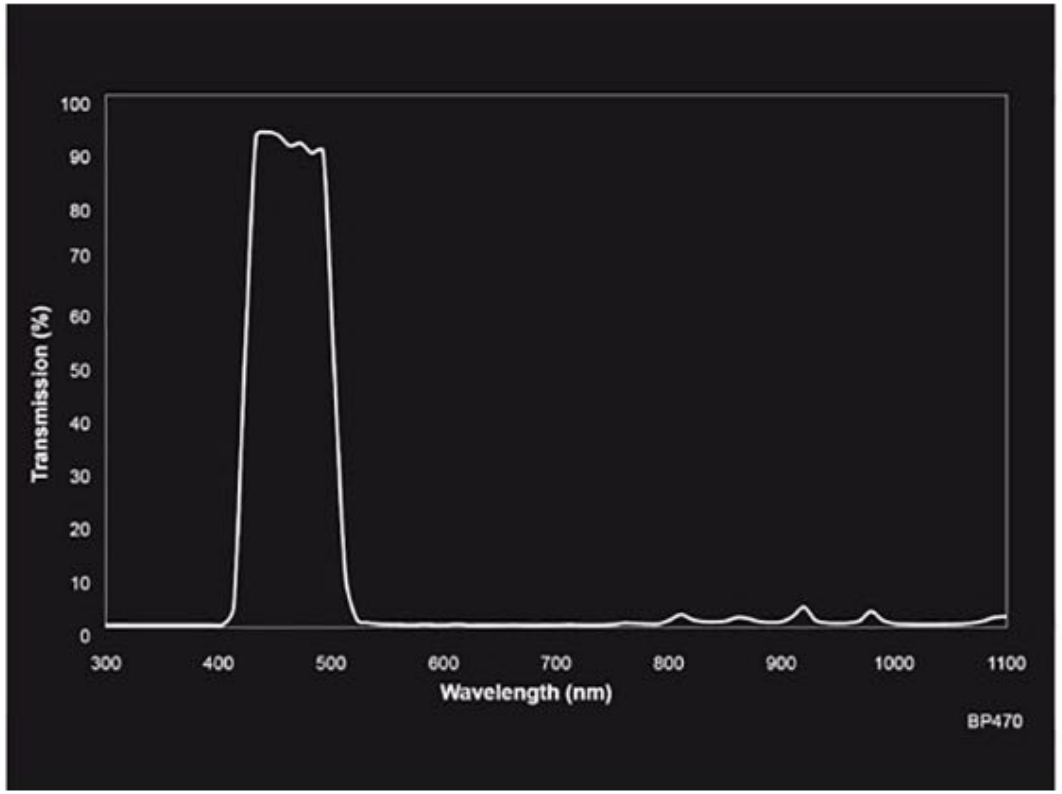


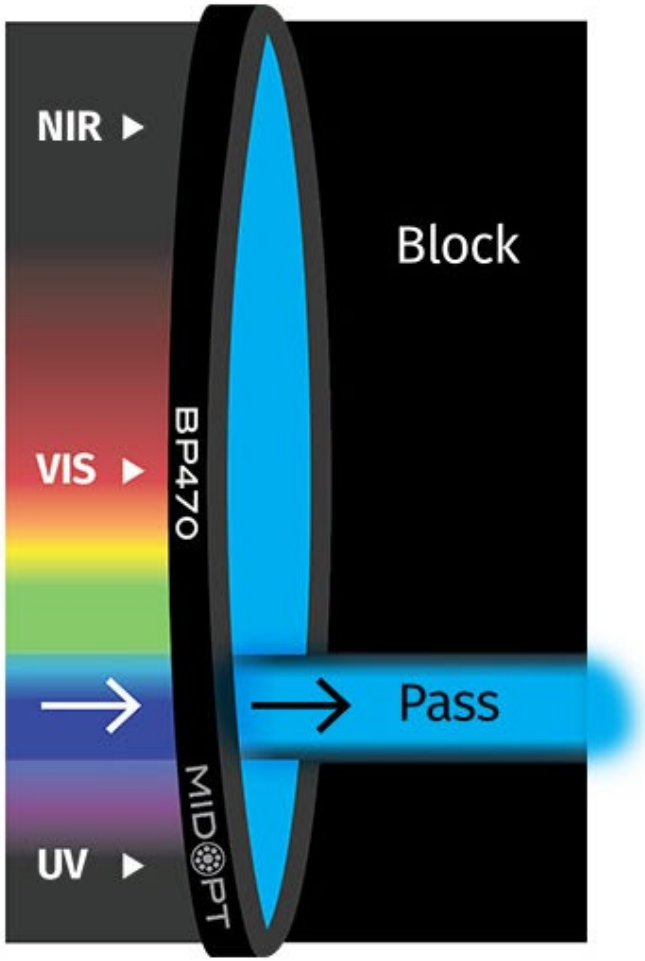
Scheme of the microspectrometer with self-focusing diffraction grating (above) and deflecting mirror (below)

片上集成色散型光谱仪产业化代表公司

德国Insion：分辨率>10 nm

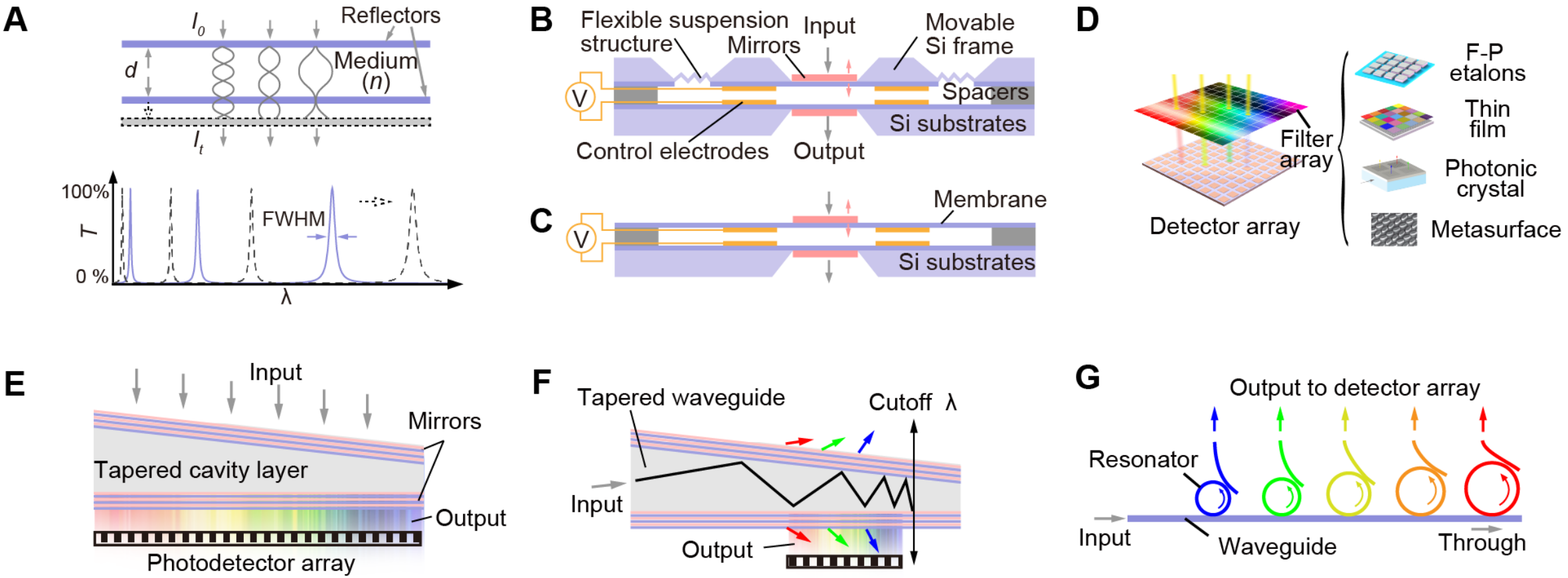
2.2 窄带滤光型微光谱仪





窄带滤光：弱水三千只取一瓢

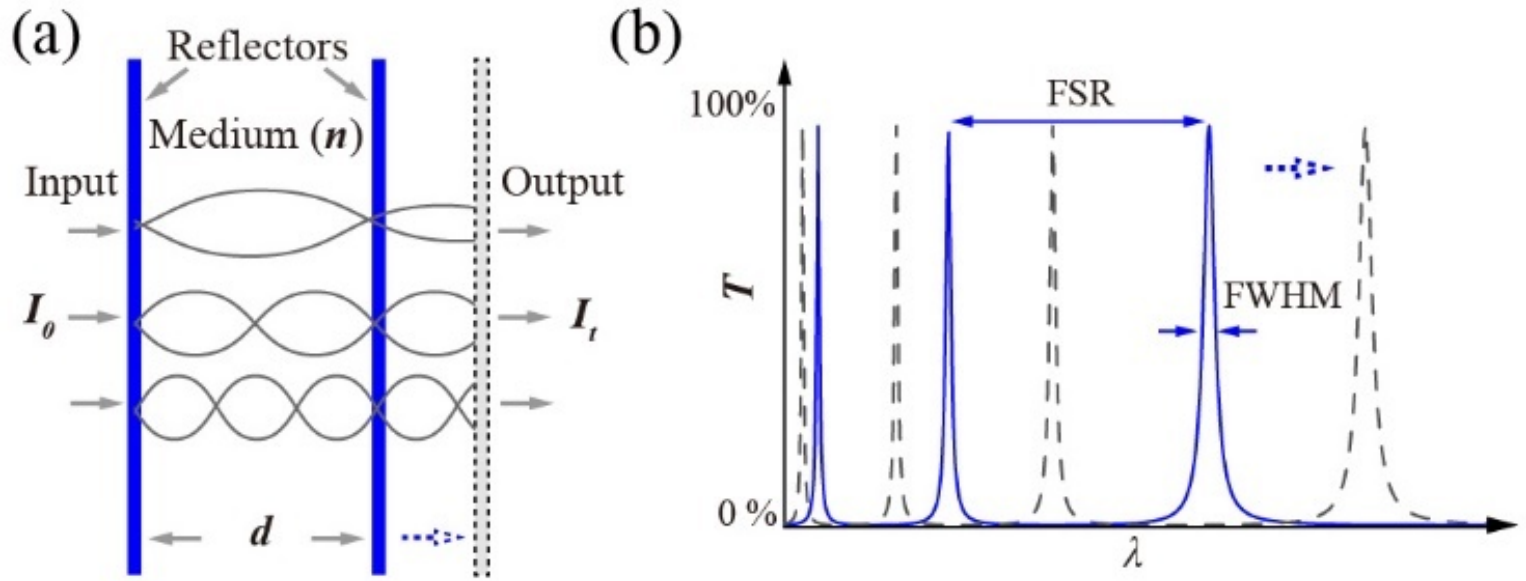
2.2 窄带滤光型微光谱仪



与色散型相比，窄带滤光型光谱仪无须长光路和光学元件的精准对准来获得高分辨率。

按照分光方式：可调滤光片，滤光片阵列，线性渐变滤光片，波导滤光

2.2 窄带滤光型微光谱仪-可调滤光片



吸收损耗

$$T = \left(1 - \frac{A}{1 - r}\right)^2 \frac{1}{1 + \frac{4r}{(1-r)^2} sin^2\left(\frac{2\pi}{\lambda} ndcos\theta - \varphi\right)}$$

镜子反射率

光程

$$r = r_1 \times r_2$$

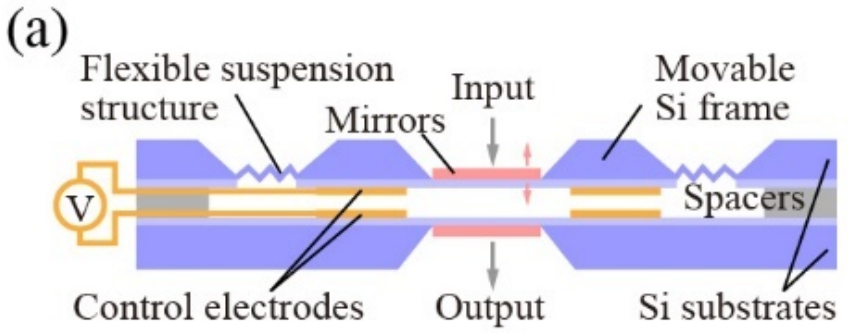
$$F_{int} = \pi\sqrt{r} / (1\text{-}r)$$

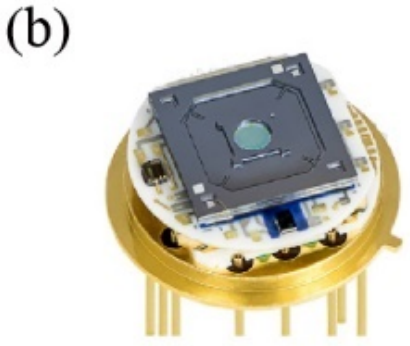
F-P谐振腔（Fabry–Pérot Interferometer）可调谐滤波器

2.2 窄带滤光型微光谱仪-可调滤光片

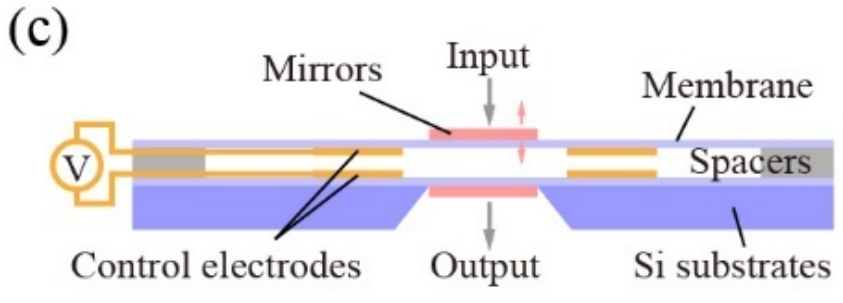
$$T = \left(1 - \frac{A}{1-r}\right)^2 \frac{1}{1 + \frac{4r}{(1-r)^2} sin^2\left(\frac{2\pi}{\lambda} ndcos\theta - \varphi\right)}$$

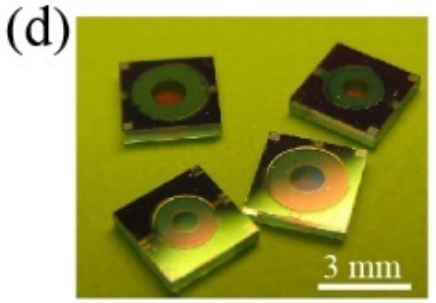
通过调腔长d调节光程





德国InfraTech的块体型F-P腔可调谐滤波器微型光谱仪



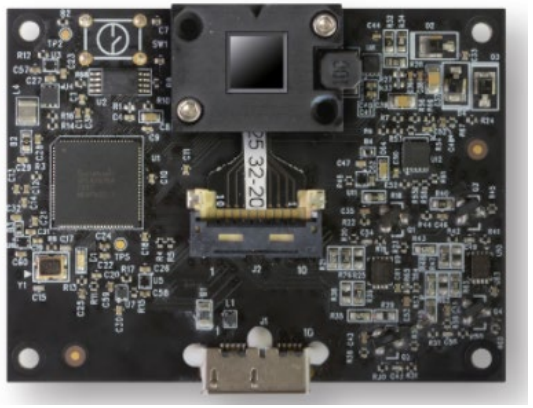


芬兰VTT开发的薄膜型F-P腔可调谐滤波器微型光谱仪

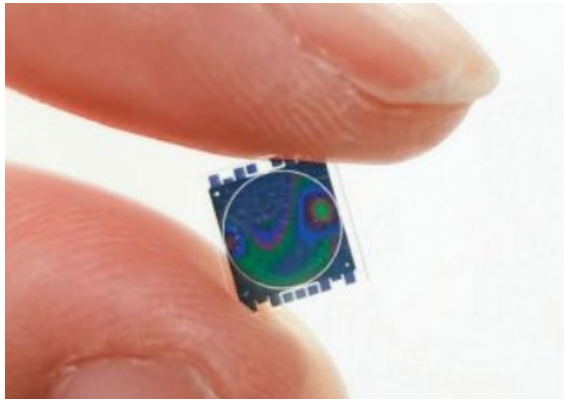
芬兰Spectral Engines 公司的F-P腔可调谐滤波器微型光谱仪

日本滨松的F-P腔可调谐滤波器微型光谱仪

2.2 窄带滤光型微光谱仪-可调滤光片



以色列Unispectral公司：静电驱动F-P腔可调谐滤波器微型成像光谱仪



深圳海谱：压电驱动F-P腔可调谐滤波器微型成像光谱仪

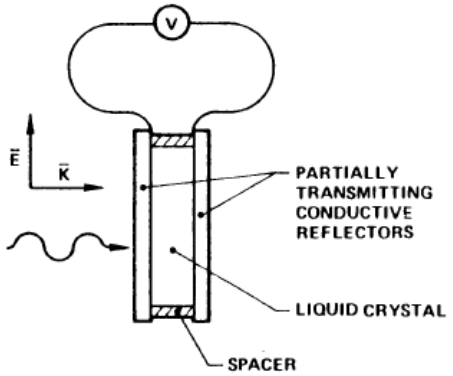
F-P腔可调谐滤波器微型成像光谱仪

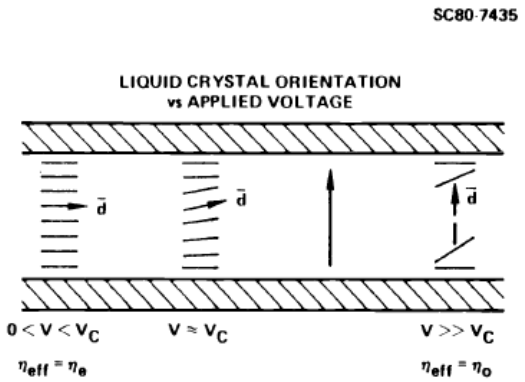
注：成像光谱仪是成像与光谱探测技术的结合，相当于图像里的每个像素都是光谱仪。成像光谱相比单点光谱是多了二维的空间信息，与二维探测器获得的普通照片相比又具有更高的维度，然而目前可获得的探测器只有空间上的两维，离光谱成像的三维数据和缺一个维度，因此成像光谱仪要么通过牺牲测量时间获得额外的光谱维度数据，要么通过牺牲空间分辨率拓展出光谱维度。

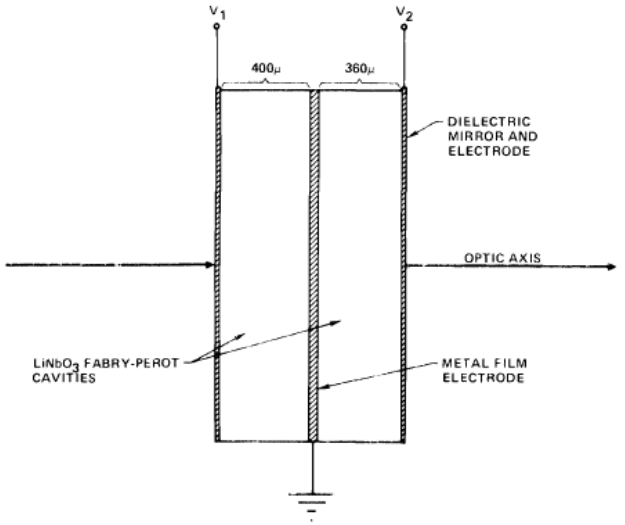
2.2 窄带滤光型微光谱仪-可调滤光片

$$T = \left(1 - \frac{A}{1-r}\right)^2 \frac{1}{1 + \frac{4r}{(1-r)^2} sin^2\left(\frac{2\pi}{\lambda} ndcos\theta - \varphi\right)}$$

通过调折射率n调节光程







W. Gunning *et al.*, ***SPIE Imaging Spectroscopy*** 268, 190 (1981)

W. Gunning *et al.*, ***SPIE Active Optical Devices*** 202, 21 (1979)

液晶可调谐F-P腔滤波器

Optic Express

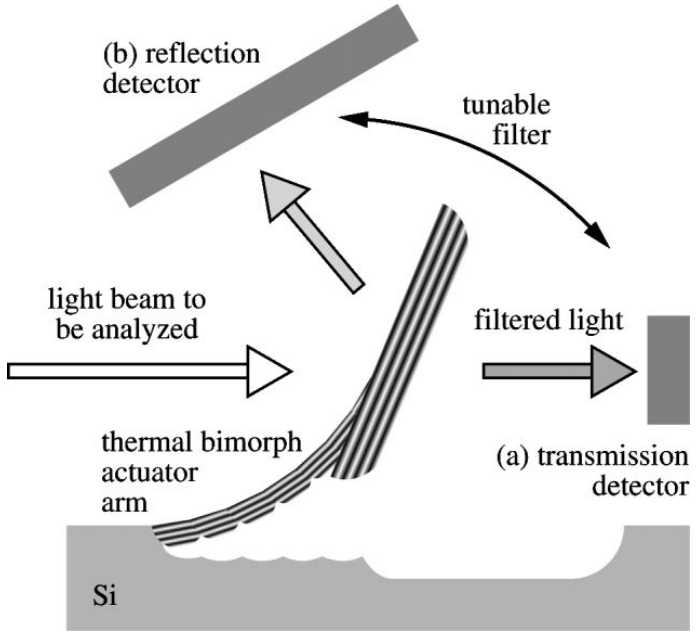
相变材料可调谐F-P腔滤波器

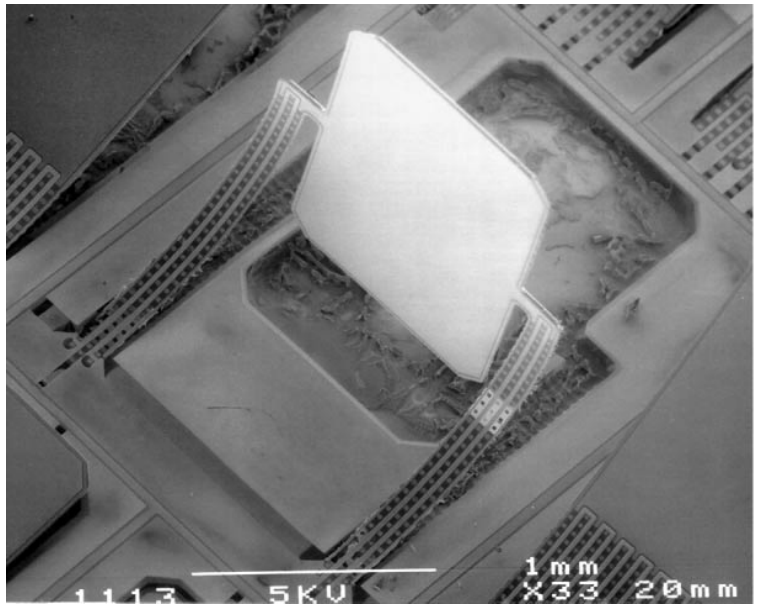
铌酸锂可调谐F-P腔滤波器

2.2 窄带滤光型微光谱仪-可调滤光片

$$T = \left(1 - \frac{A}{1-r}\right)^2 \frac{1}{1 + \frac{4r}{(1-r)^2} sin^2\left(\frac{2\pi}{\lambda} ndcos\theta - \varphi\right)}$$

通过调入射角θ调节光程

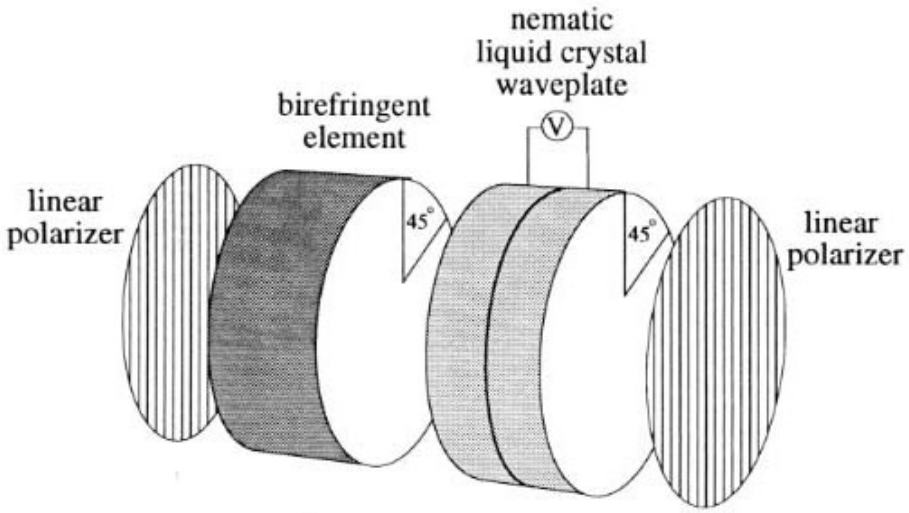




G. Lammel *et al.*, ***Sensors and Actuators A*** 92, 52 (2001)

角度可调谐F-P腔滤波器

2.2 窄带滤光型微光谱仪-可调滤光片

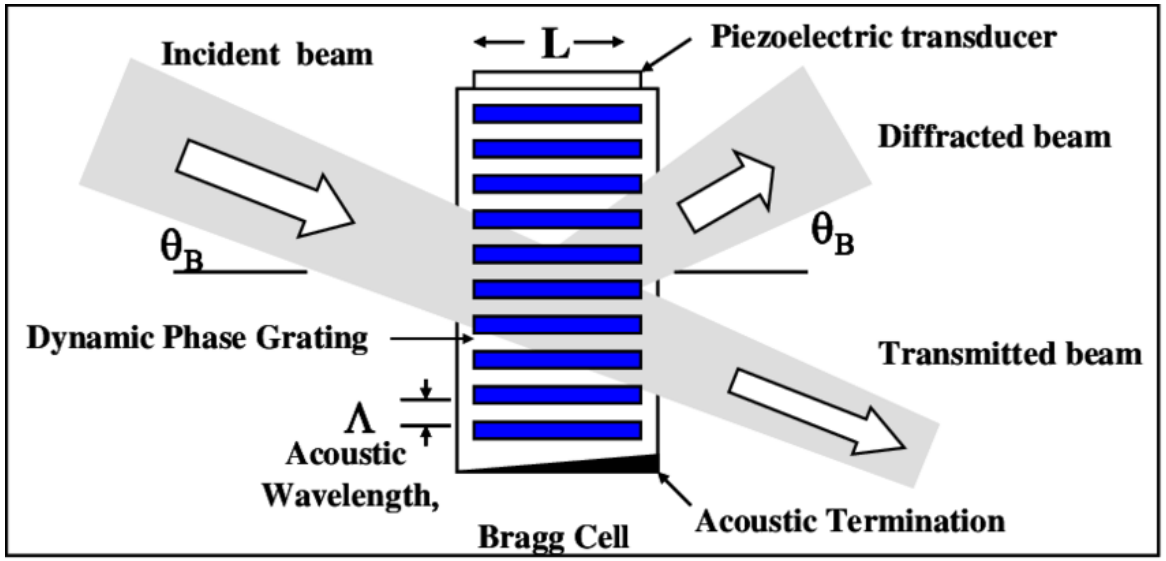


通过双折射晶体时，o光（ordinary） and e光（extraordinary）具有不同的折射率和不同的相速度，从而导致经过双折射晶体后光的偏振方向发生变化（类似于波片），导致无法从下一个偏振片中出射。只有当o光和e光光程差为波长整数倍时才能顺利通过下一个偏振片。调节液晶波片的角度就可以选择出需要透过的波长。

深圳中达瑞和公司的液晶可调滤波器成像光谱仪，分辨率10 nm

液晶可调谐滤波器微型成像光谱仪

2.2 窄带滤光型微光谱仪-可调滤光片



声光可调谐滤光片利用声场在固态双折射晶体中产生周期性波动的折射率，类似于可调谐衍射光栅。

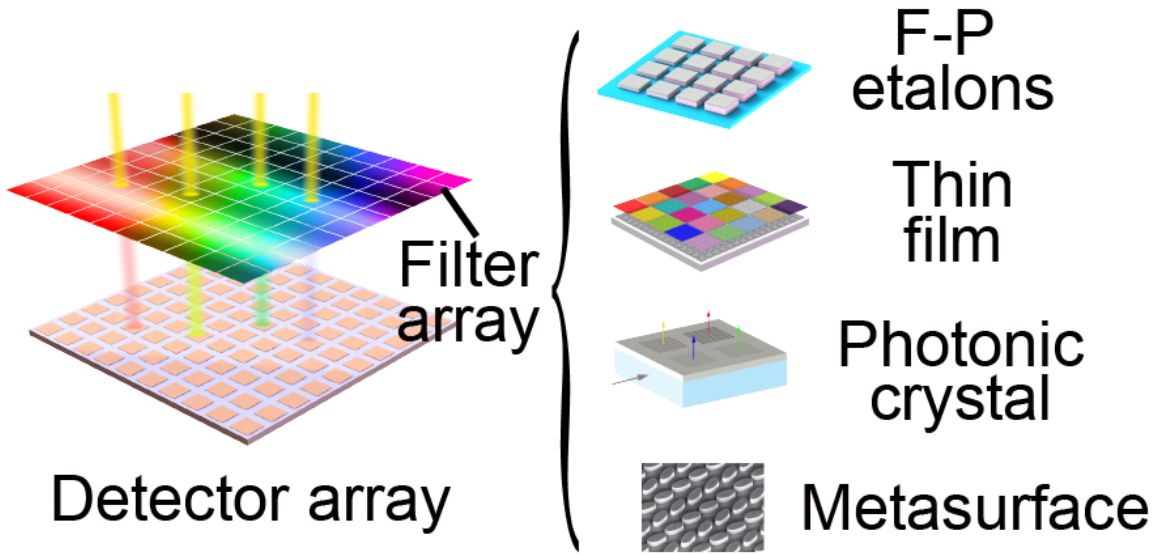
美国NASA的AOTF成像光谱仪

嘉兴中科光电工程中心的AOTF成像光谱仪（航天技术产业转化）

声光可调滤光片（Acousto-optic tunable filter, AOTF）成像光谱仪

2.2 窄带滤光型微光谱仪-滤光片阵列

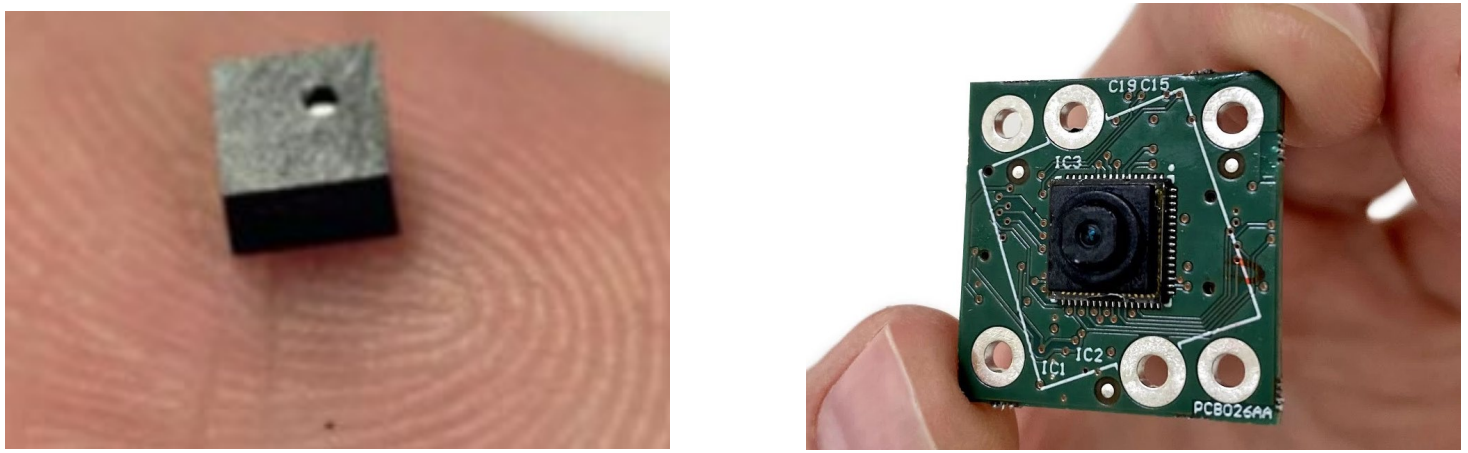
可调滤光片是通过牺牲时间换取光谱，而滤光片阵列的方案是通过二维空间换光谱，可以做到实时检测。



窄带滤光片阵列基于滤光原理分别有F-P腔滤光片，薄膜滤光片，纳米光学滤光片等

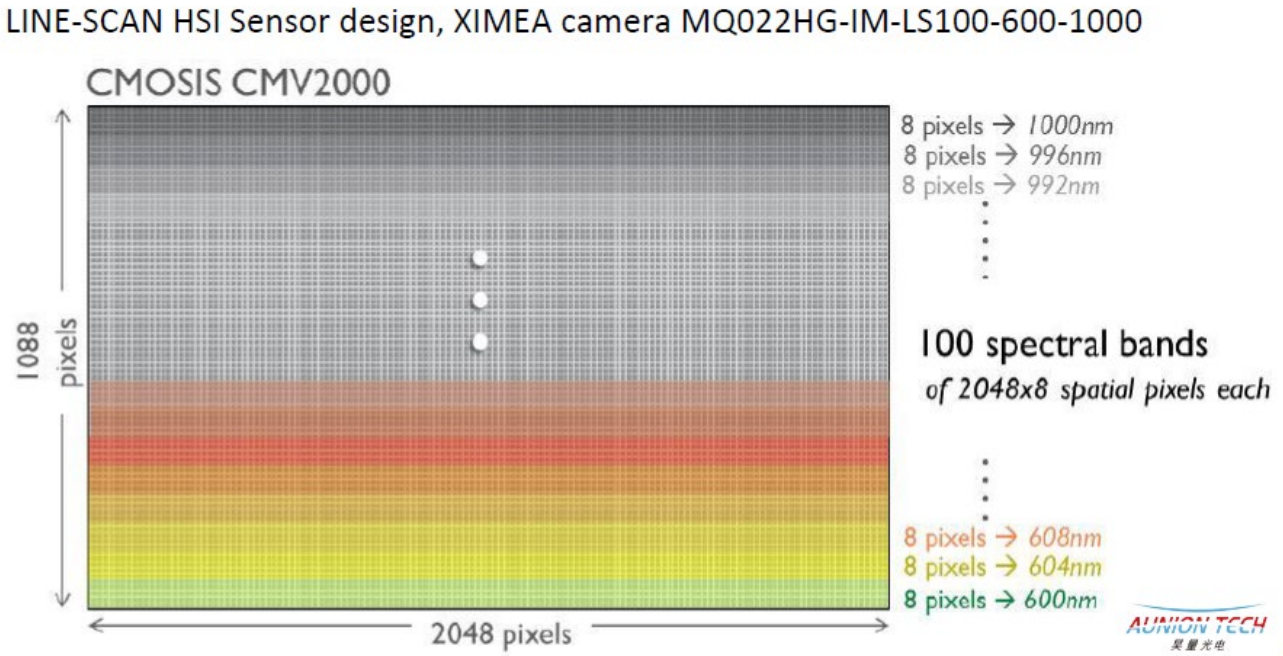
2.2 窄带滤光型微光谱仪-F-P腔滤光片阵列

CMOS芯片上集成F-P腔滤光片示意图（左），F-P滤光片阵列示意图（中）和透过光谱曲线（右）



比利时Spectricity公司基于F-P腔滤光片阵列的光谱仪（左）和成像光谱仪（右）

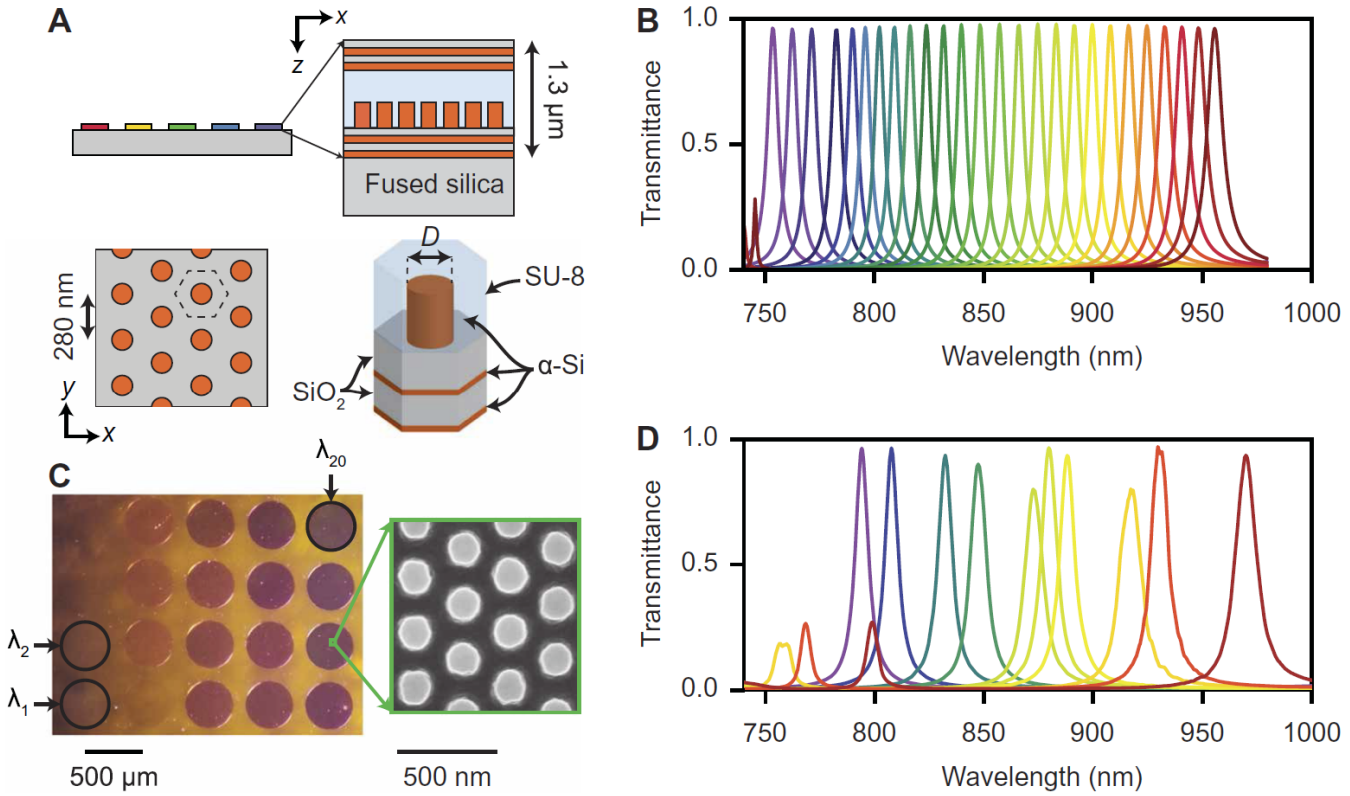
2.2 窄带滤光型微光谱仪-F-P腔滤光片阵列



比利时imec和Ximea公司合作开发基于F-P滤光片阵列和F-P渐变滤光片的成像光谱仪

2.2 窄带滤光型微光谱仪-F-P腔和超材料结合滤光片阵列

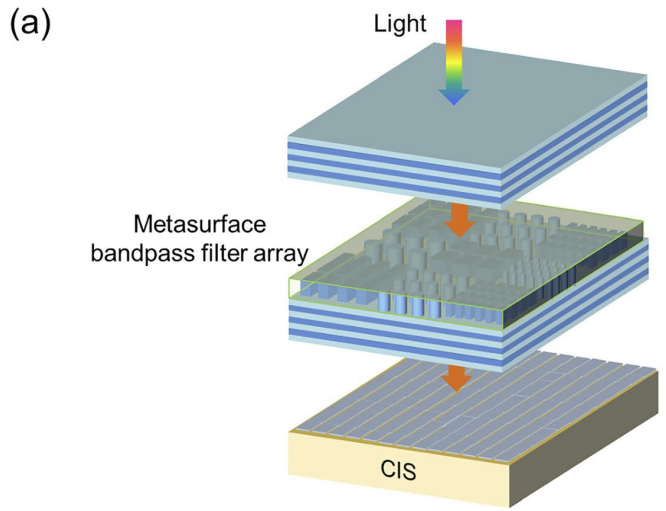
改变F-P腔介质层厚度需要多次曝光，沉积和刻蚀步骤，很难大规模生产。

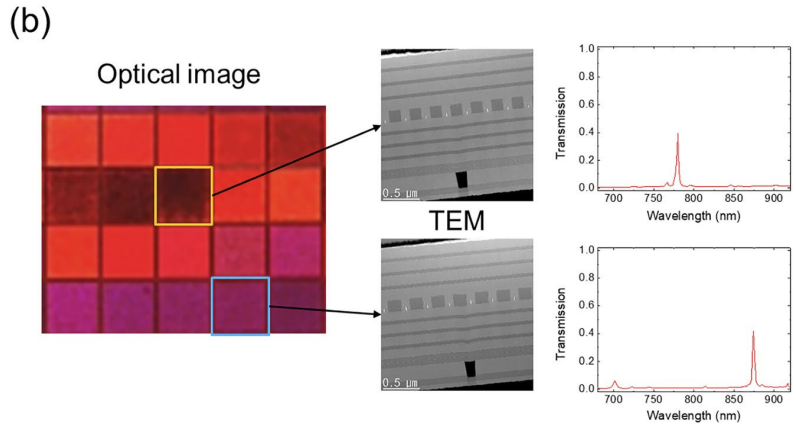


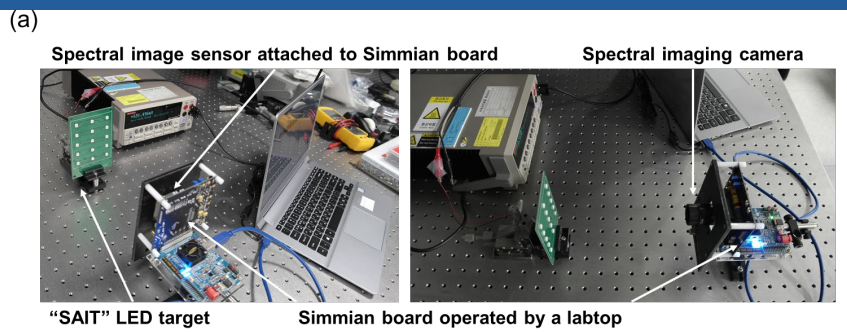
A. McClung *et al.*, ***Science Advances*** 6, eabc7646 (2020)

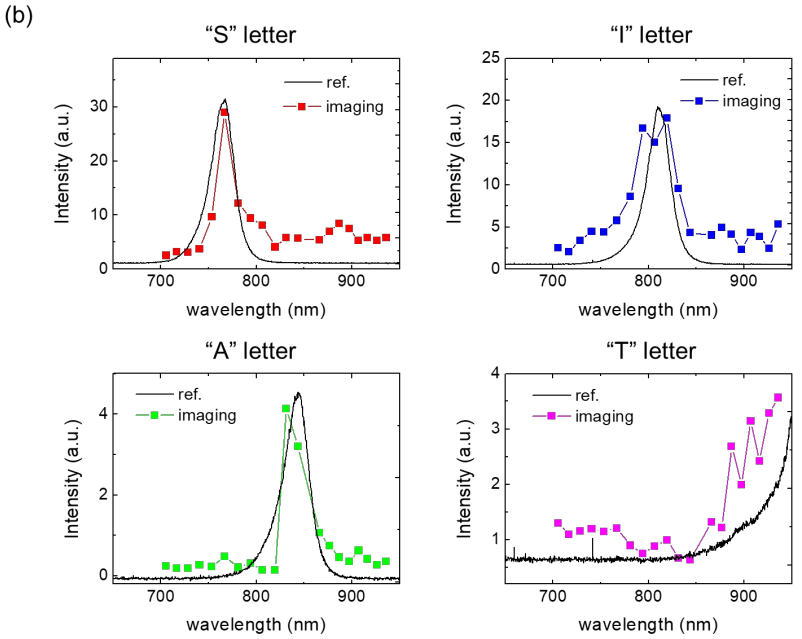
F-P腔中增加超材料，通过超材料调节透过光波长，只需一次光刻，无须加工不同厚度的介质层

2.2 窄带滤光型微光谱仪-F-P腔和超材料结合滤光片阵列





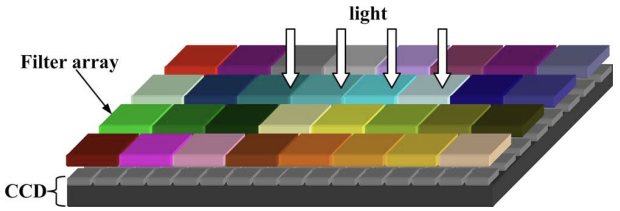




J. Lee *et al.*, ***Nanophotonics*** (2022)
DOI: 10.1515/nanoph-2021-0706

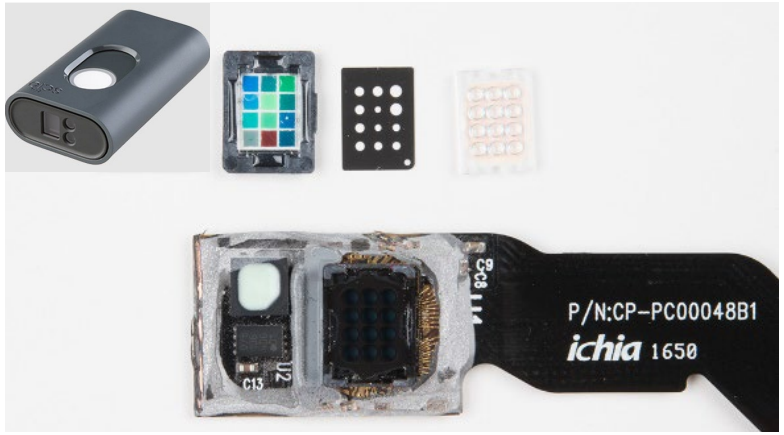
三星高级技术研究所光子器件实验室报道了基于F-P腔与超材料阵列结合的微型光谱成像仪

2.2 窄带滤光型微光谱仪-薄膜滤光片阵列



S. Wang *et al.*, ***Optics Letters*** 32, 632 (2007)

上海技物所的组合刻蚀/镀膜滤光片阵列光谱仪



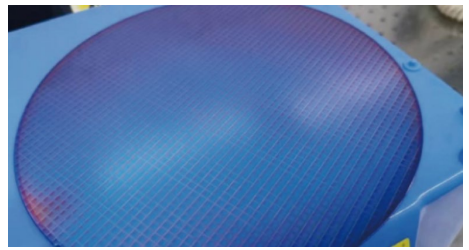
以色列SCIO（原名Consumer Physics）滤光片阵列光谱仪

AMS公司的9通道滤光片阵列光谱检测仪AS734L1

长春求是光谱的薄膜滤光片阵列光谱仪

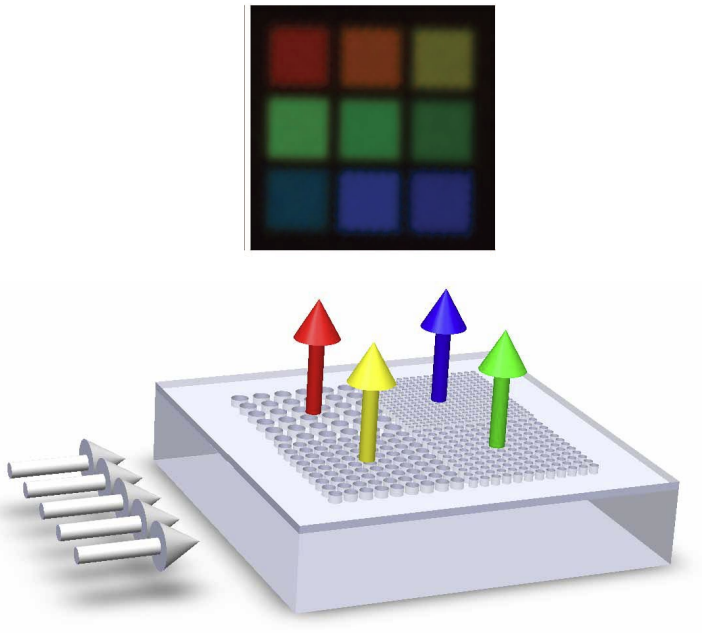
Spectral DEVICES INC.

加拿大Spectral Devices公司多光谱滤光片成像光谱仪



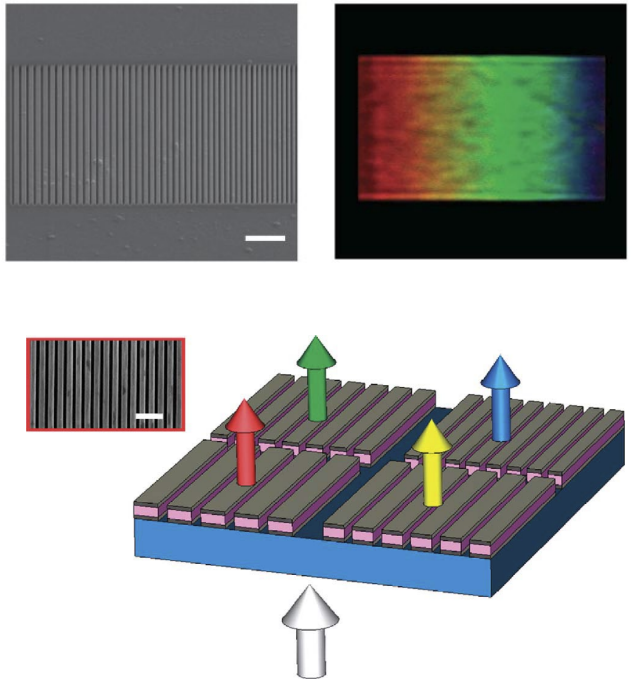
苏州多感科技等

2.2 窄带滤光型微光谱仪-纳米光学滤光片阵列



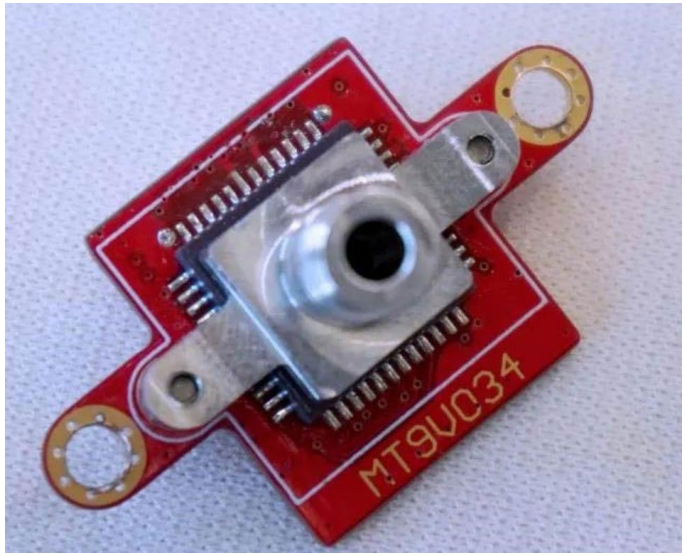
N. Pervez *et al.*, ***Optics Express*** 18, 8277 (2010)

光子晶体滤光片阵列光谱仪



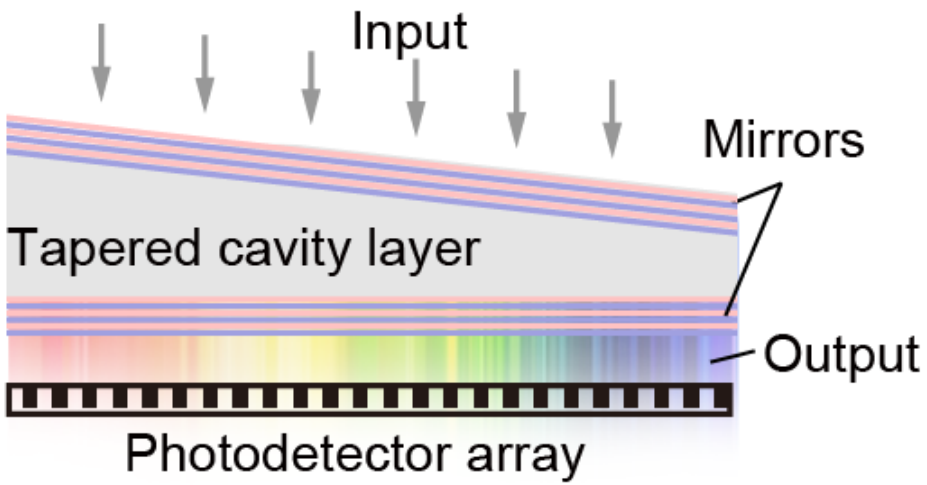
T. Xu *et al.*, ***Nature Communications*** 1, 59 (2010)

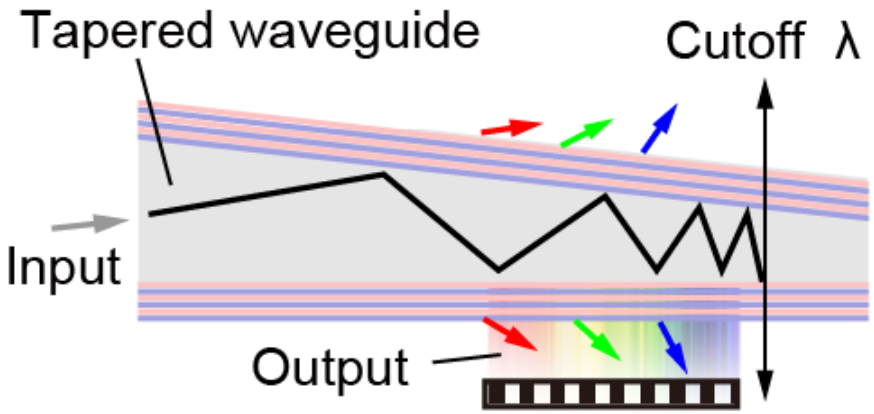
等离子体滤光片阵列光谱仪



南京微纳科技研究院的纳米光子结构滤光片阵列光谱仪

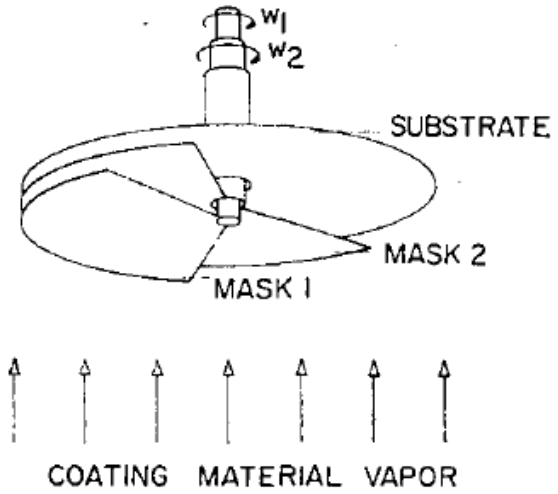
2.2 窄带滤光型微光谱仪-线性渐变滤光片

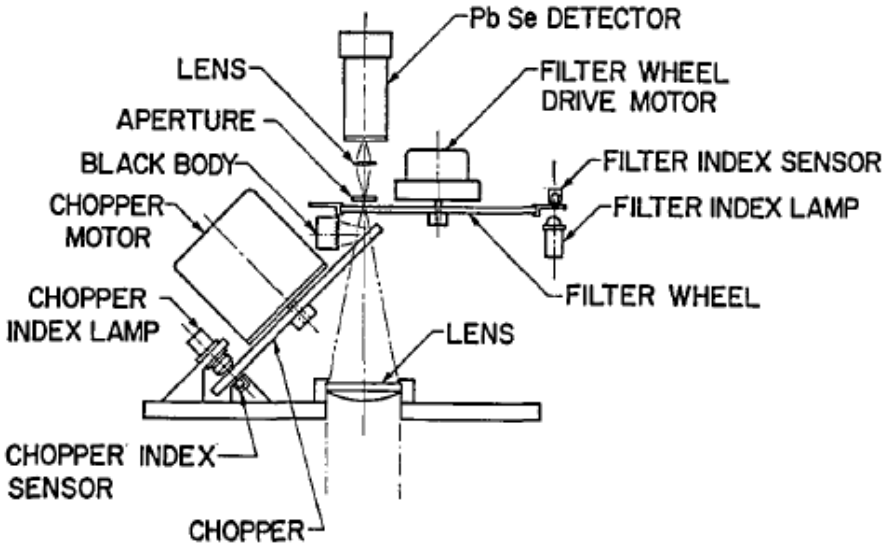


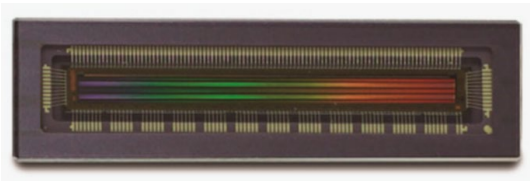


线性渐变滤光片（LVF）的两种工作方式：透过式和泄漏式

2.2 窄带滤光型微光谱仪-线性渐变滤光片







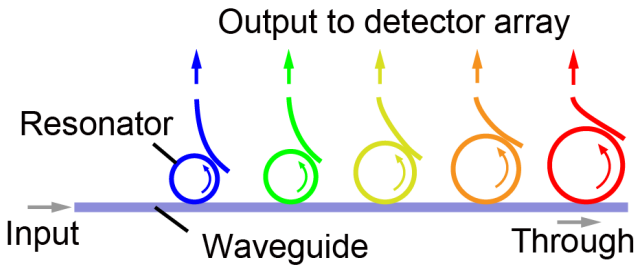
Alfred Thelen, ***Applied Optics*** 4, 977 (1965)

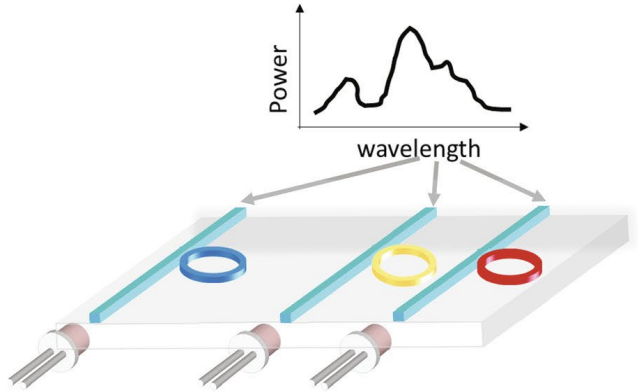
W.A. Hovis *et al*., ***Applied Optics*** 6, 1057 (1967)

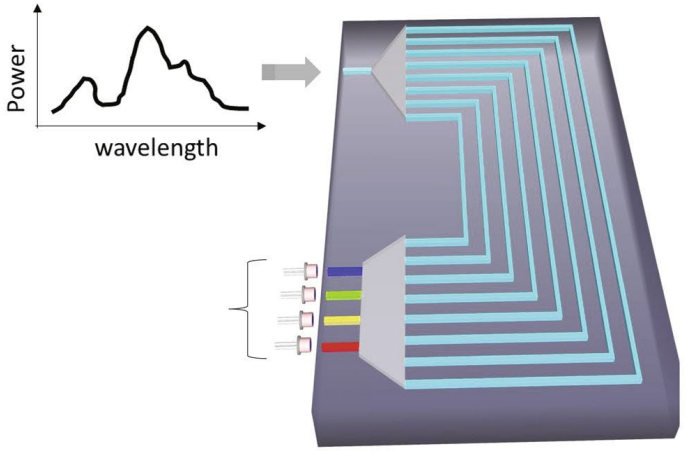
1960年代渐变窄带滤光片镀膜方法和便携式光谱仪设计

长光辰谱的线性渐变滤光片光谱仪

2.2 窄带滤光型微光谱仪-波导滤光



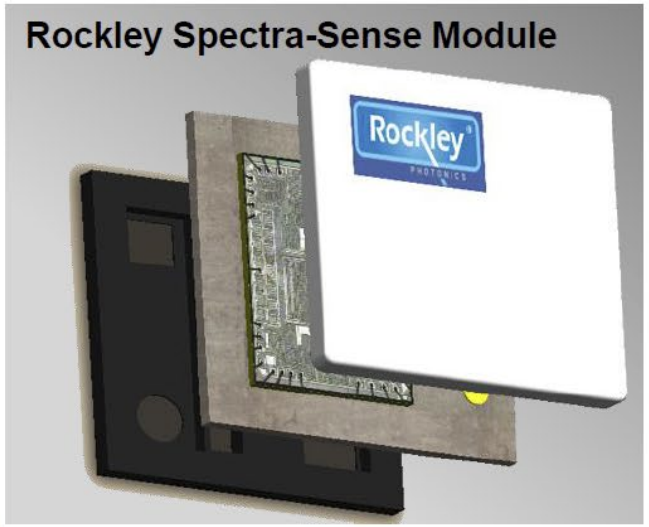




A. Li *et al.*, ***Nature Communications*** 12, 2704 (2021)

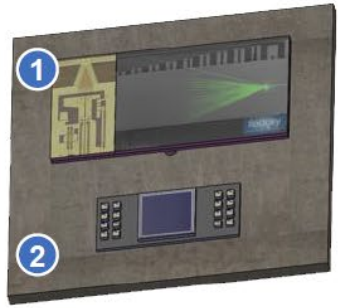
片上集成滤波器阵列光谱仪

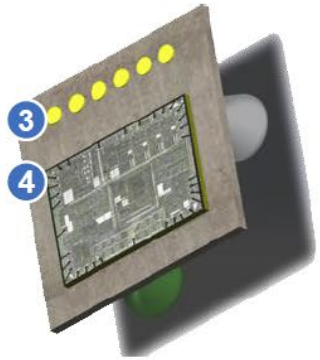
2.2 窄带滤光型微光谱仪-波导滤光



Optical Sensing Frontside

Photonic Integrated Circuits
1. Infrared
2. Visible Spectrum





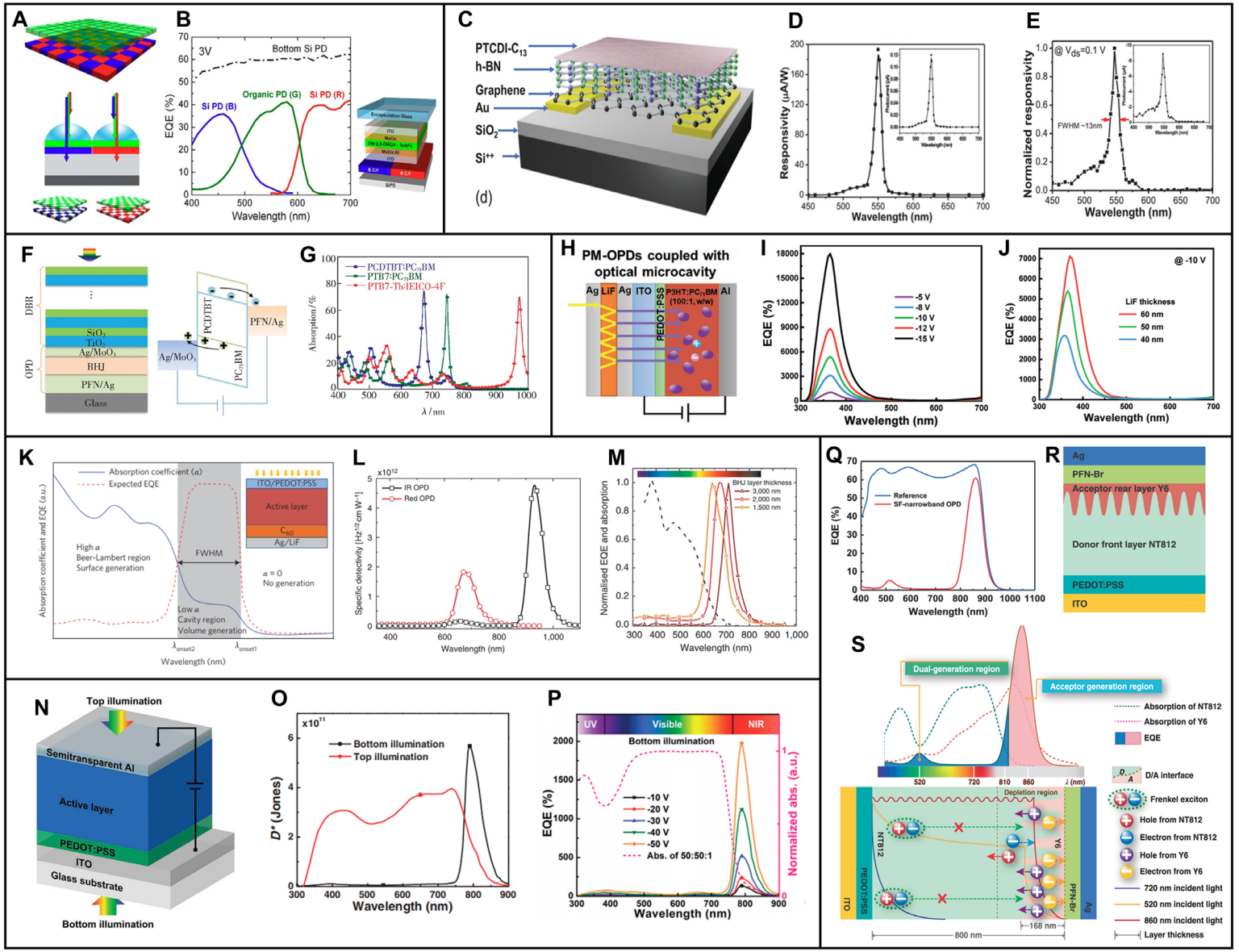
Electrical Interface Backside

3. Electronic Pin Contacts
4. ASIC Controller, MUX, AFE, Drivers

Rockley's proprietary clinic-on-the-wrist module condenses traditional tabletop technology onto a wearable chip

Rockley的片上集成滤波器阵列光谱仪

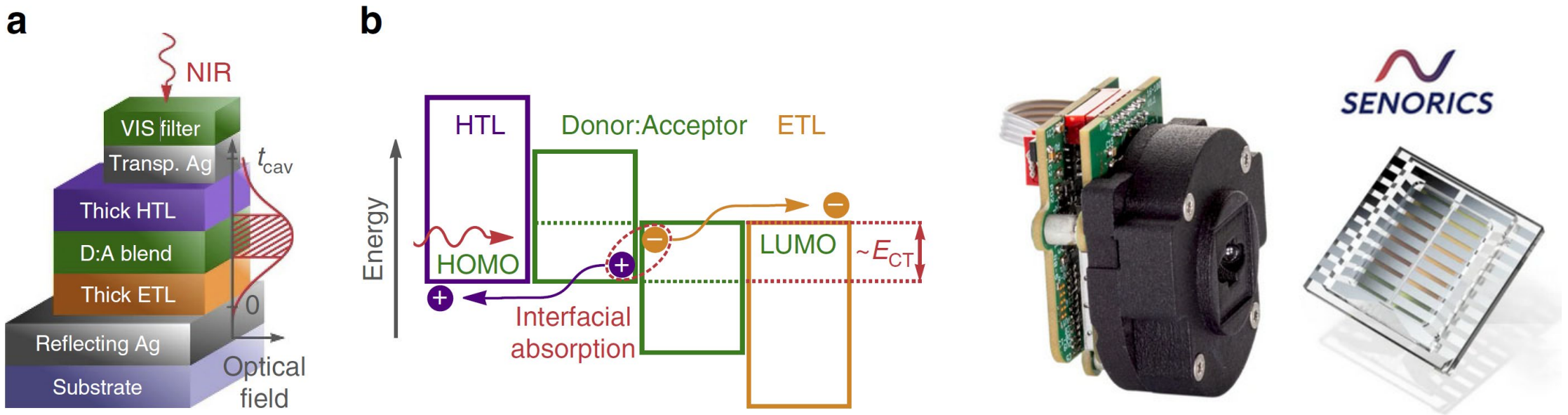
2.2 窄带滤光型微光谱仪-窄带响应探测器



D. Wu *et al.*, ***Adv. Devices Instrum.*** (2023) DOI: 10.34133/adi.0006

除了设计窄带滤光片，另一种思路是直接将探测器做出窄带响应，参考上面列的review paper

2.2 窄带滤光型微光谱仪-材料窄带光谱响应



B. Siegmund *et al.*, ***Nature Communications*** 8, 15421 (2017)

德国Senorics公司 通过将聚合物探测器与FP谐振腔耦合实现窄带光谱响应，覆盖短波红外16波段

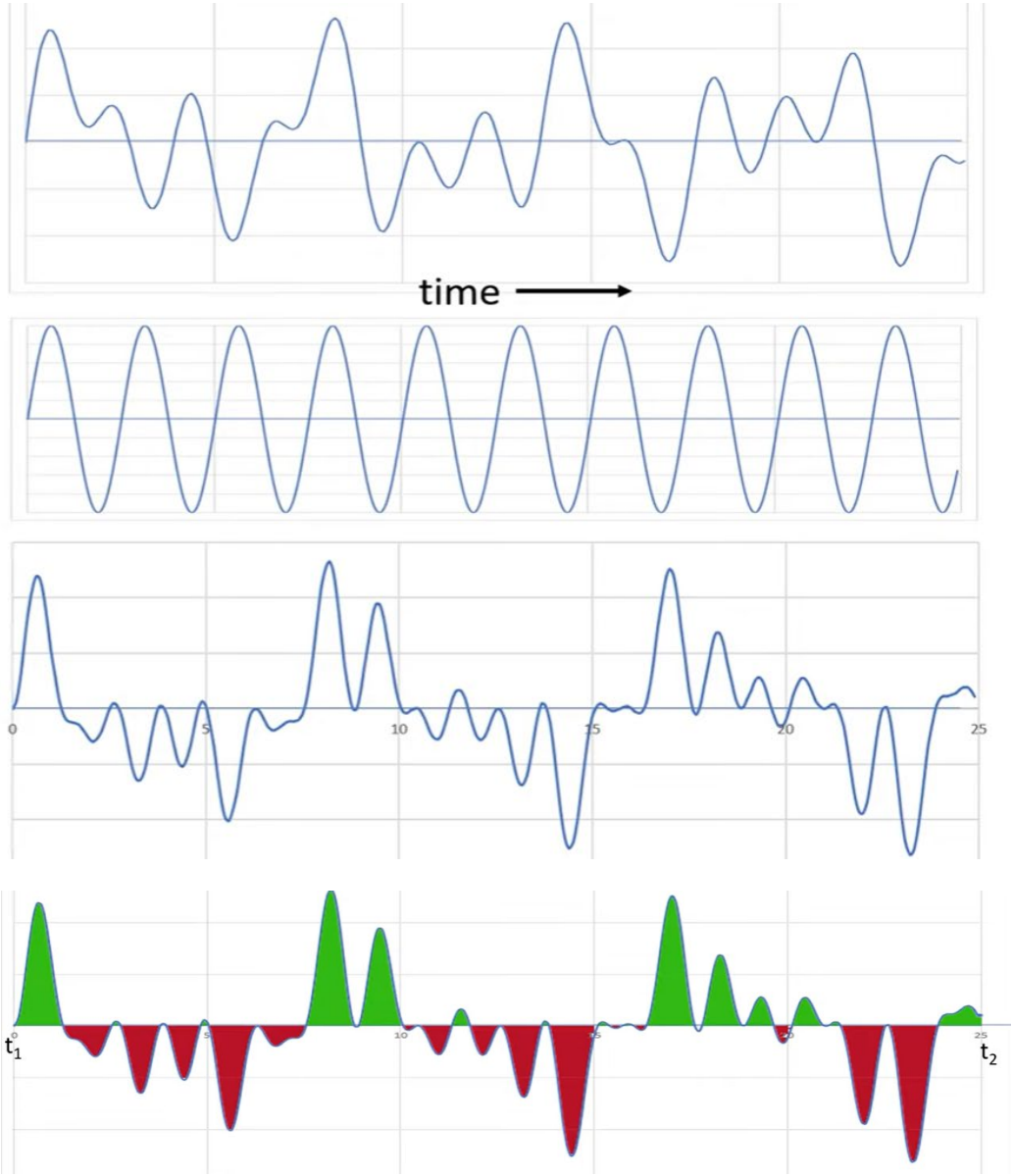
2.3 傅里叶型微光谱仪 傅里叶变换原理

Signal

X

Frequency f_1

=



Integrate all the values over time.

If the frequency under examination is not present in the signal, the sum will be zero because the positive and negative values will cancel each other out over time.

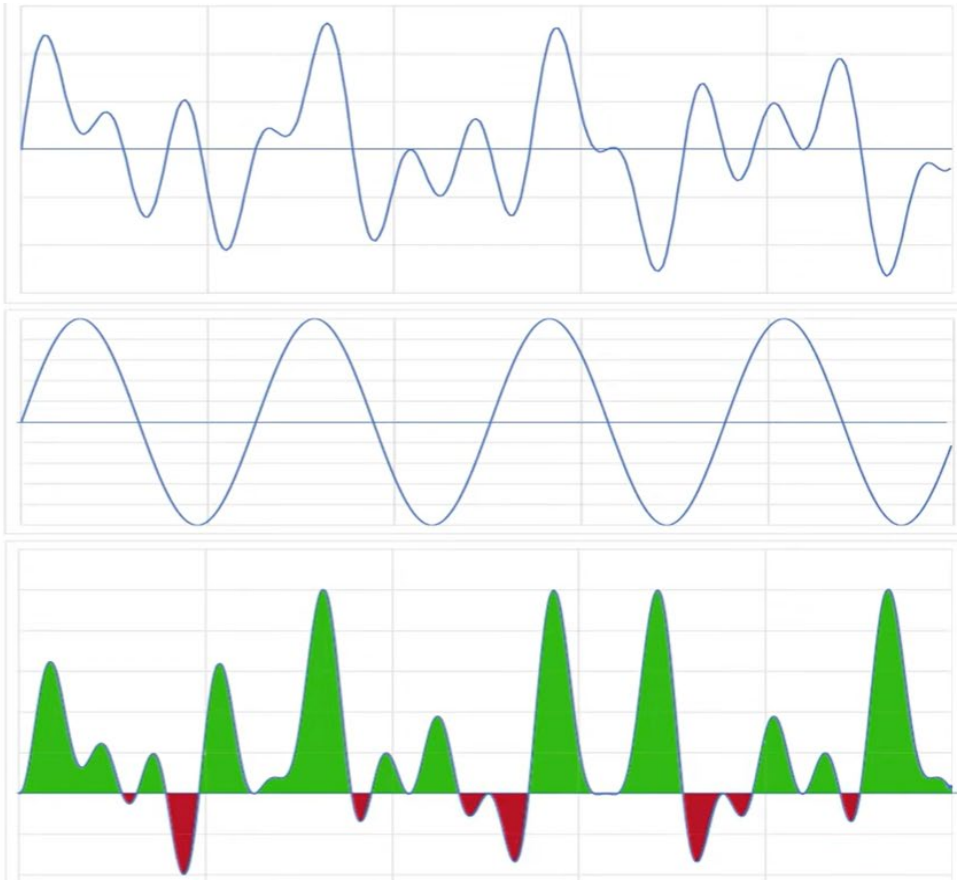
2.3 傅里叶型微光谱仪 傅里叶变换原理

Same signal

X

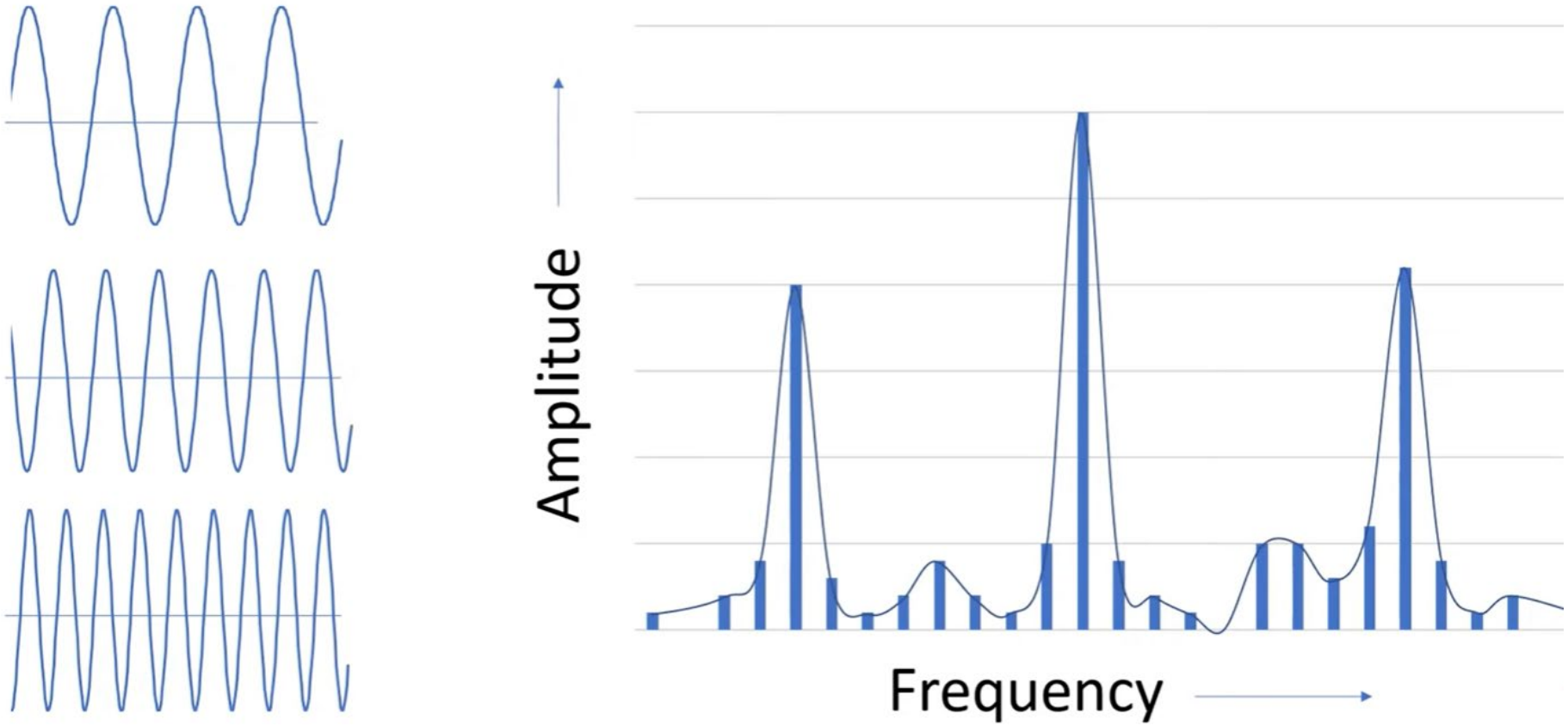
Frequency f_2

=



Do the same integration for a frequency which actually is present in the signal, the net value will be much larger.

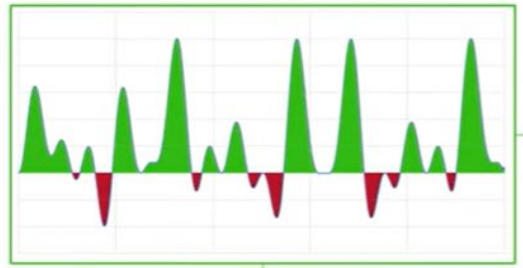
2.3 傅里叶型微光谱仪 傅里叶变换原理



By repeating this multiplication and integration process for a large number of frequencies, we can construct a frequency spectrum.

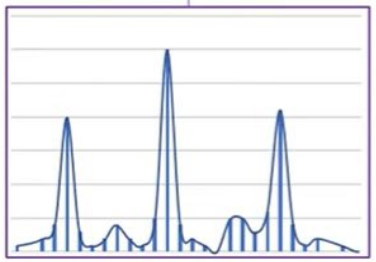
2.3 傅里叶型微光谱仪 傅里叶变换原理

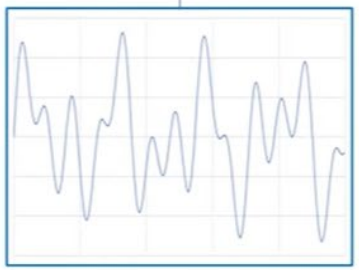
This process can be described by the Fourier Transform formula.

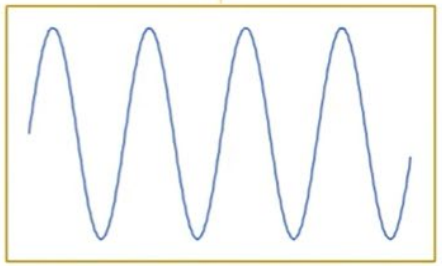


This multiplication is integrated over time.

$$G(f) = \int_{-\infty}^{\infty} g(t) \cdot e^{-i2\pi ft}\, dt$$







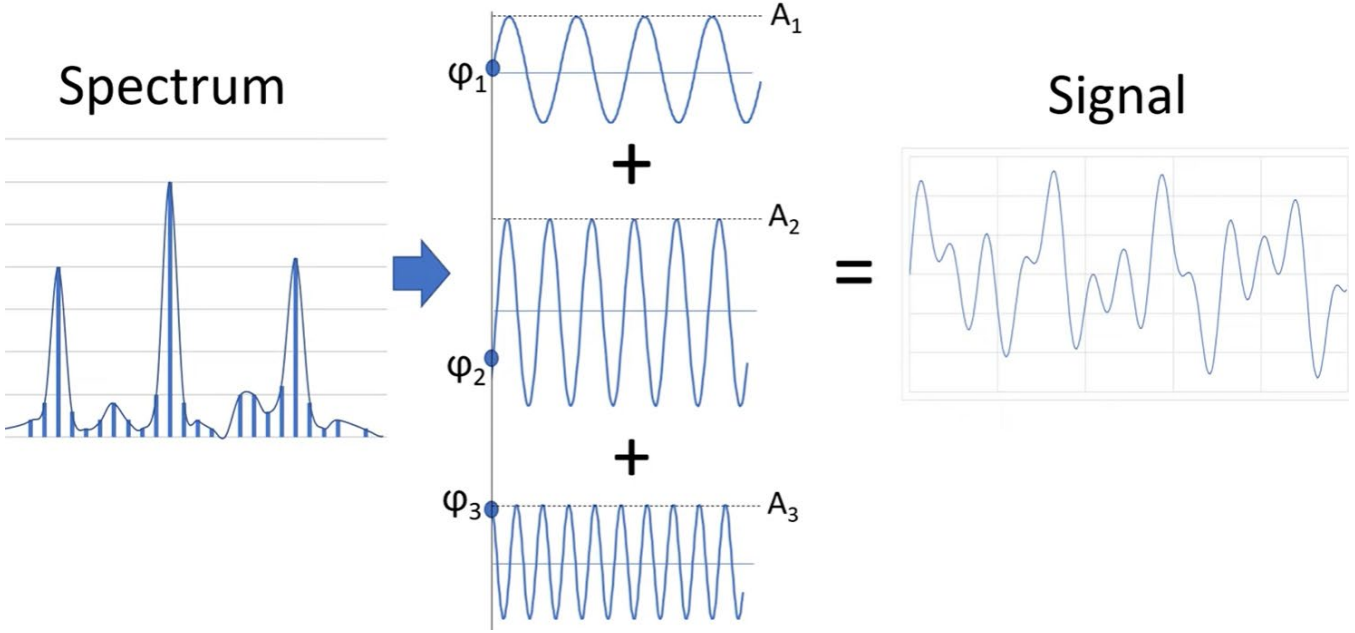
Euler's formula!

Spectrum as a function of the frequency **G(*f*)**

Signal as a function of time **g(*t*),** which includes phase φ of each frequency

Complex harmonic function with frequency ***f*** , also include phase information

2.3 傅里叶型微光谱仪 傅里叶变换原理



$$F(\omega) = \int_{-\infty}^{\infty} f(t) \exp(-i\omega t)\, dt$$

Fourier Transform

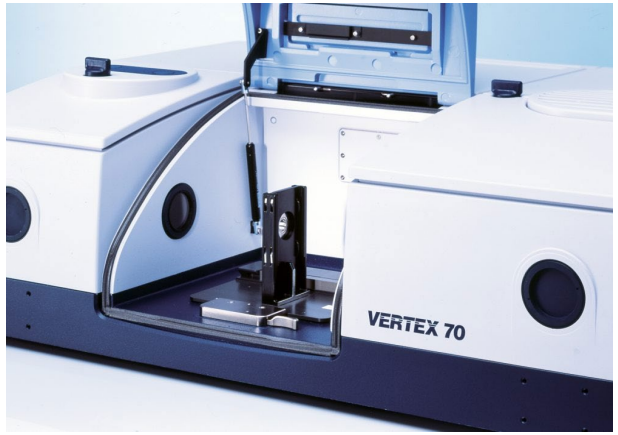
So, we can transform to the frequency domain and back. Interestingly, these transformations are very similar.

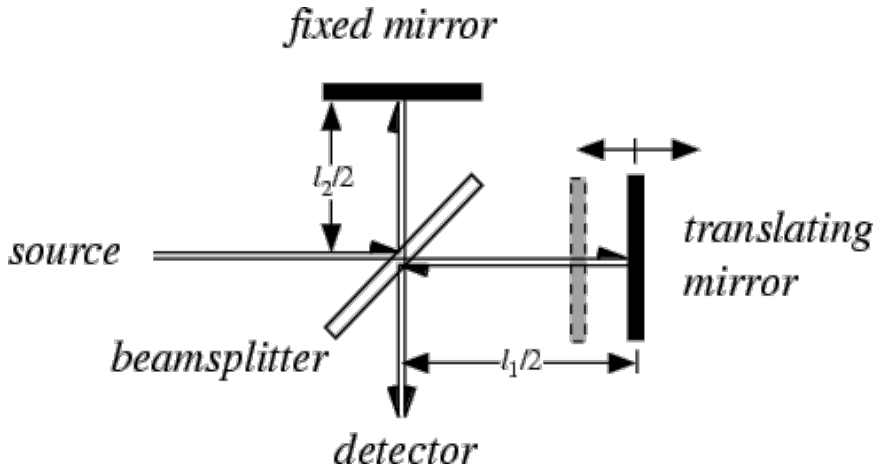
$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) \exp(i\omega t)\, d\omega$$

Inverse Fourier Transform

These equations together are Fourier's Theorem.

2.3 傅里叶型微光谱仪 原理

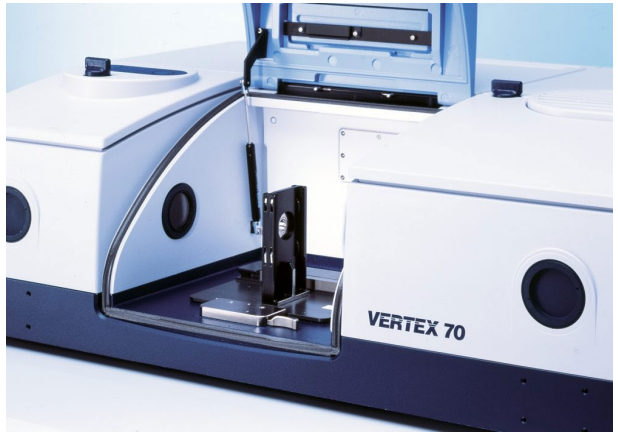


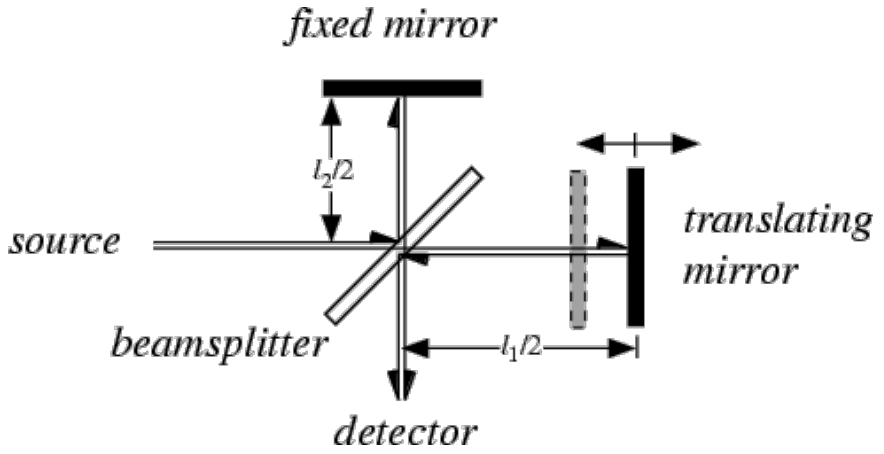


对于单色光来说，若其入射波长为λ，波数为σ（σ=1/λ），入射强度为2B(σ)，经过半透半反镜后，光束分为强度相等（B(σ)）的两束，假设两路光程差为x，当x改变时，其干涉强度关于波数呈余弦函数关系，则像面上两点的强度可以表示为：

$$I(x) = 2B(\sigma)[1 + \cos(2\pi x\sigma)]$$ 干涉公式

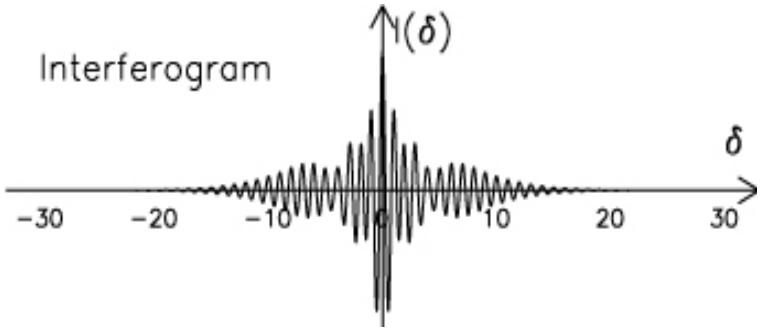
2.3 傅里叶型微光谱仪 原理



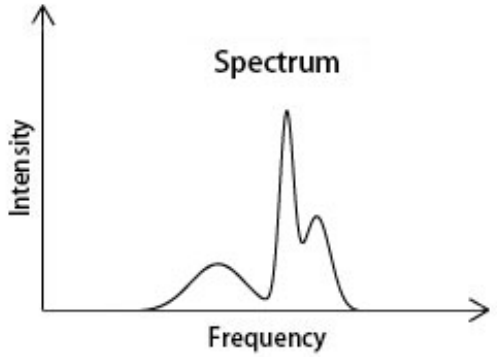


当光源中含有多种频率的光时，像点的光强可表示为

$$I(x) = \int_0^\infty 2B(\sigma)[1 + \cos(2\pi x\sigma)]d\sigma$$

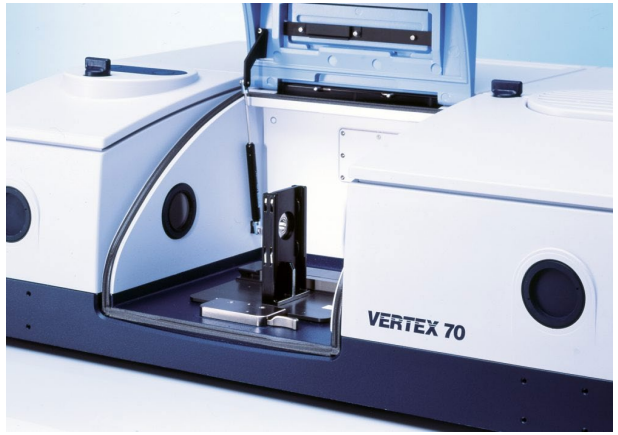


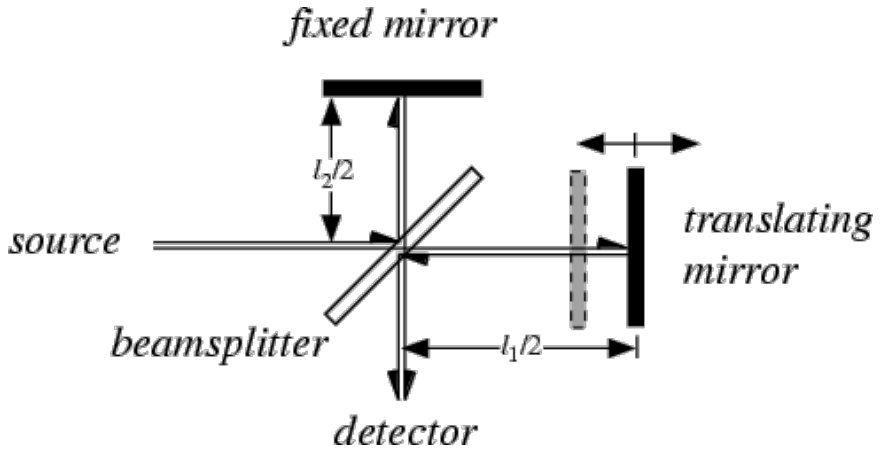
探测器上的光强度为 $I(x)$



入射强度为 $2B(\sigma)$

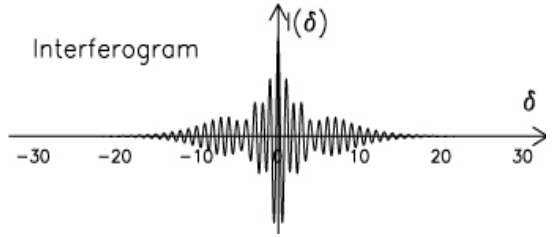
2.3 傅里叶型微光谱仪 原理





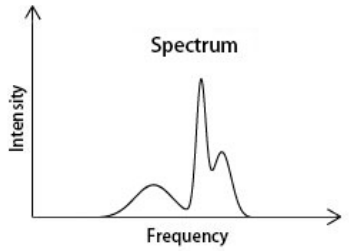
减去强度的平均值，得到

$$I(x) = \int_0^\infty 2B(\sigma)\cos(2\pi x\sigma)\,d\sigma$$

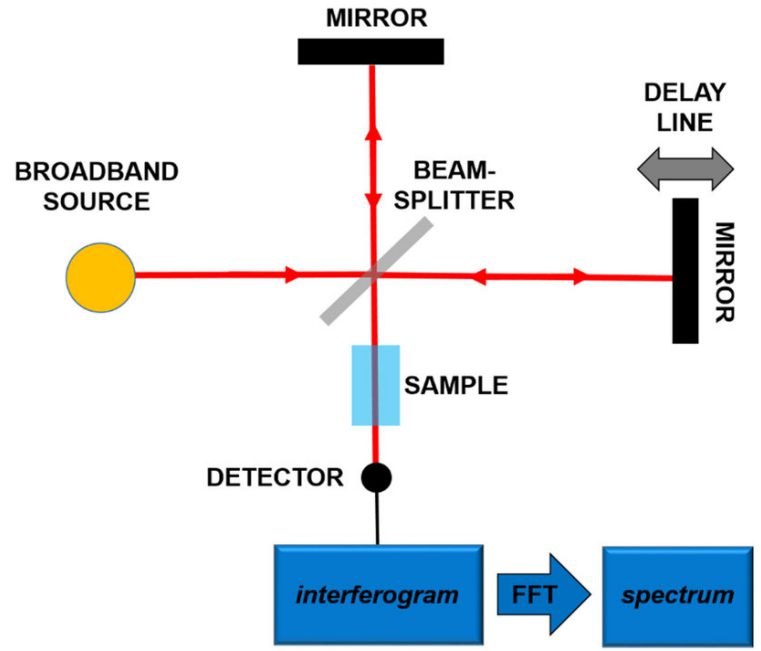


利用傅里叶反变换算法得到原始光谱，由于B(σ)系数不影响其光谱的相对成分，因此可将2B(σ)替换为B(σ)，光谱还原公式为

$$B(\sigma) = \int_0^\infty I(x)\cos(2\pi\sigma x)dx$$



2.3 傅里叶型微光谱仪



傅里叶变换光谱仪通常用于检测红外波段的吸收或发射光谱，工作原理是使用干涉仪来调制到达单个探测器上的光，再通过傅里叶变换将探测器上收集到的干涉图转换为与波长相关的光谱。傅里叶变换型光谱仪具有杂散光低、分辨率较高、光通量大、精度高等优点，使得微型化的同时满足高光通量和高分辨率等性能的要求。

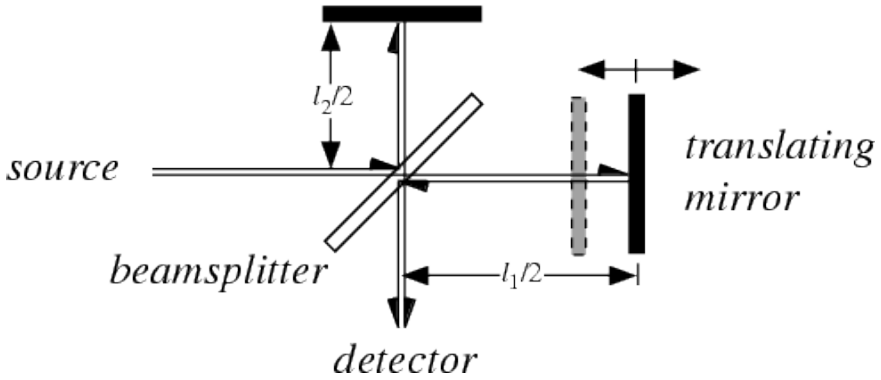
思考：为什么红外光谱仪大部分是傅里叶型的，很少用光栅色散?

Fellgett’ s advantage: 所有光谱同时到达探测器，能量比色散/滤光后的强，减少探测时间，噪声较低。

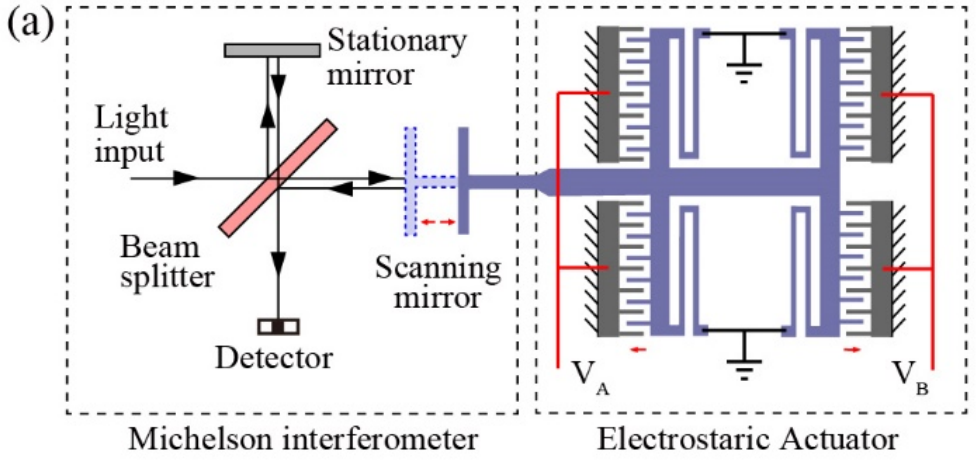
Jacquinot advantage: 光路简单，不需要狭缝，整体进光亮较高。

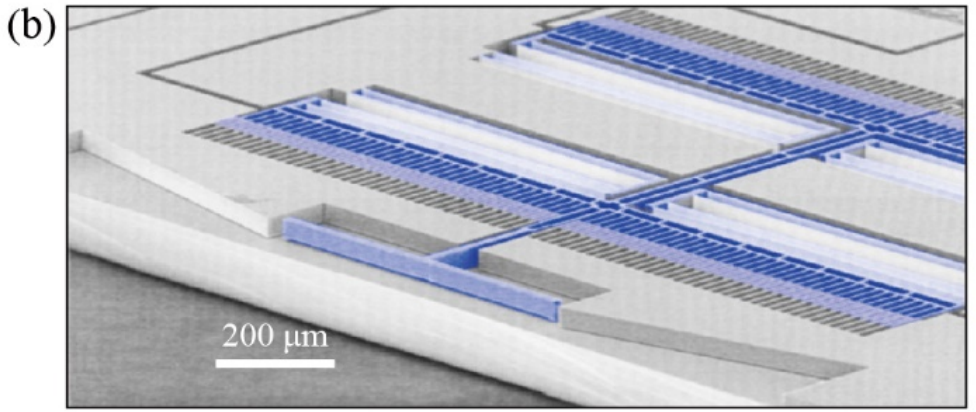
微型傅里叶光谱仪可分为两类：一类是**时间调制**型；另一类是**空间调制**型。

2.3 傅里叶型微光谱仪-时间调制



传统FTIR也是时间调制型，由驱动器控制迈克尔逊干涉仪的反射镜运动以获得干涉曲线。

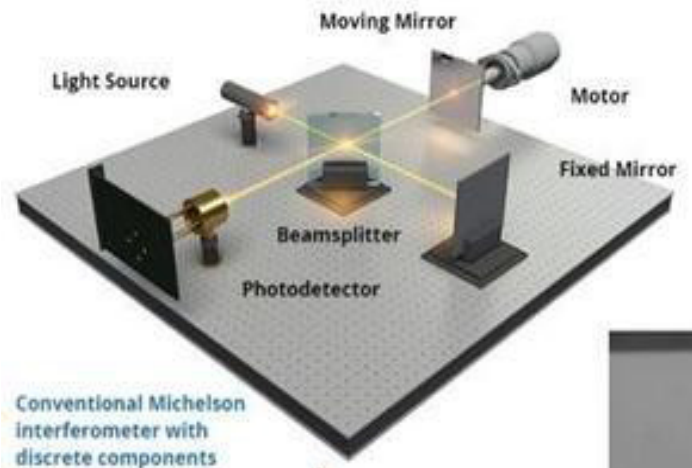




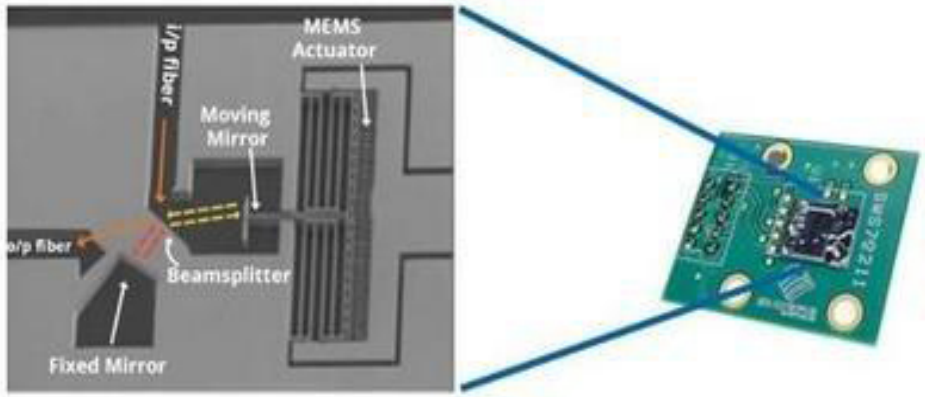
M. Herzig et al., ***Opt. Lett.*** 24, 1705 (1999)

通过MEMS工艺做成片上傅里叶型微光谱仪。动镜通过静电、电磁力、电热驱动移动。

2.3 傅里叶型微光谱仪-时间调制



Conventional Michelson interferometer with discrete components



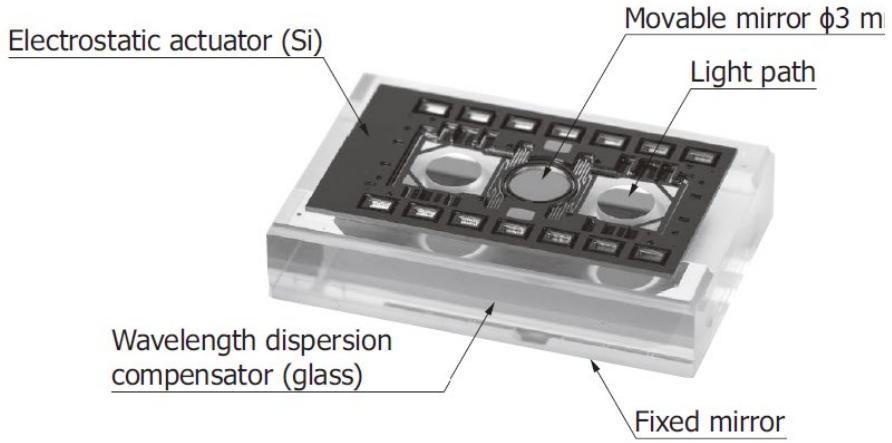
Monolithic Michelson interferometer chip enable by SIMOST

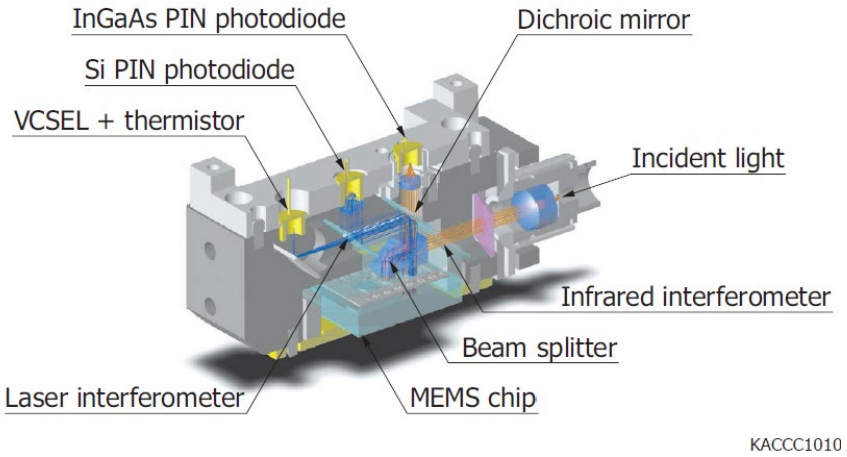
美国Si-Ware公司的时间调制傅里叶型微光谱仪

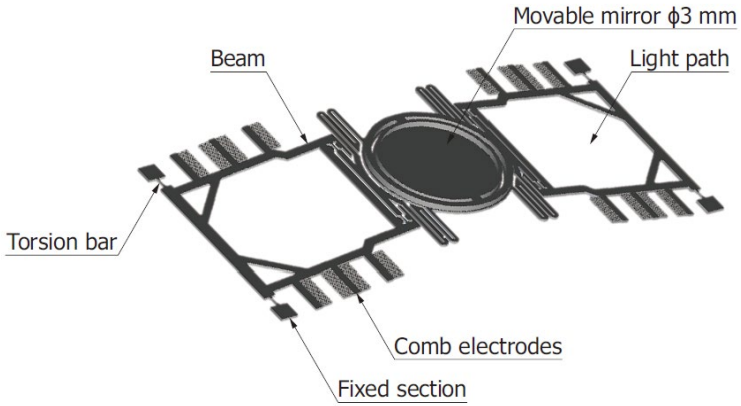
2.3 傅里叶型微光谱仪-时间调制

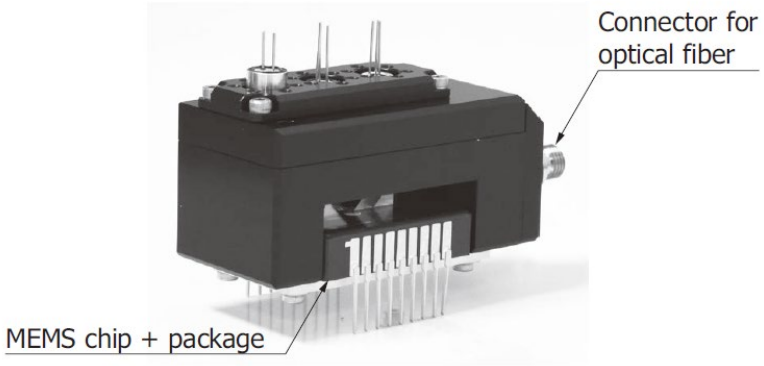
尺寸49×57×76 mm

分辨率5.7 nm





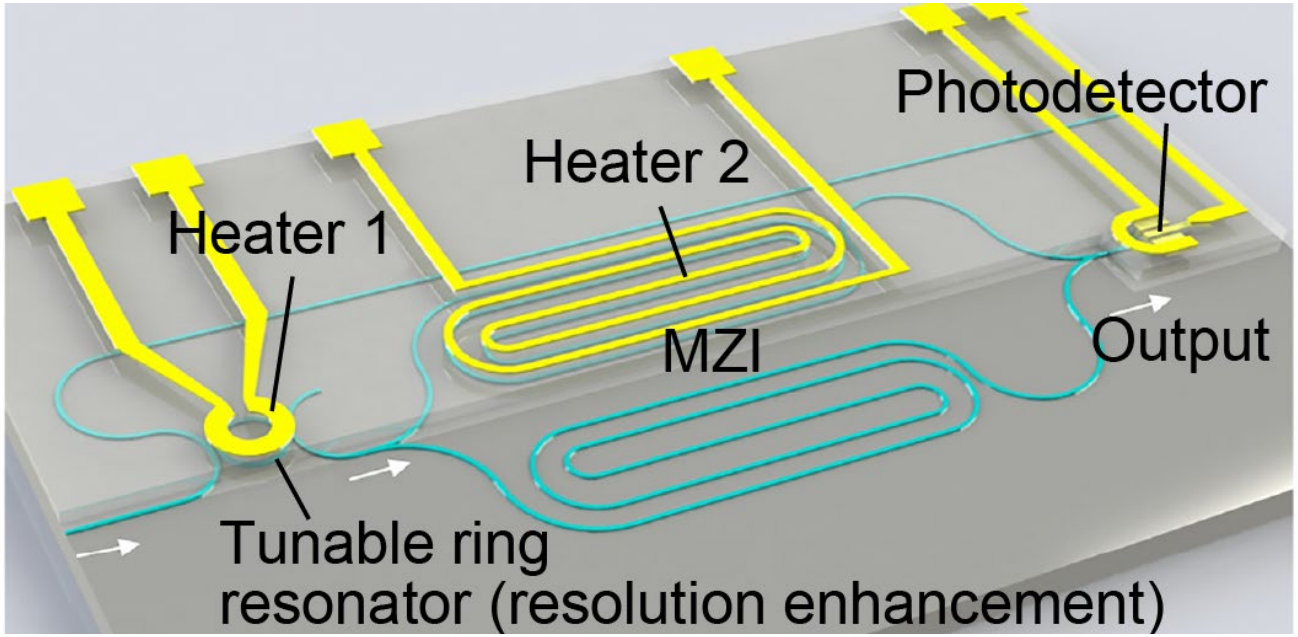




日本滨松公司的时间调制傅里叶型微光谱仪C15511-01

移动距离有限，分辨率有限，稳定性差

2.3 傅里叶型微光谱仪-时间调制



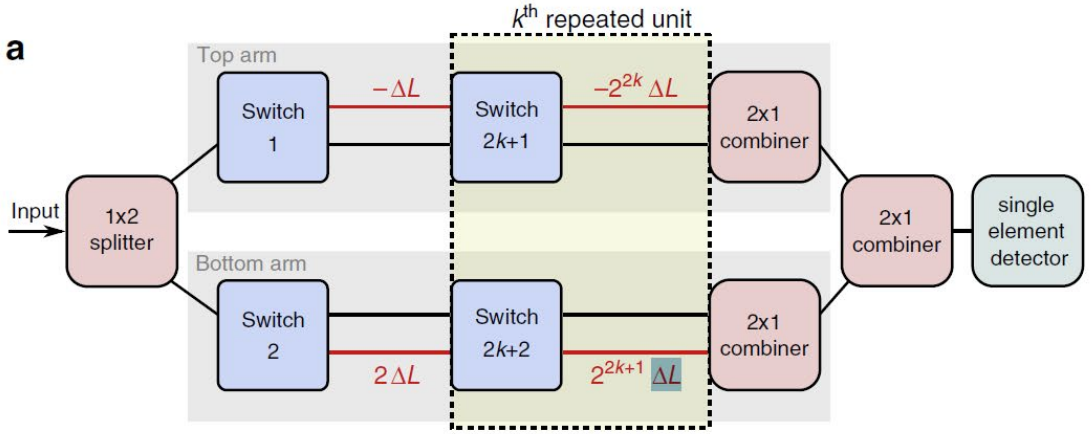
M. Souza *et al.*, ***Nature Communications*** 9, 665 (2018)

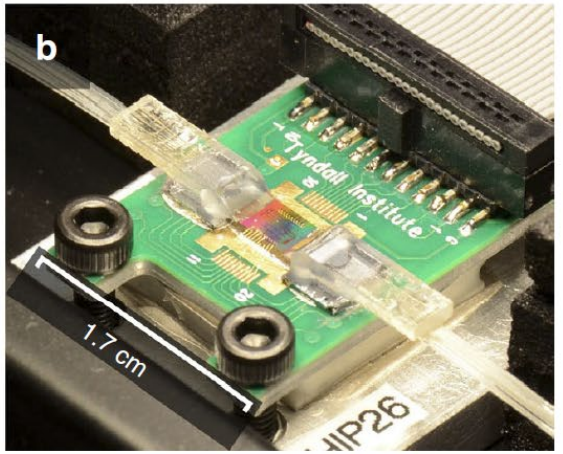
没有动镜的基于集成波导的平面片上傅里叶变换光谱仪。这些系统没有采用迈克尔逊结构，而是基于马赫-曾德尔干涉仪(MZIs)，Mach-Zehnder干涉仪中光被分裂到至少两个单向通道，然后进入同一个探测器并被重新组合，由于光在不同通道的光程差不同，因此相位差也不同，便产生了干涉。

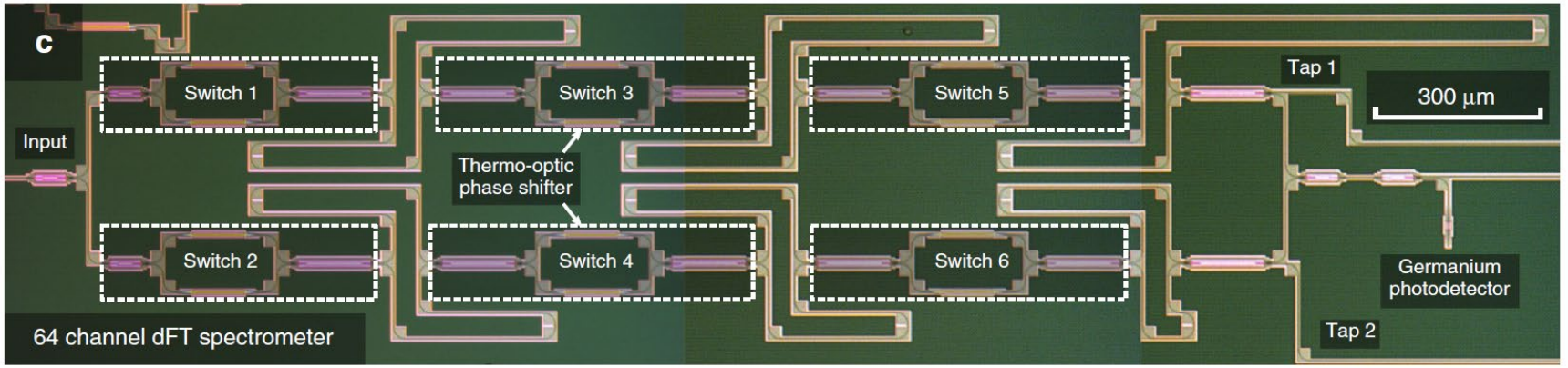
连续调制的Mach-Zehnder干涉仪可以通过电光调制铌酸锂波导实现，也可以通过热光学效应实现，比如在集成光路附近嵌入微型加热器。

MZI 干涉仪，通过加热其中一个波导臂改变光程差，无移动部件，稳定

2.3 傅里叶型微光谱仪-时间调制





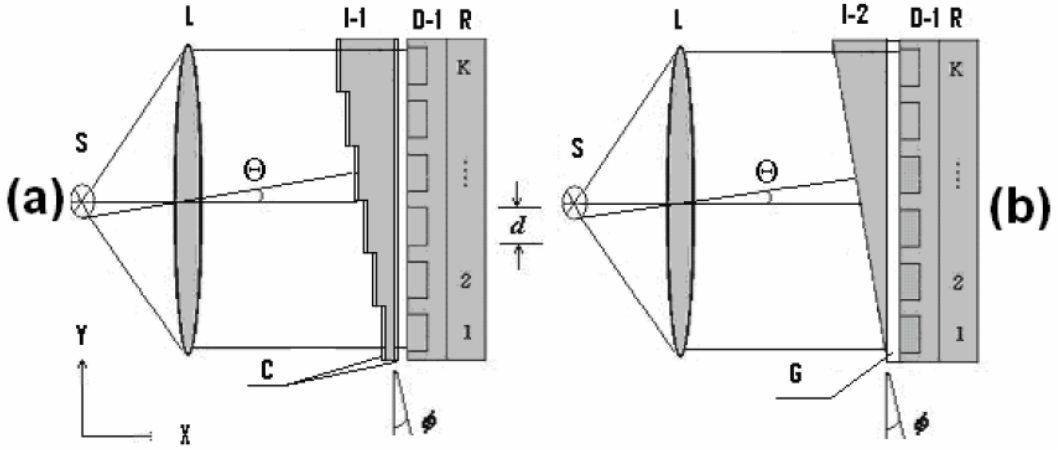


数字傅里叶变换光谱仪使用了一个具有许多光开光的光子电路，将光信号沿着不同长度的路径传递。在此，光谱分辨率取决于路径的数量或光谱通道数，其与光开关的数量成指数关系。

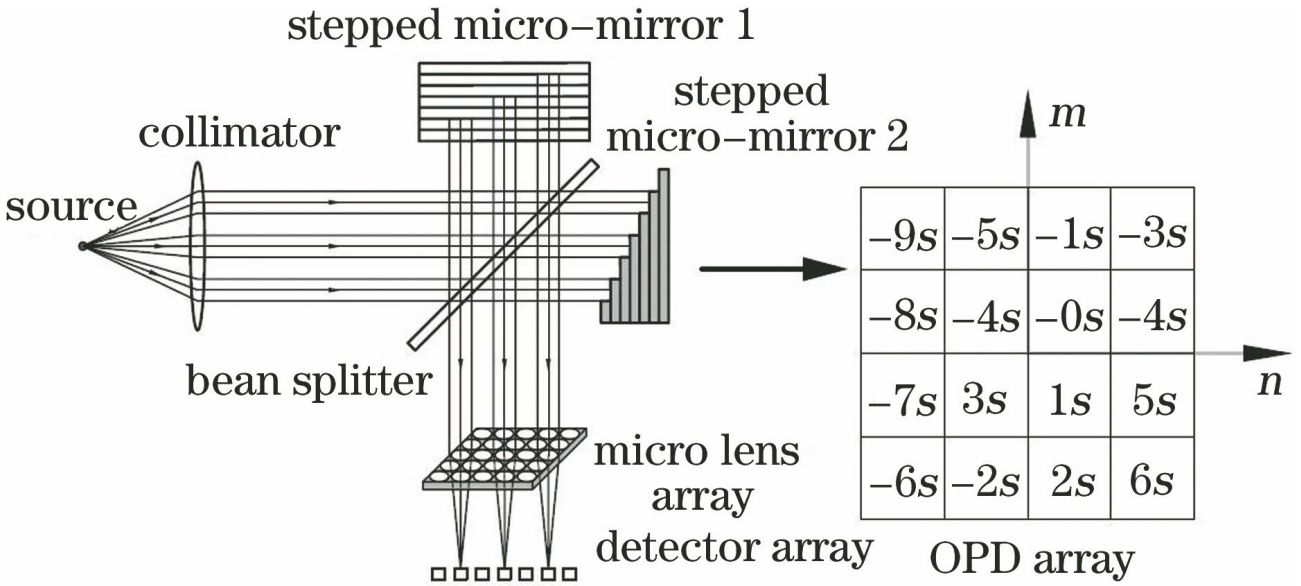
D. Kita *et al.*, ***Nature Communications*** 9, 4405 (2018)

MIT的数字MZI 干涉仪，通过多个光开关调节光路从而调节光程差，无移动部件

2.3 傅里叶型微光谱仪-空间调制



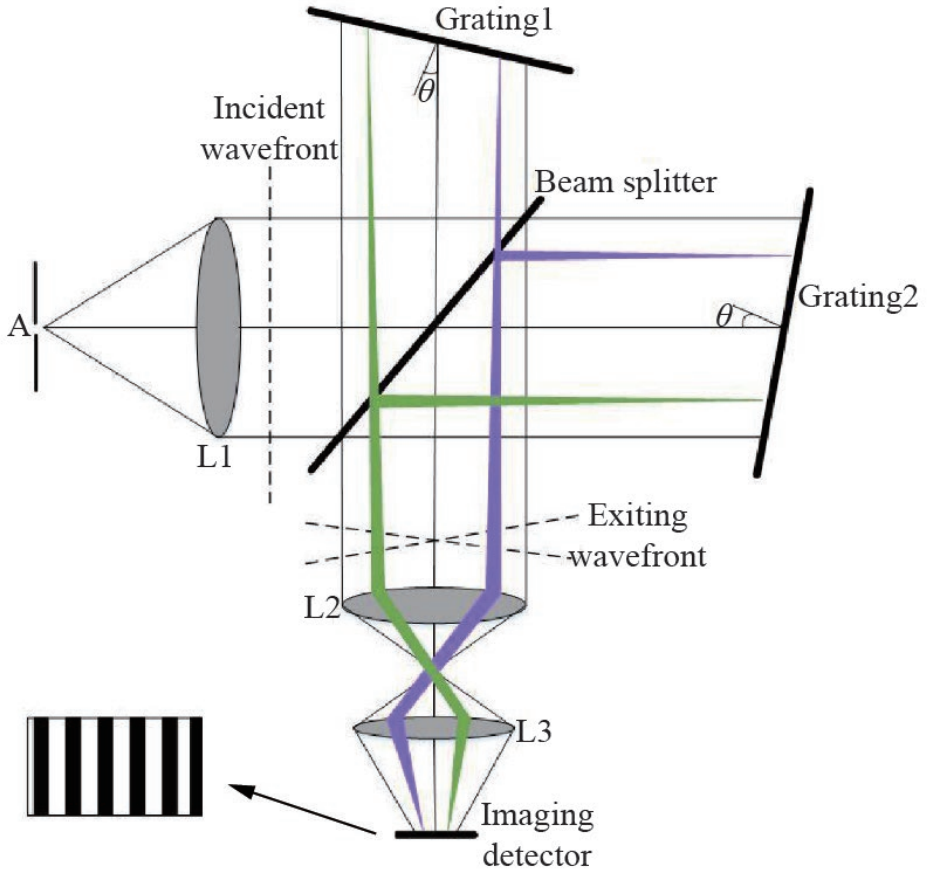
A. Manuilskiy *et al.*, ***SPIE*** 6395, 639504 (2006)



吕金光 等. *光学学报*, 38, 0230001 (2018)

静态傅里叶型光谱仪，通过空间位置的改变得到光程差序列，无须扫描实现干涉信号的空间调制。

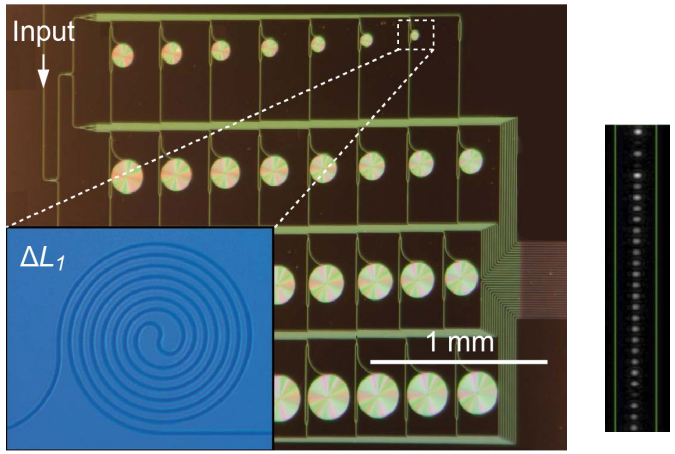
2.3 傅里叶型微光谱仪-空间调制



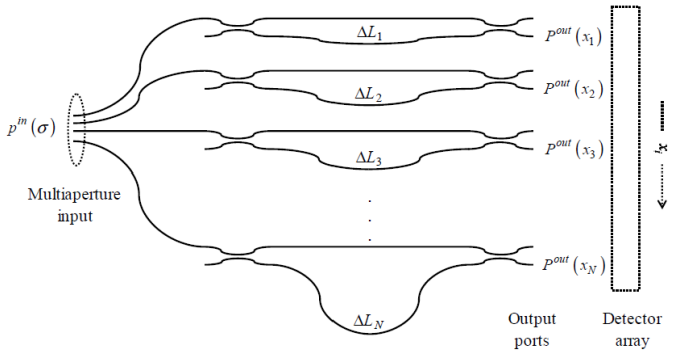
王新强 等. *应用光学*, 41, 462 (2020)

空间外差型，通过两个光栅对不同波长的路径差，实现干涉信号的空间调制。

2.3 傅里叶型微光谱仪-空间调制



A. Velasco *et al.*, ***Optics Letters*** 38, 706 (2013)



M. Florjanczyk *et al.*, ***Optics Express*** 26, 18178 (2007)

基于多个Mach-Zehnder干涉仪阵列，形成空间调制。例如，螺旋状的波导阵列，波导的长度线性变化，从而引起光程差。

但是，此类系统的光谱分辨率受到最大光程差以及Mach-Zehnder干涉仪数量的限制，这都限制了光谱仪在保持高性能的同时尽可能的微型化。

Table 1. ANN Classification: Best Results Confusion Matrix

	Predicted			
Actual	Class 1	Class 2	Class 3	Class 4
Class 1	1782	383	161	1
Class 2	90	1869	318	49
Class 3	0	676	1647	1
Class 4	1	39	62	2222

Table 2. SVM Classification: Best Results Confusion Matrix

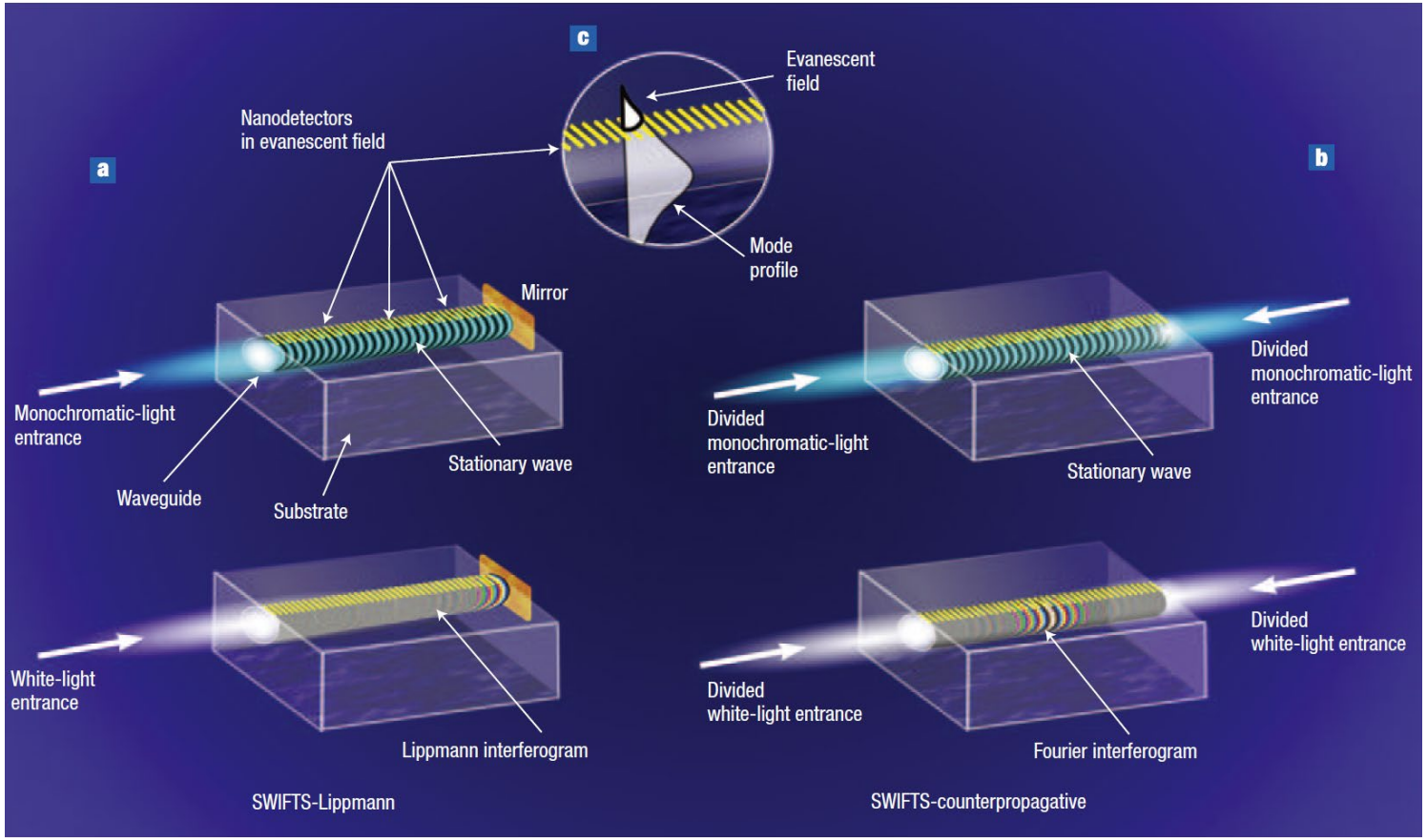
	Predicted			
Actual	Class 1	Class 2	Class 3	Class 4
Class 1	1523	343	461	0
Class 2	361	1622	343	0
Class 3	133	5	2186	0
Class 4	0	0	0	2324

A. Bermello *et al.*, ***Optics Letters*** 44, 5840 (2019)

不同光程差的多个Mach-Zehnder干涉仪阵列空间调制。

机器学习提高光谱范围和分辨率。

2.3 傅里叶型微光谱仪-空间调制



E. Coarer *et al*., ***Nature Photonics*** 1, 473 (2007)

驻波集成傅里叶变换光谱仪

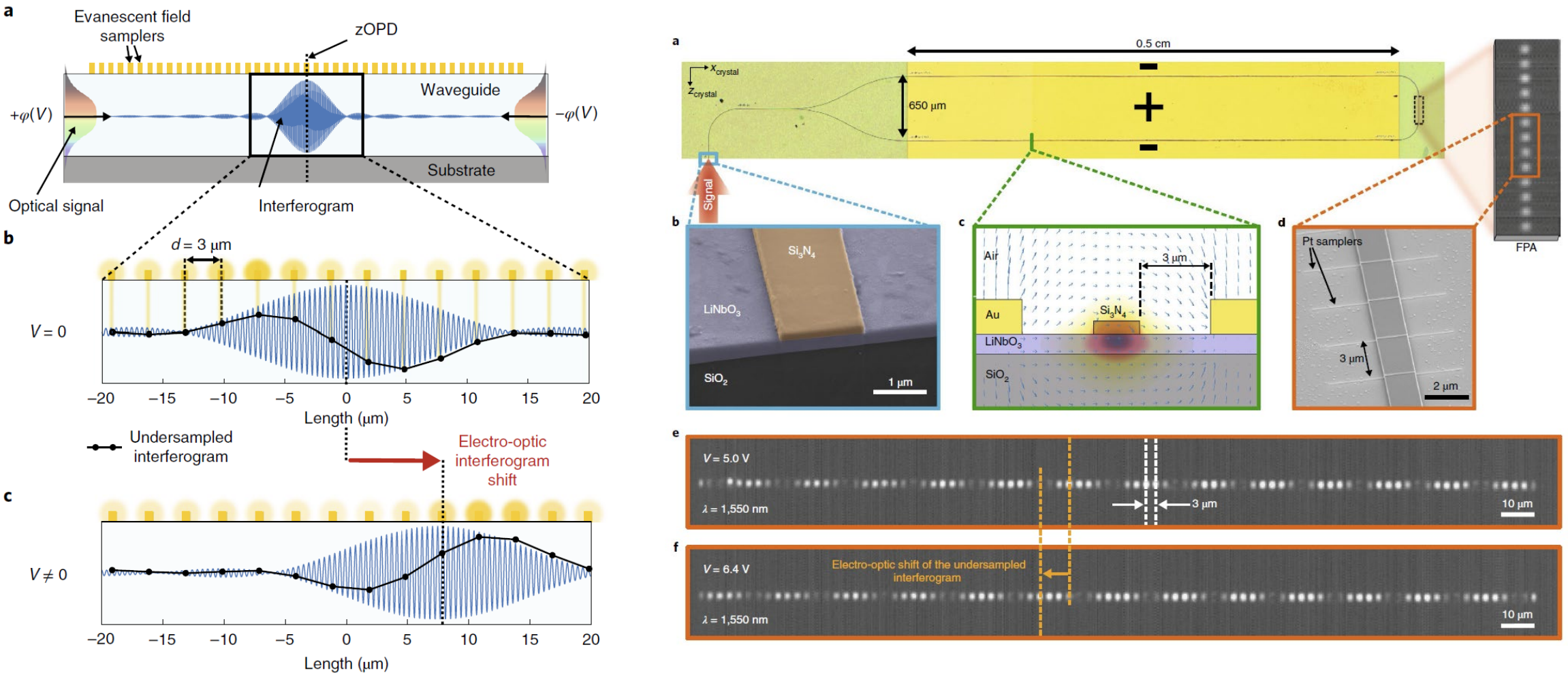
驻波集成傅里叶变换光谱仪有两种工作方式：

一种为单反式，即进入波导的光经反射镜反射后与入射光干涉，形成驻波；

另一种为相遇式，即从波导的两边分别入射光波，干涉形成驻波。将微型探测器集成到波导上，每个像元就可以采集到一定的干涉信号。

探测器像元尺寸限制了系统的光谱分辨率。

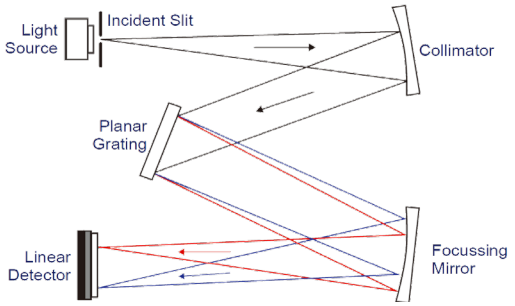
2.3 傅里叶型微光谱仪-空间调制

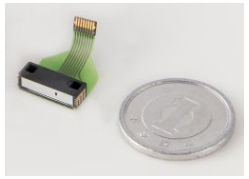


D. Pohl *et al.*, ***Nature Photonics*** 14, 24 (2020)

干涉位置通过电压可调，解决驻波集成傅里叶变换光谱仪欠采样问题，提高分辨率

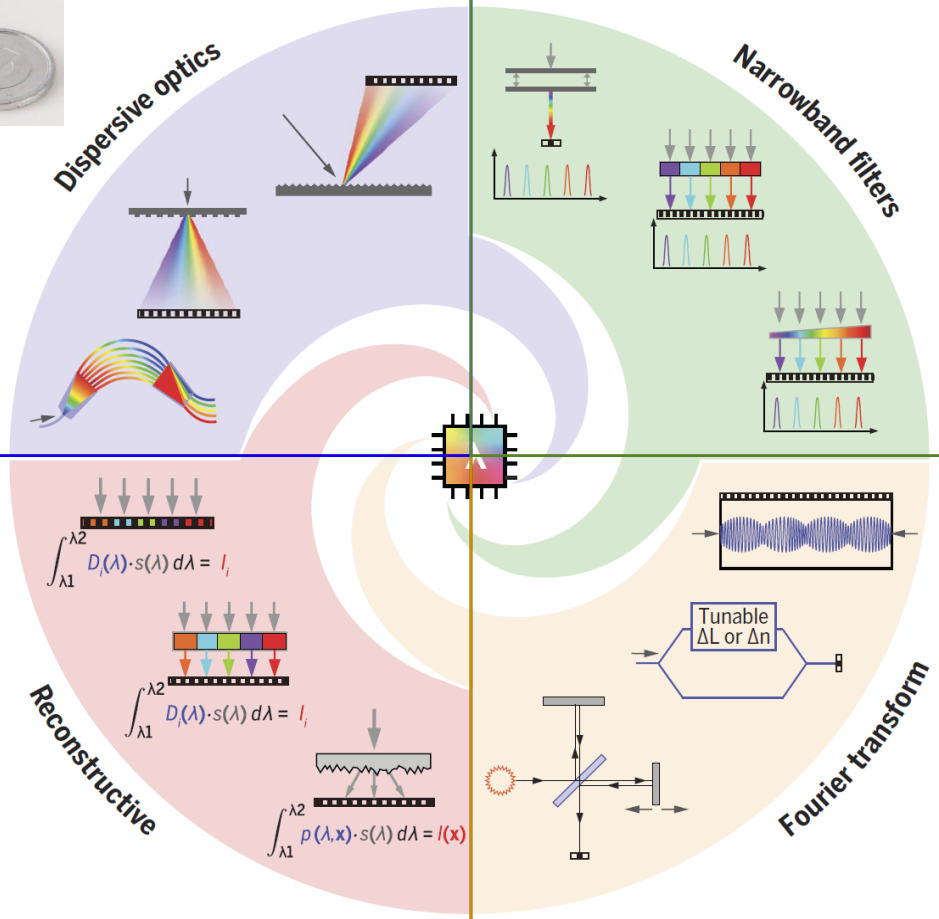
光谱仪微型化策略

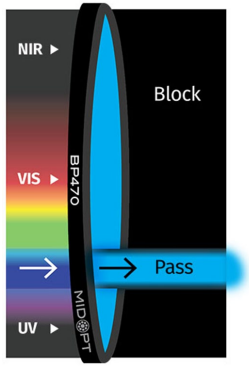




传统光谱仪体积庞大：光栅、探测器阵列、准直光路

缩小尺寸，**性能大幅下降**



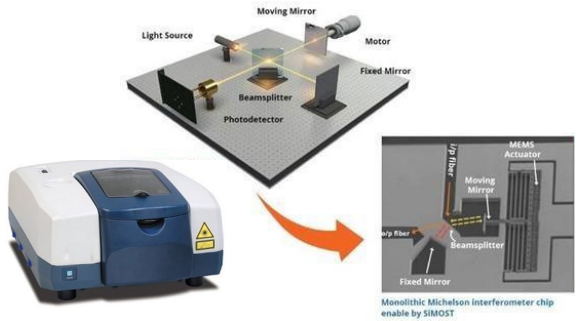


高分辨率与滤光片数量和成本矛盾

前三种路线在性能和成本方面有难以克服的挑战。

计算光谱用算力增强光学元件，能进一步减小器件的尺寸和成本。

计算光谱⁺ 是大趋势



运动部件导致稳定性差，分辨率低；
波导通光量低，灵敏度低

Z. Y. Yang *et al.*, ***Science*** 371, eabe0722 (2021)

2.4 计算重构型微光谱仪

前三类微型光谱仪在本质上是按比例缩小的传统光谱仪，这也是微型光谱仪发展初期的主要研究内容。它们都存在着各自的局限性，主要表现在光谱分辨率受光程路径的限制、光谱分辨率受通道数限制、刻蚀沉积等工艺的复杂性。这三种技术路线的微型化在分辨率和成本等方面存在许多难以克服的挑战，以至于几十年来全世界都没能基于这三种方案做出一款光谱分辨率高，成本又低的微型化产品，基础技术和概念也没有发生重大变化。

随着计算机的计算能力的提高，以及人们对算法研究的日渐成熟，新型的光谱获取方式——计算重构光谱被提出来。计算光谱利用高速的计算去部分替代物理分光元件的工作负荷，所以能进一步减小器件的尺寸和重量，是未来微型光谱发展的趋势。

计算光谱主要可分为光谱的编码和解码两个步骤，编码涉及实验标定，而解码则设计反问题的求解。光谱信息的编码方式主要分为两类：光谱空间映射编码和光谱响应编码。

2.4 计算重构型微光谱仪

Computational spectrometers involving two processes: **encoding** and **decoding**.

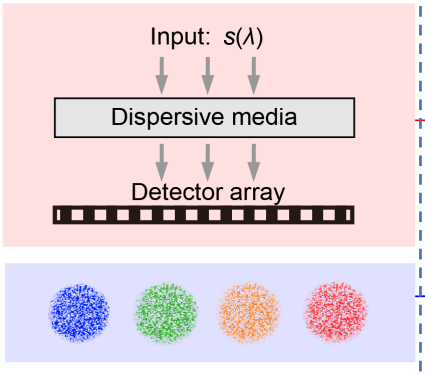
计算光谱主要分为光谱**编码**和**解码**两个过程。编码可分为：光谱**空间映射编码**和**光谱响应编码**。

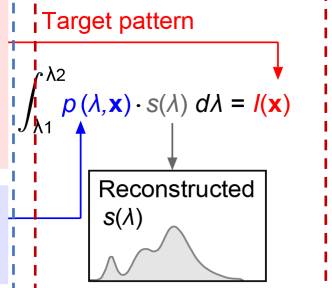
Two encoding strategies

Decoding

空间映射编码

1. Spectral to spatial: A disordered material is used to project the original spectrum to specific spatial patterns.



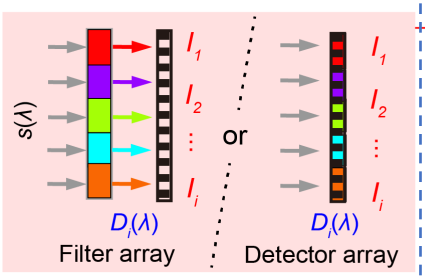


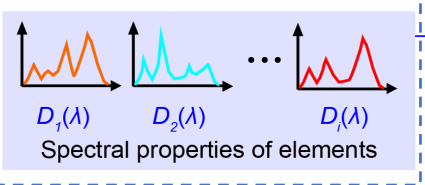
光谱响应编码：

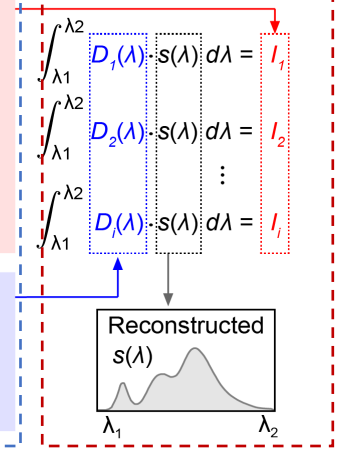
2. Spectral response: Converse spectrum into multiplexing signals

1.滤光编码分别需要滤光片和探测器。

2.探测器编码同时完成探测和编码功能。

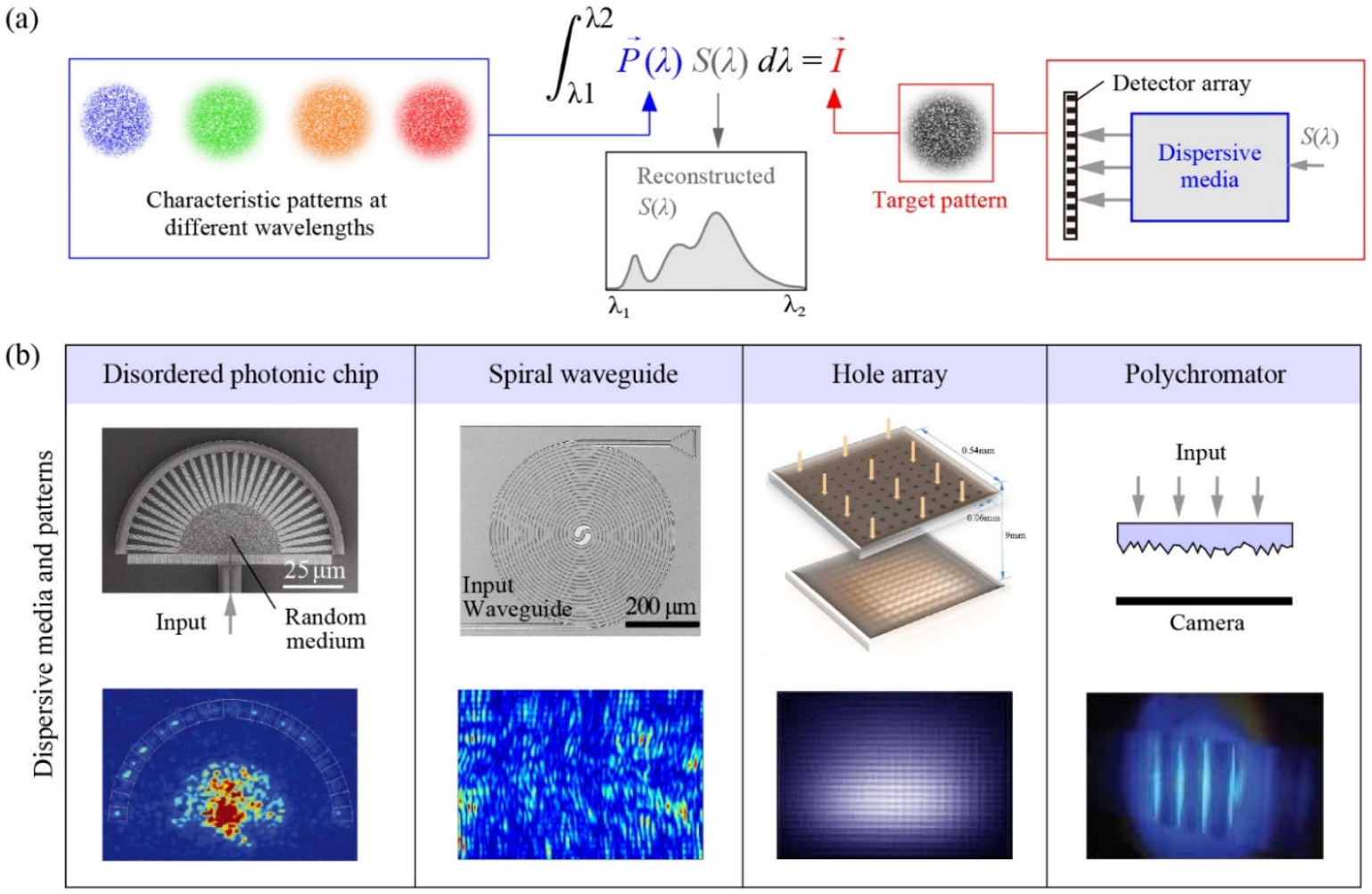






Z. Y. Yang *et al.*, ***Science*** 371, eabe0722 (2021)

2.4 计算重构型微光谱仪-空间映射编码



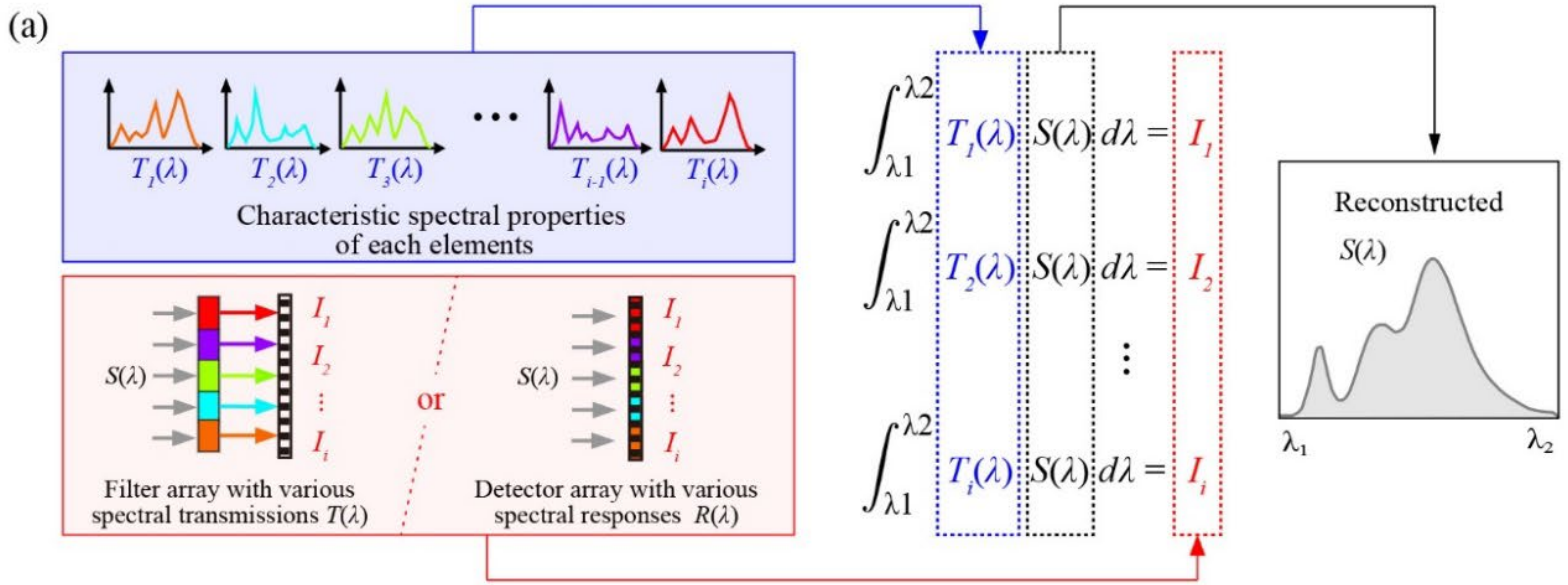
当单色光穿过诸如长多模光纤之类的色散元件时，由于光纤导模之间的串扰，它将在光纤的输出端创建与波长相关的特征图案。因此，当任意多色光通过多模光纤时，输出将是每个单色光创建的特征图案的叠加。要重构的目标光谱本质上是与这些特征图案相对应的比例权重的集合。

此外，还可以通过将光入射到诸如无序光子芯片、螺旋波导、色散孔阵列或多色器之类的小型光学元件来生成特征图案

2.4 计算重构型微光谱仪-空间映射编码

空间映射特征图案容易受到环境干扰，比如振动、温度变化的影响。光谱分辨率越高，影响就越大。基本无法产业化。

2.4 计算重构型微光谱仪-光谱编码

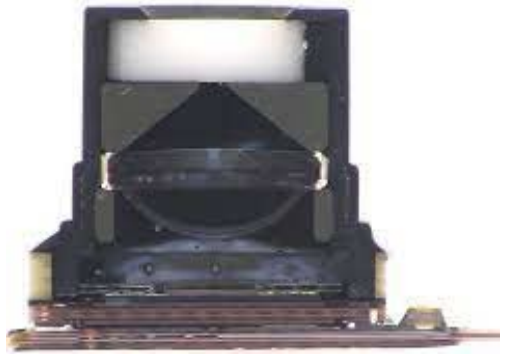


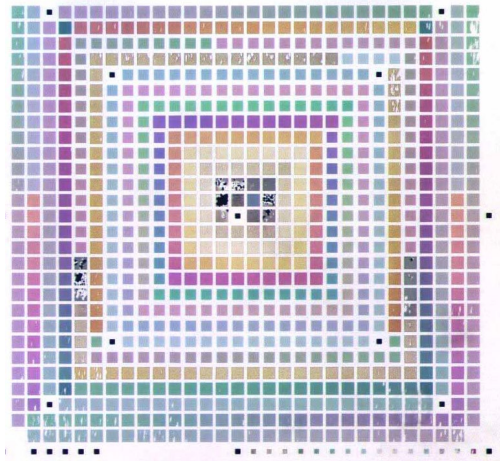
实现重建光谱仪的第二种方法是为每个探测器定制不同的光谱响应编码。

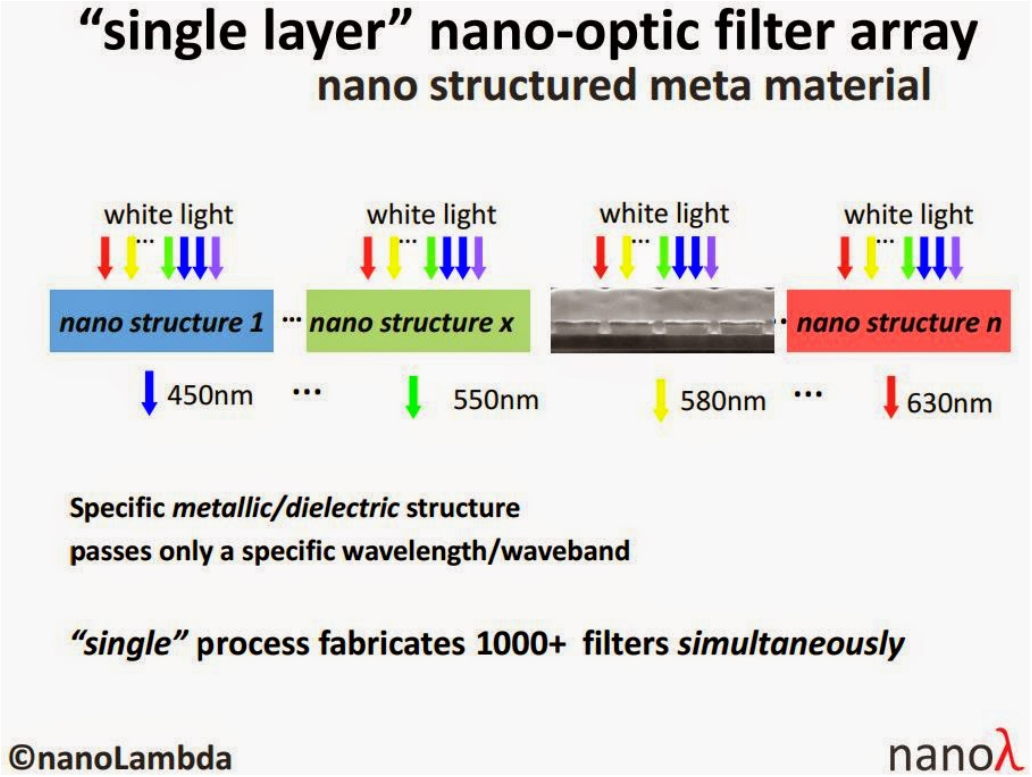
光谱响应编码又分为滤光片编码、探测器响应编码、照明编码

2.4 计算重构型微光谱仪-光谱滤光编码

最早的光谱编码计算型光谱仪可以追溯到2007年韩国NanoLambda公司申请的美国专利。他们用的是表面等离子体/超材料随机滤光阵列。



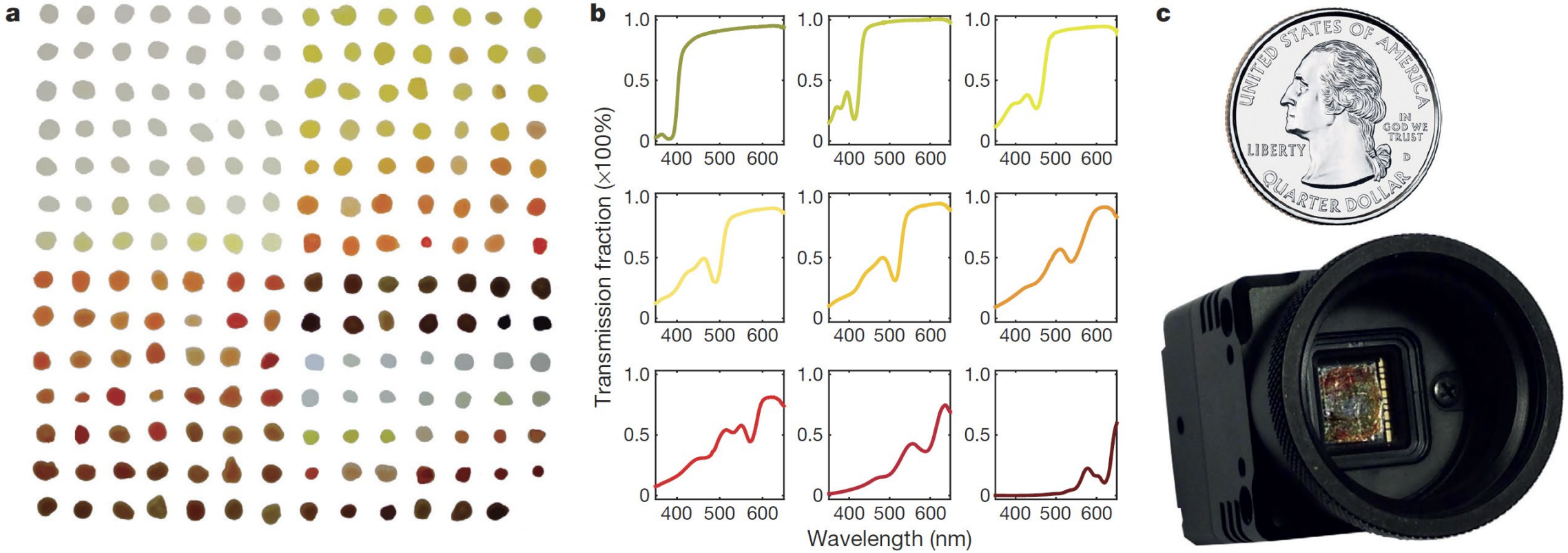




韩国Nanolambda公司的等离子体共振滤光计算光谱仪

2.4 计算重构型微光谱仪-光谱滤光编码

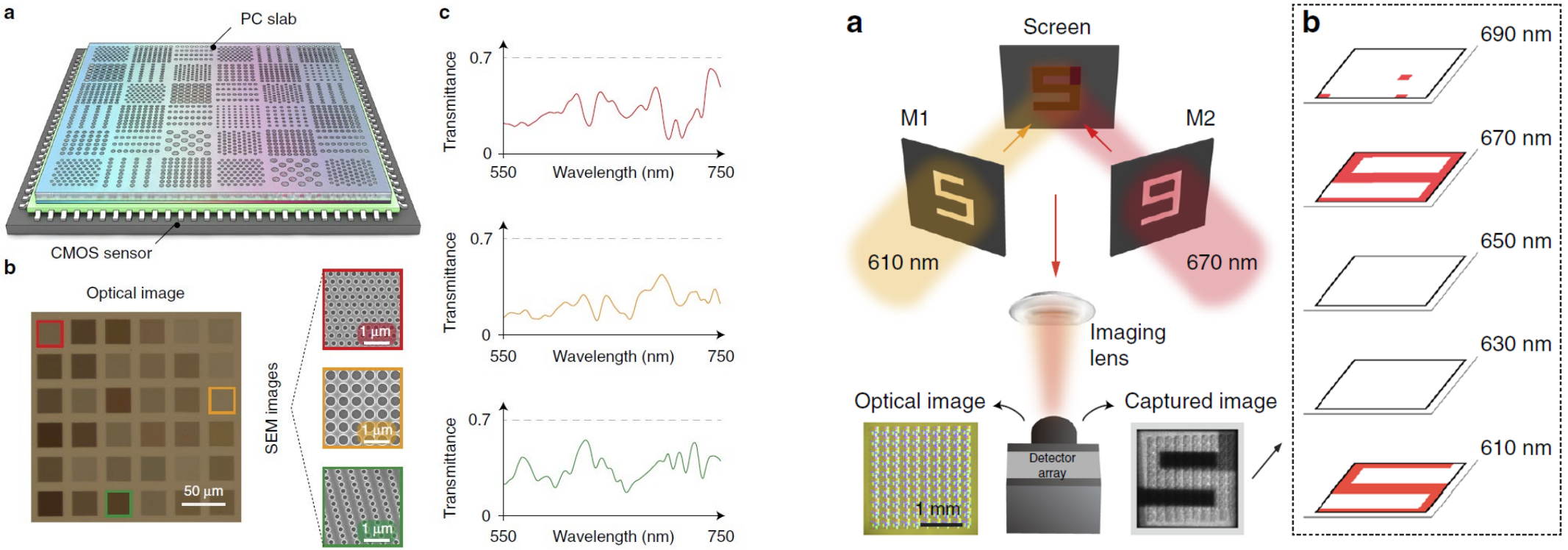
此后相继出现了各种滤光结构，几乎任何能产生不同光谱响应函数的滤光元件都可以应用于计算光谱仪系统，如量子点、等离子体、纳米光学、超材料、光子晶体板、液晶、薄膜、标准具阵列等。



鲍捷老师的量子点光谱仪

J. Bao *et al.*, ***Nature*** 523, 67 (2015)

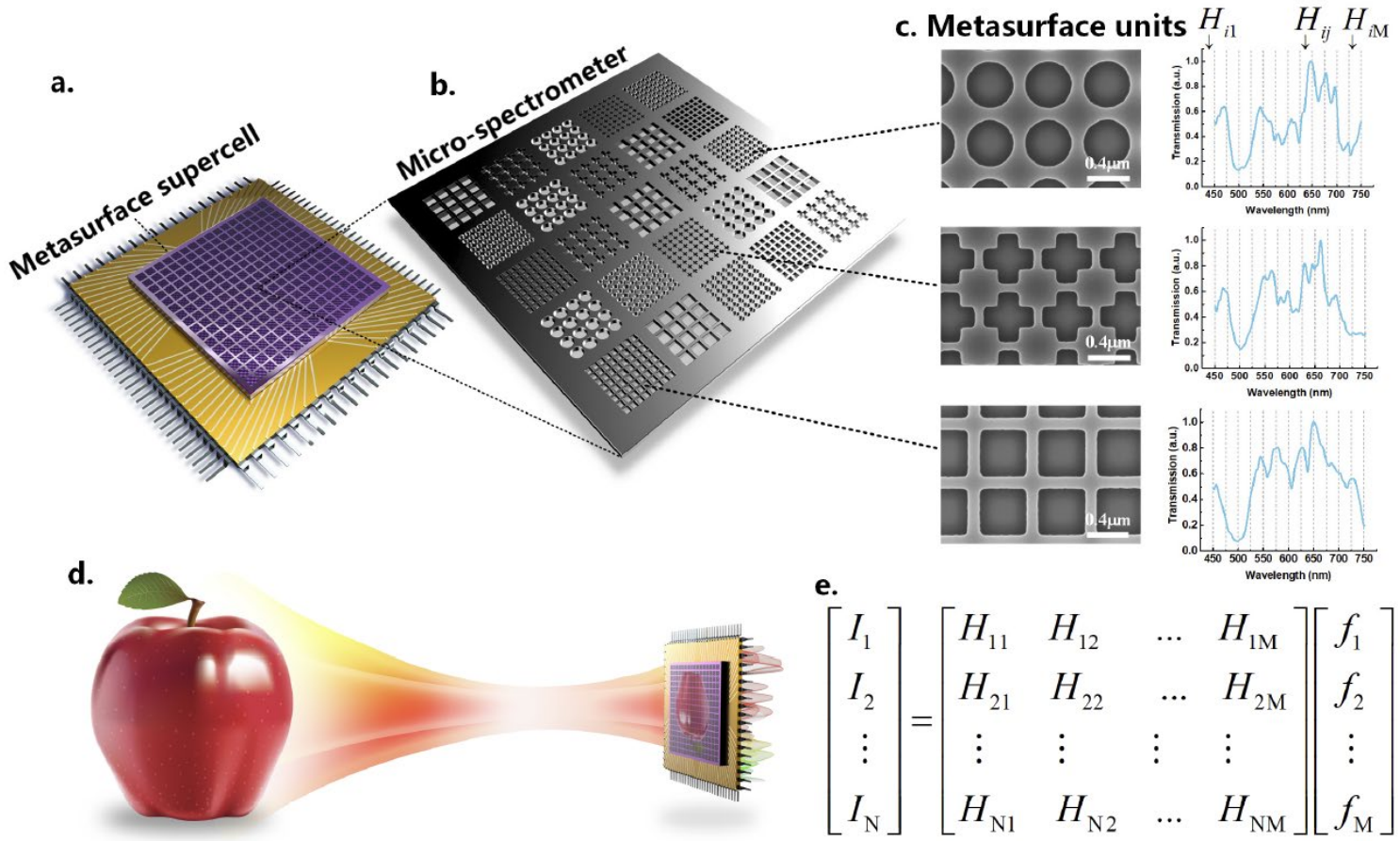
2.4 计算重构型微光谱仪-光谱滤光编码



Z. Wang *et al*., ***Nature Communications*** 10, 1020 (2019)

威斯康辛的Zongfu Yu老师的光子晶体计算光谱仪+成像光谱（10×10 像素）

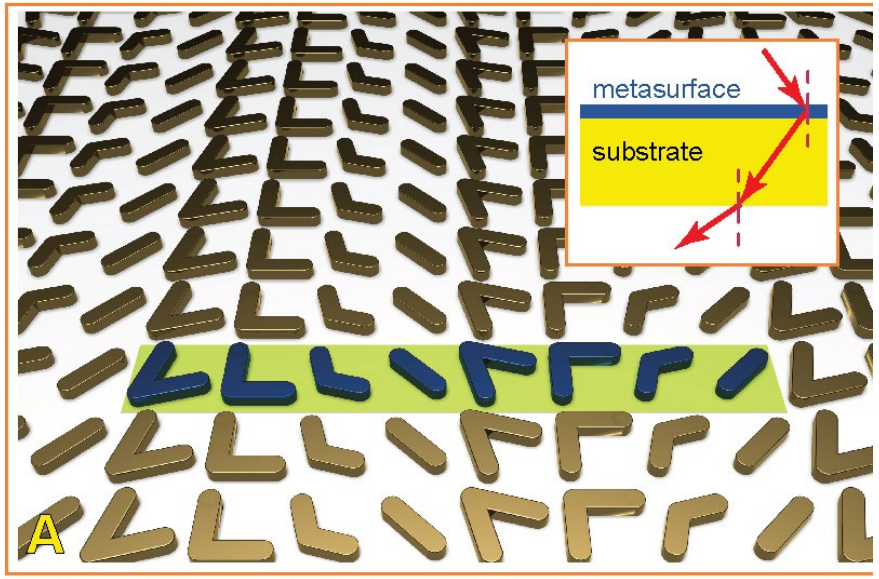
2.4 计算重构型微光谱仪-光谱滤光编码



清华与光科技（Seetrum）采用的是硅基的超材料，和Nanolambda相比通光量会有所提高。

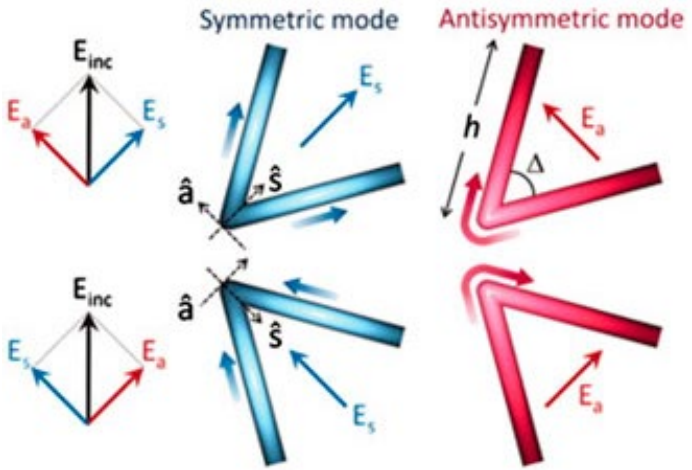
与光科技的超材料计算光谱仪+计算成像光谱仪

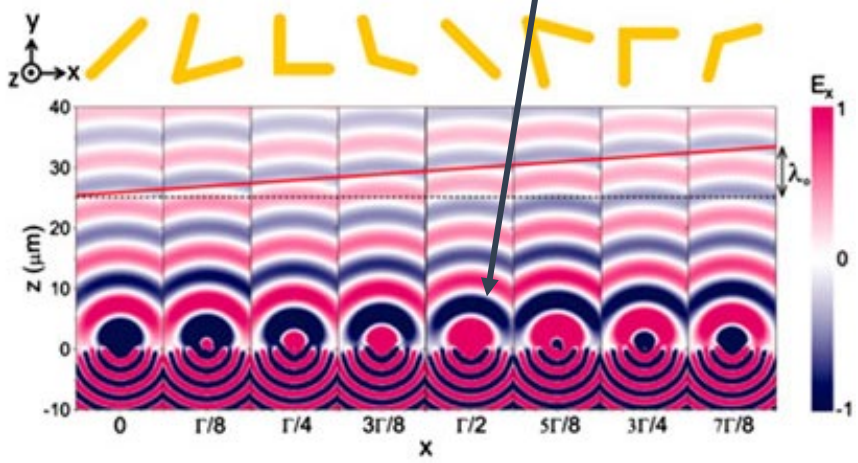
How does metasurface works?



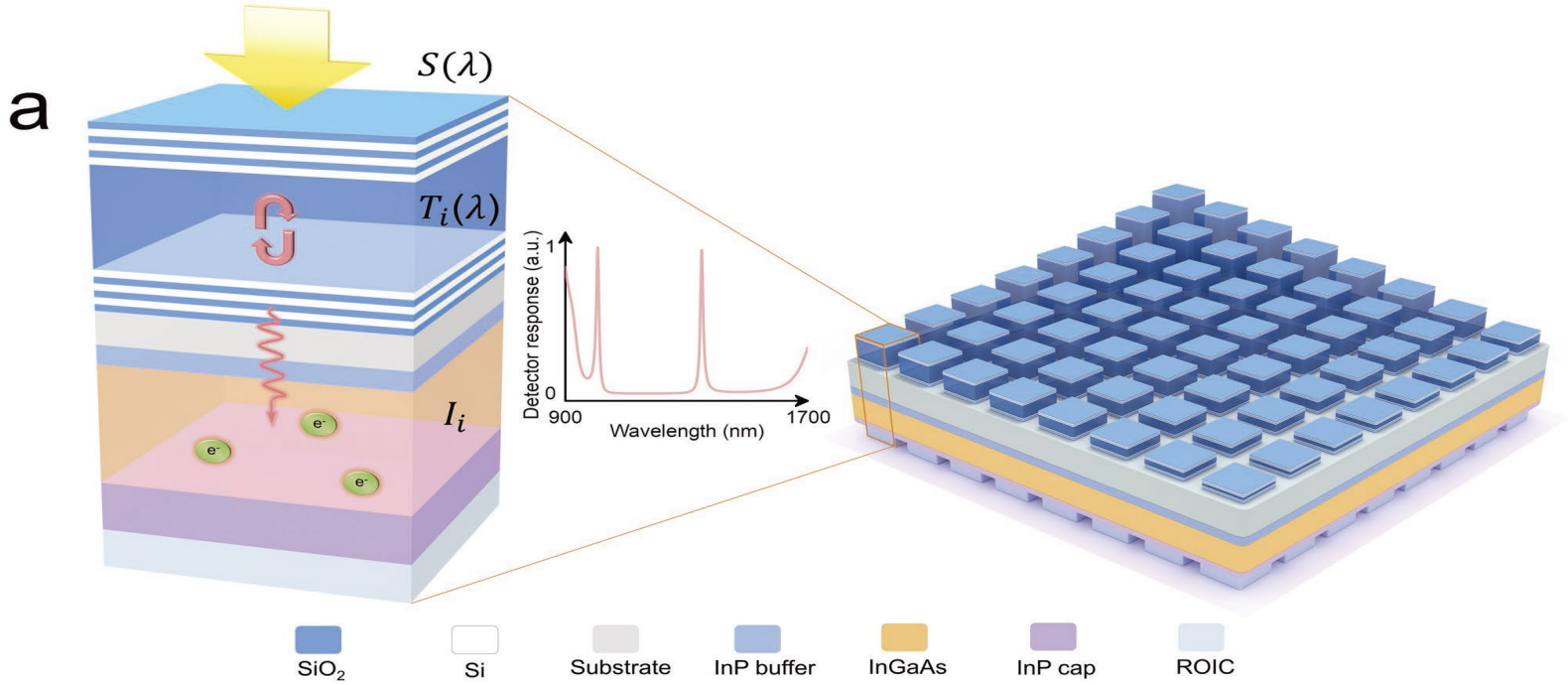
The effective permeability (μ) and permittivity (ε) are governed by different aspects of the shape of metallic unit cells.

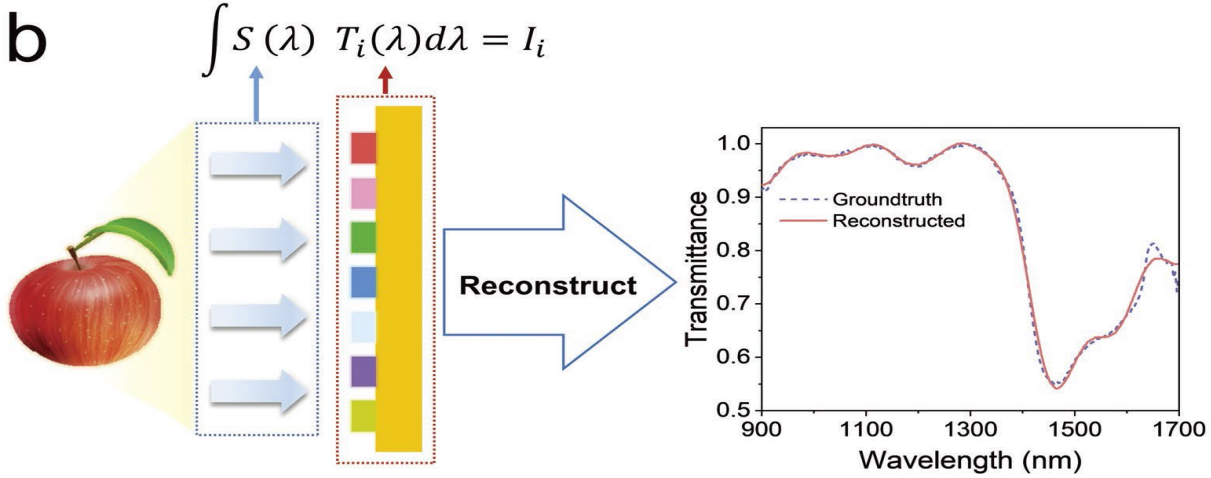
Light phase is modulated.





2.4 计算重构型微光谱仪-光谱滤光编码

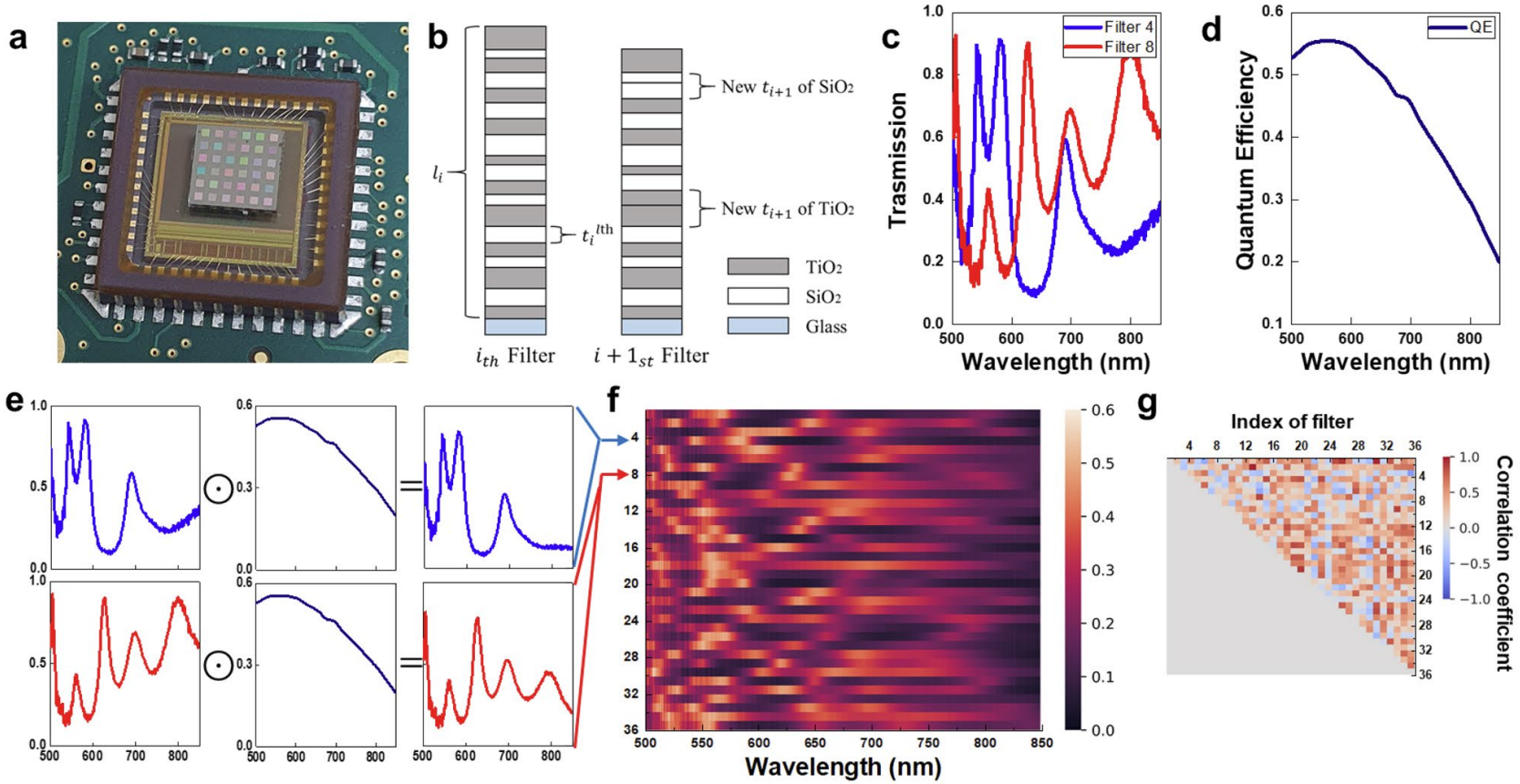




Z. Xuan, ***Adv. Optical Mater.*** 2200284 (2022)

王少伟老师基于FP腔阵列计算光谱

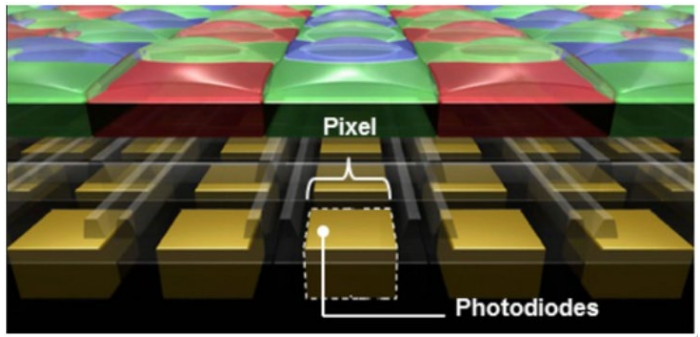
2.4 计算重构型微光谱仪-光谱滤光编码



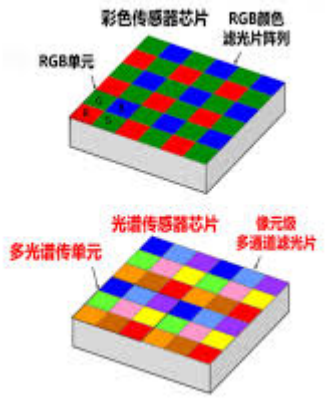
C. Kim, ***Scientific Reports*** 12,4053 (2022)

干涉薄膜编码型计算光谱仪

2.4 计算重构型微光谱仪-光谱滤光编码



从传统RGB拜尔滤光片到马赛克滤光片阵列

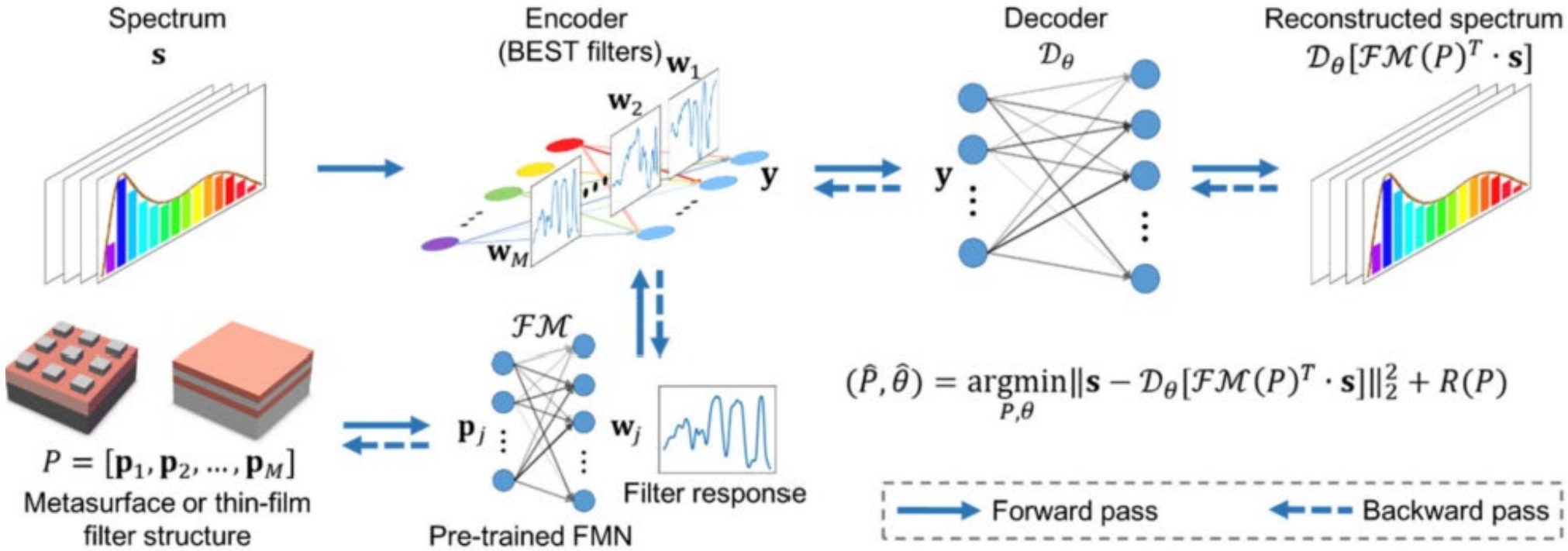


分立滤光片阵列无法覆盖所有波长

计算重构出全谱段

华为Mate 70 红枫摄像头工作原理

2.4 计算重构型微光谱仪-光谱滤光编码优化算法



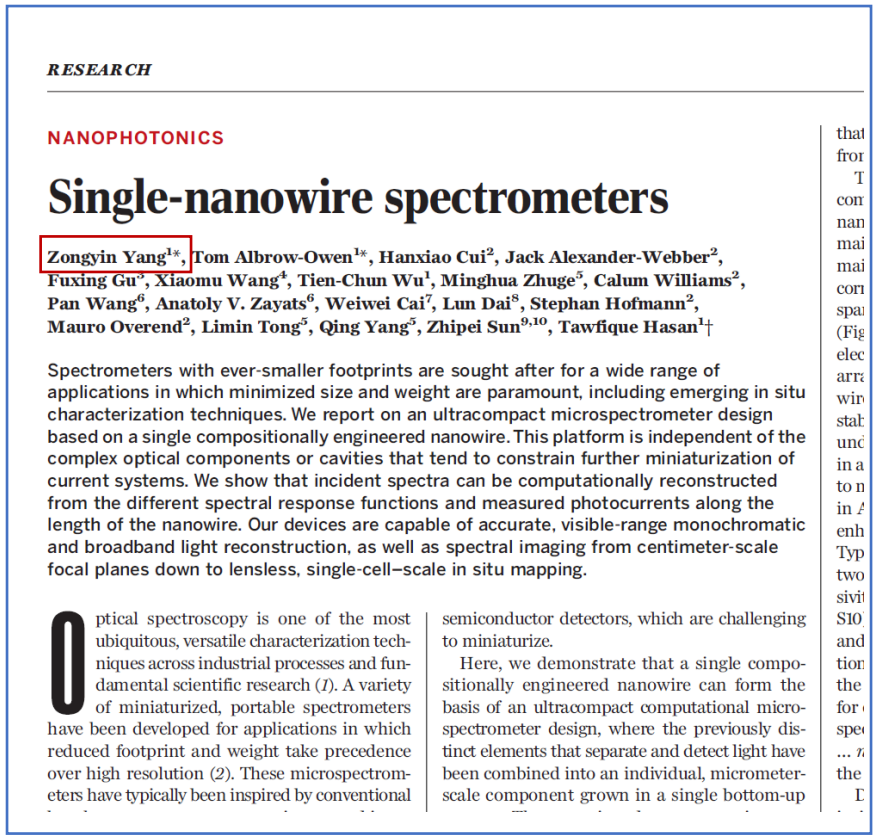
H. Song *et al.*, ***Advanced Theory and Simulations*** 4, 2000299 (2021)
DOI: 10.1002/adts.202000299

不能完全随机，浙大郝翔老师的随机滤光元件逆向迭代优化设计

2.4 计算重构型微光谱仪-光谱响应编码

滤光型光谱编码需要单独制造滤光片和探测器阵列，这增加了制造的复杂性并限制了小型化。光谱响应编码纳米结构可以整合滤光和探测两种功能。例如，基于成分梯度合金化半导体纳米线的计算光谱仪，它可以沿轴向分成若干个部分(探测器)。这些探测器的响应函数由于合金成分的变化而变化。尽管分辨率仍然不高(约5-10nm)，但这种纳米线光谱仪将波长选择和光电检测功能集成到单个纳米结构中，将尺寸缩小为数十微米，比其他任何的计算光谱仪系统低两个数量级。

2.4 计算重构型微光谱仪-光谱响应编码



RESEARCH

NANOPHOTONICS

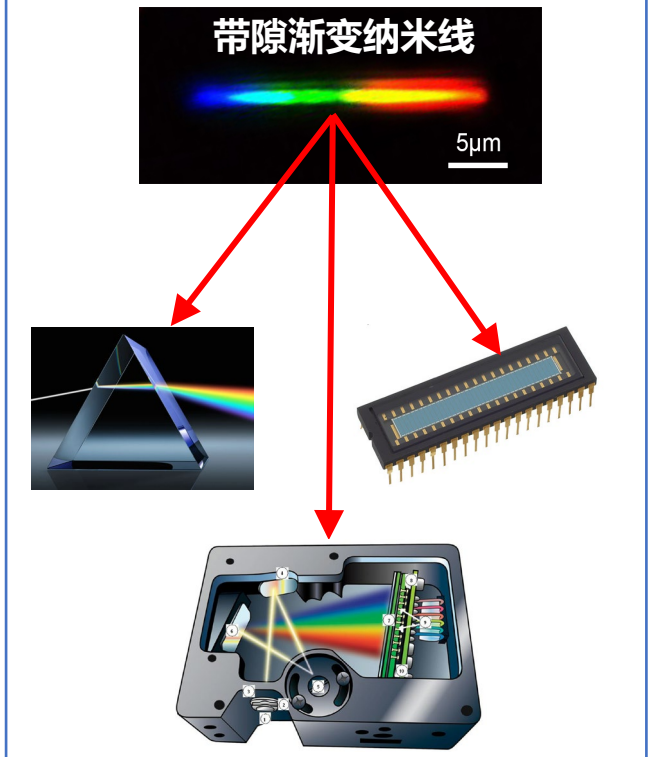
Single-nanowire spectrometers

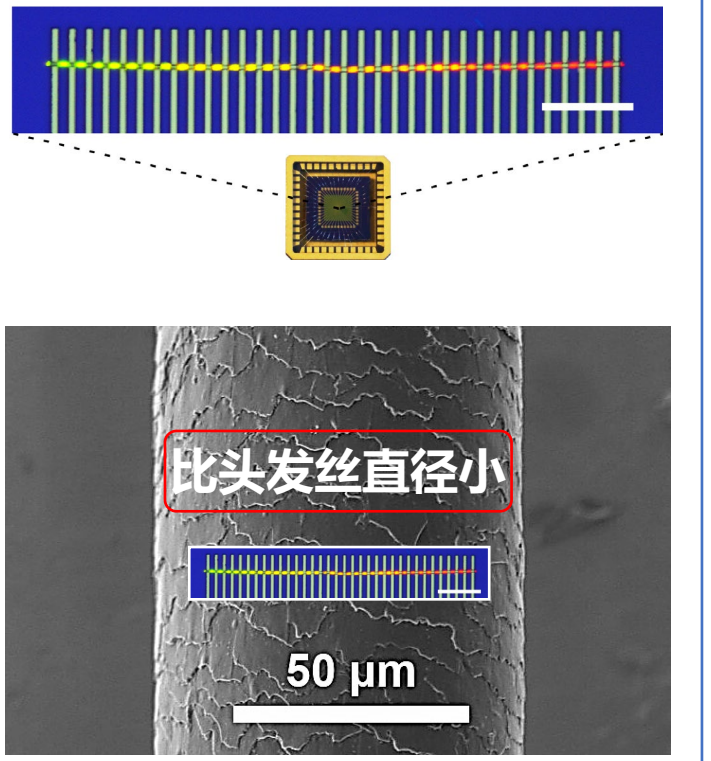
Zongyin Yang[1]*, Tom Albrow-Owen[1]*, Hanxiao Cui[2], Jack Alexander-Webber[2], Fuxing Gu[3], Xiaomu Wang[4], Tien-Chun Wu[1], Minghua Zhuge[5], Calum Williams[2], Pan Wang[6], Anatoly V. Zayats[6], Weiwei Cai[7], Lun Dai[8], Stephan Hofmann[2], Mauro Overend[2], Limin Tong[5], Qing Yang[5], Zhipei Sun[9,10], Tawfique Hasan[1]†

Spectrometers with ever-smaller footprints are sought after for a wide range of applications in which minimized size and weight are paramount, including emerging in situ characterization techniques. We report on an ultracompact microspectrometer design based on a single compositionally engineered nanowire. This platform is independent of the complex optical components or cavities that tend to constrain further miniaturization of current systems. We show that incident spectra can be computationally reconstructed from the different spectral response functions and measured photocurrents along the length of the nanowire. Our devices are capable of accurate, visible-range monochromatic and broadband light reconstruction, as well as spectral imaging from centimeter-scale focal planes down to lensless, single-cell–scale in situ mapping.

Optical spectroscopy is one of the most ubiquitous, versatile characterization techniques across industrial processes and fundamental scientific research (*1*). A variety of miniaturized, portable spectrometers have been developed for applications in which reduced footprint and weight take precedence over high resolution (*2*). These microspectrometers have typically been inspired by conventional semiconductor detectors, which are challenging to miniaturize.

Here, we demonstrate that a single compositionally engineered nanowire can form the basis of an ultracompact computational microspectrometer design, where the previously distinct elements that separate and detect light have been combined into an individual, micrometer-scale component grown in a single bottom-up



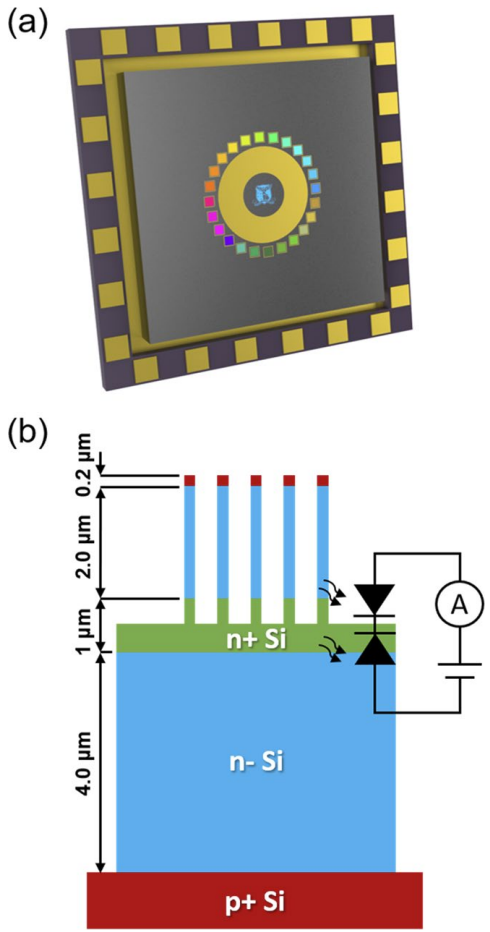


创新意义：首次提出探测器编码新机理，用一根纳米线替代了光栅、探测器和准直光路

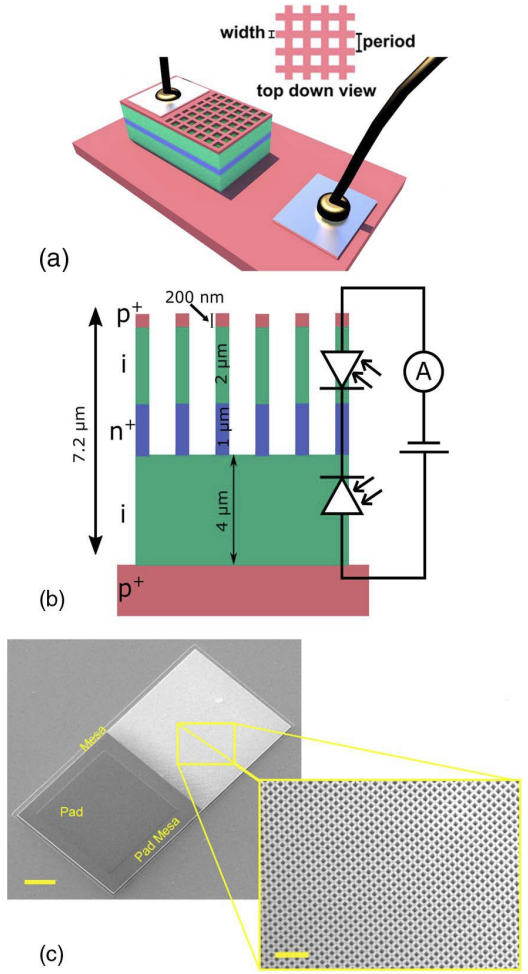
世界最小光谱仪，有望用于解决卡脖子技术

Z. Y. Yang *et al.*, ***Science*** 365, 1017 (2019)

2.4 计算重构型微光谱仪-光谱响应编码



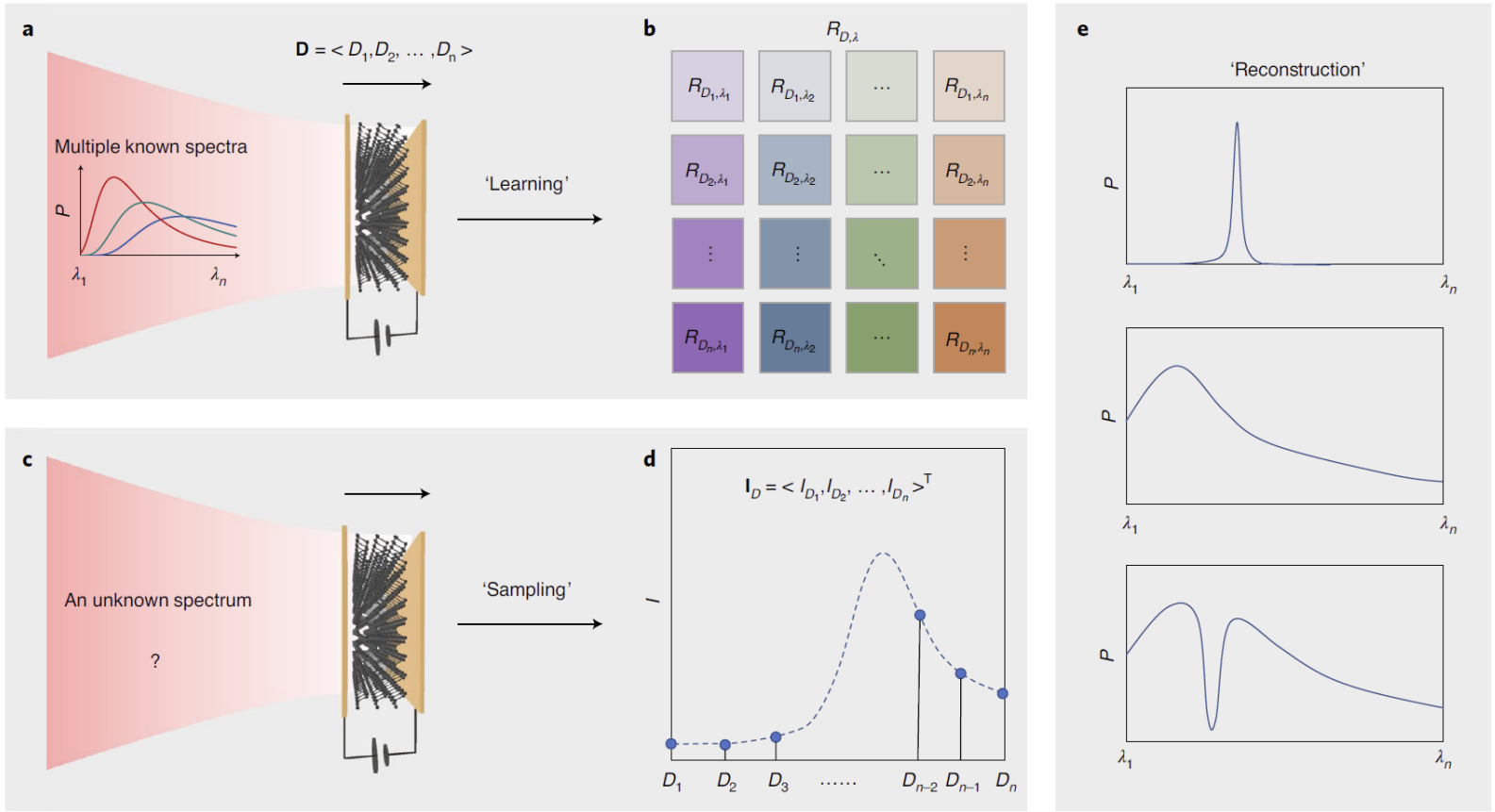
J. Meng *et al*., ***Nano Letters*** 20, 320 (2020)



J. Cadusch *et al*., ***Optica*** 6, 1171 (2019)

纳米线阵列计算光谱仪，光谱编码+探测集成在纳米线阵列中

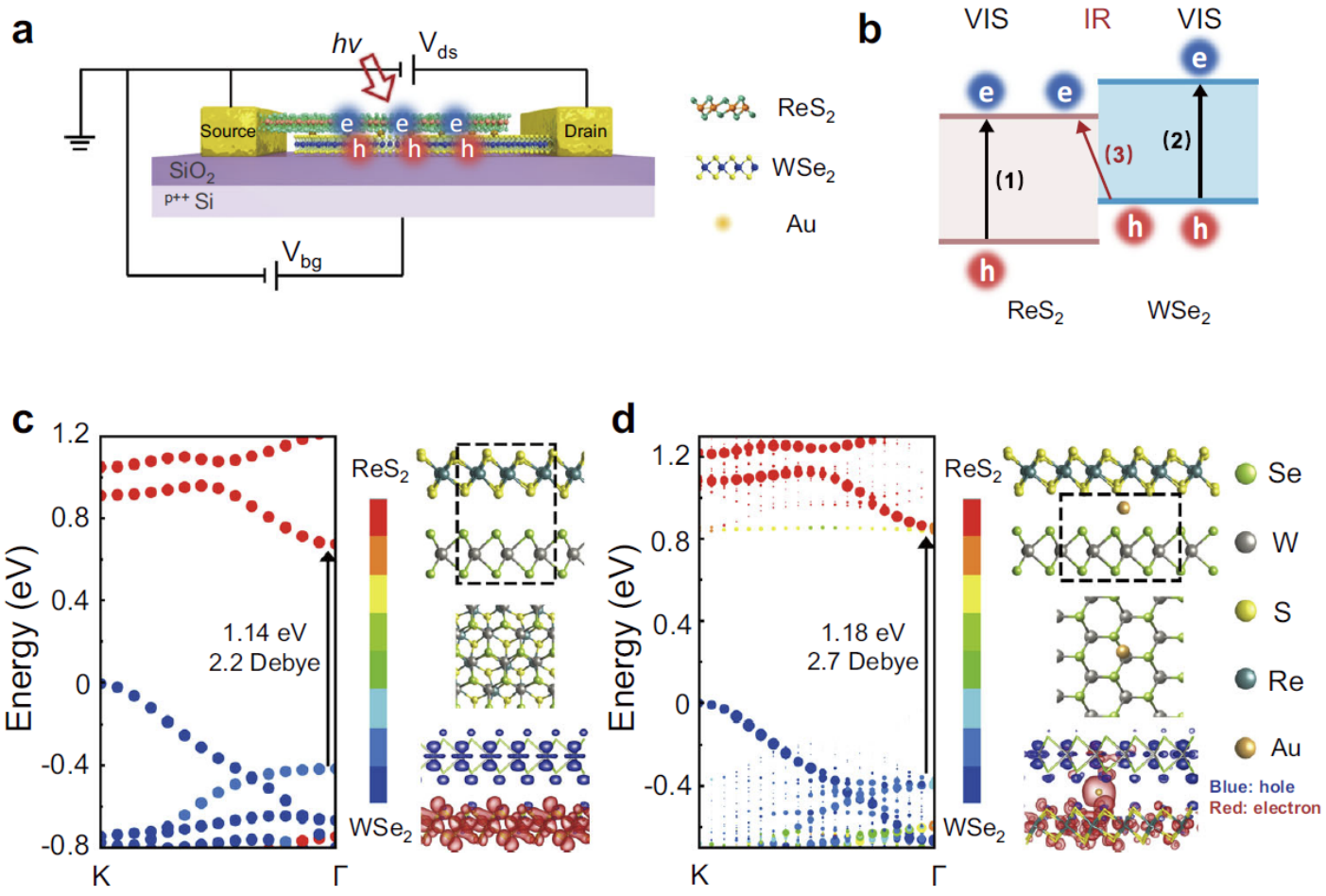
2.4 计算重构型微光谱仪-光谱响应编码



S. Yuan, ***Nature Photonics***15,601 (2021)

基于二维材料的红外计算光谱仪

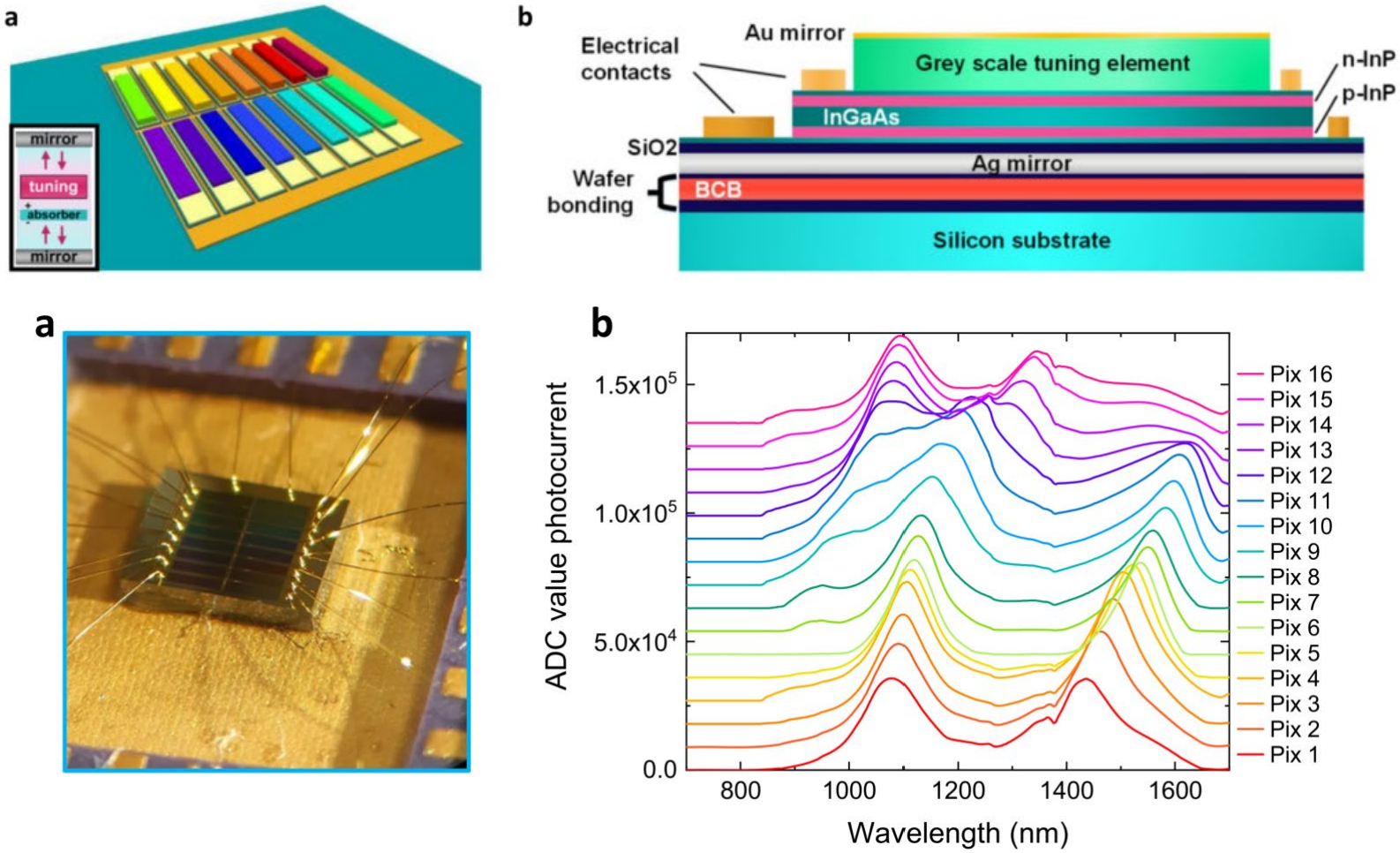
2.4 计算重构型微光谱仪-光谱响应编码



W. Deng, ***Nature Communications*** 13,4627 (2022)

基于二维材料层间激子的红外计算光谱仪

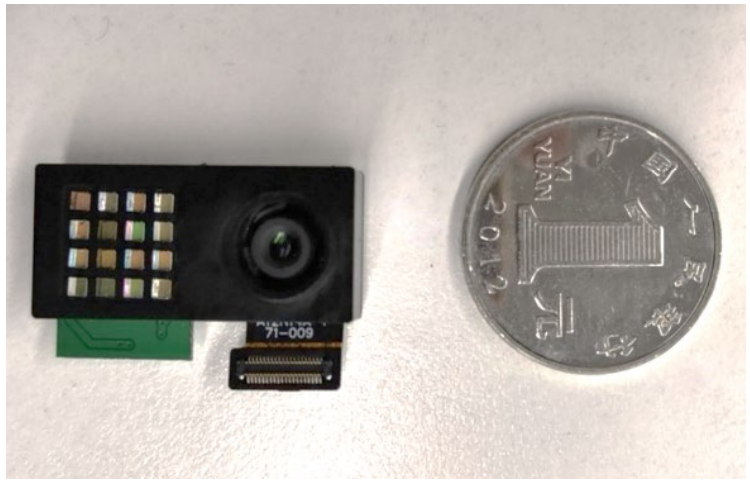
2.4 计算重构型微光谱仪-光谱响应编码

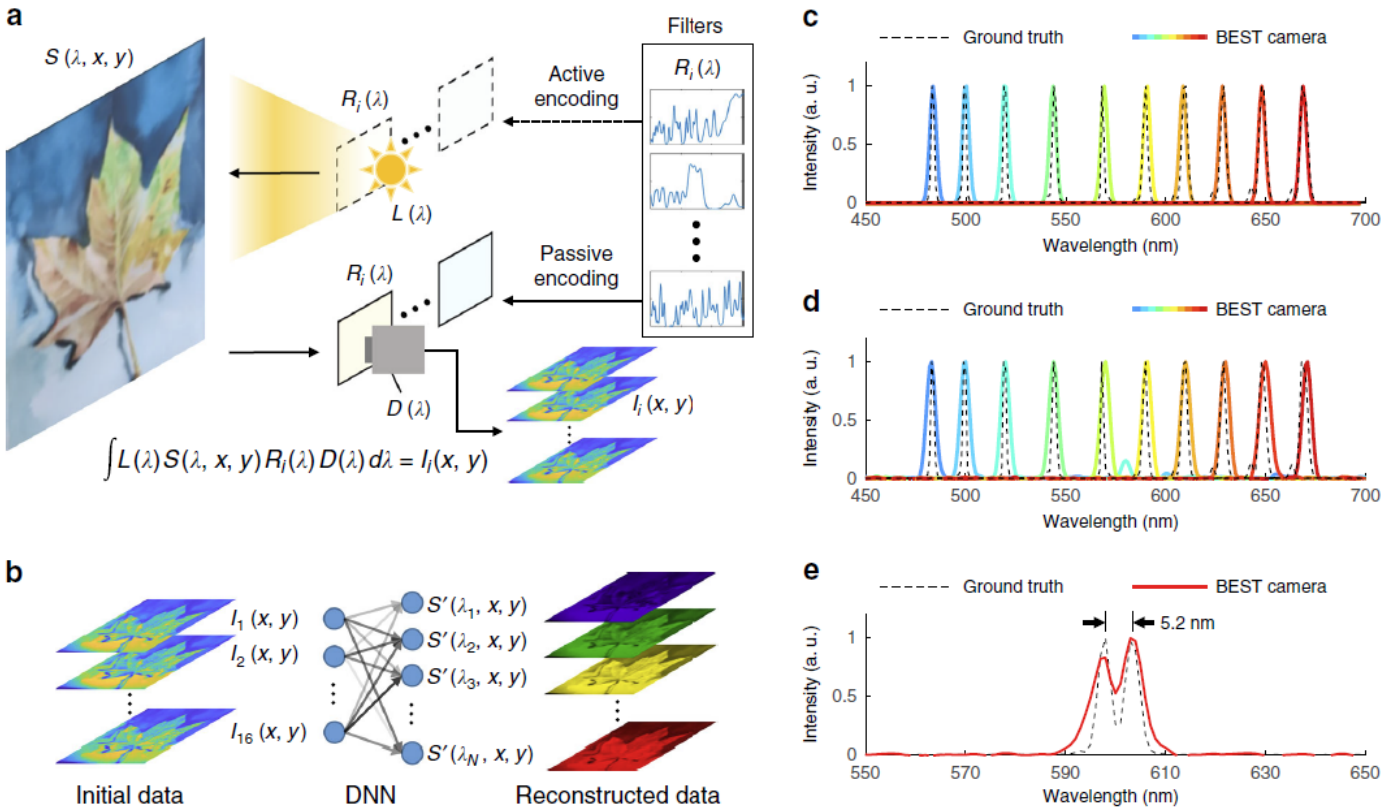


K. Hakkel *et al*., ***Nature Communications*** 13, 103 (2022)

荷兰MantiSpectra公司基于谐振腔响应加强计算光谱仪，覆盖短波红外

2.4 计算重构型微光谱仪-光谱照明编码

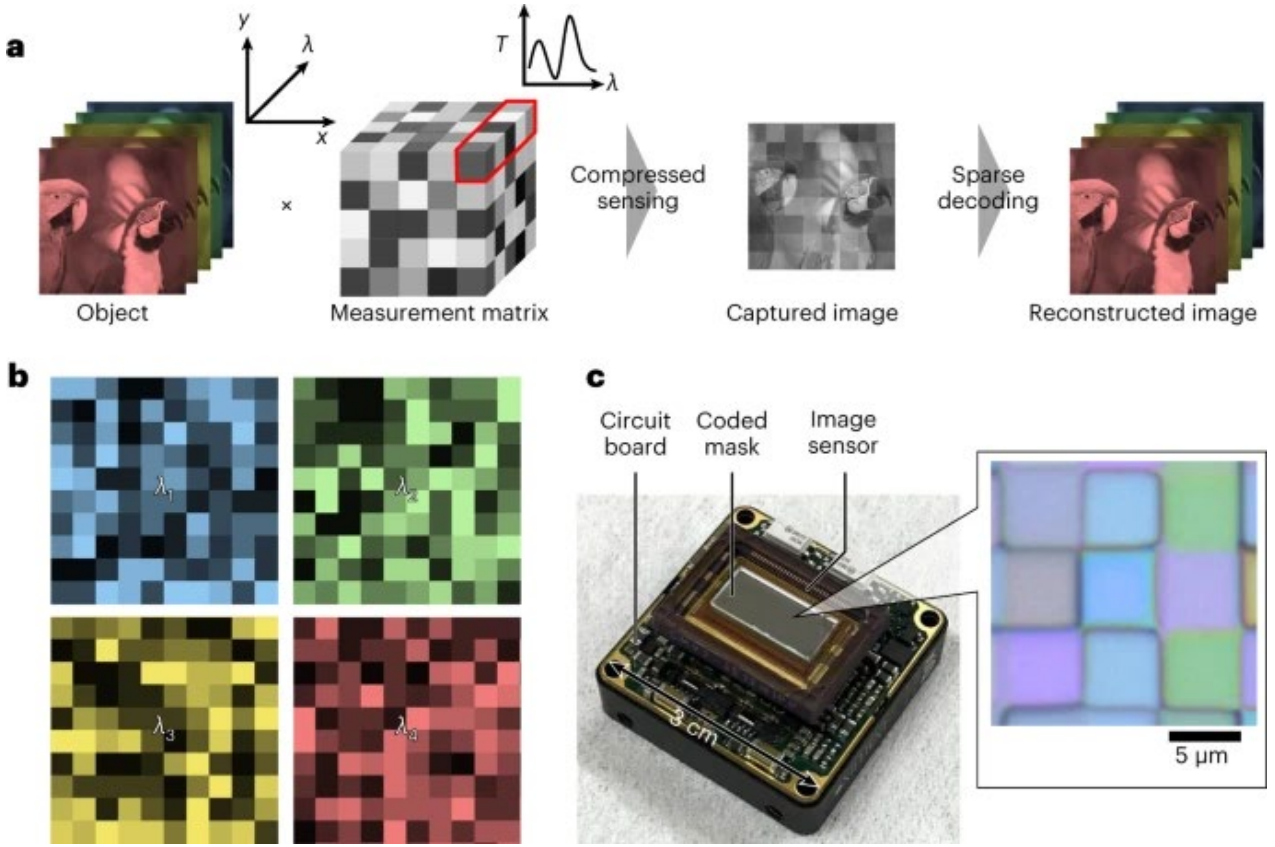




W. Zhang *et al.*, ***Light: Science&Applications*** 10, 108 (2021)

浙大郝翔老师与舜宇光学合作的基于滤光片编码的主动照明式计算成像光谱仪

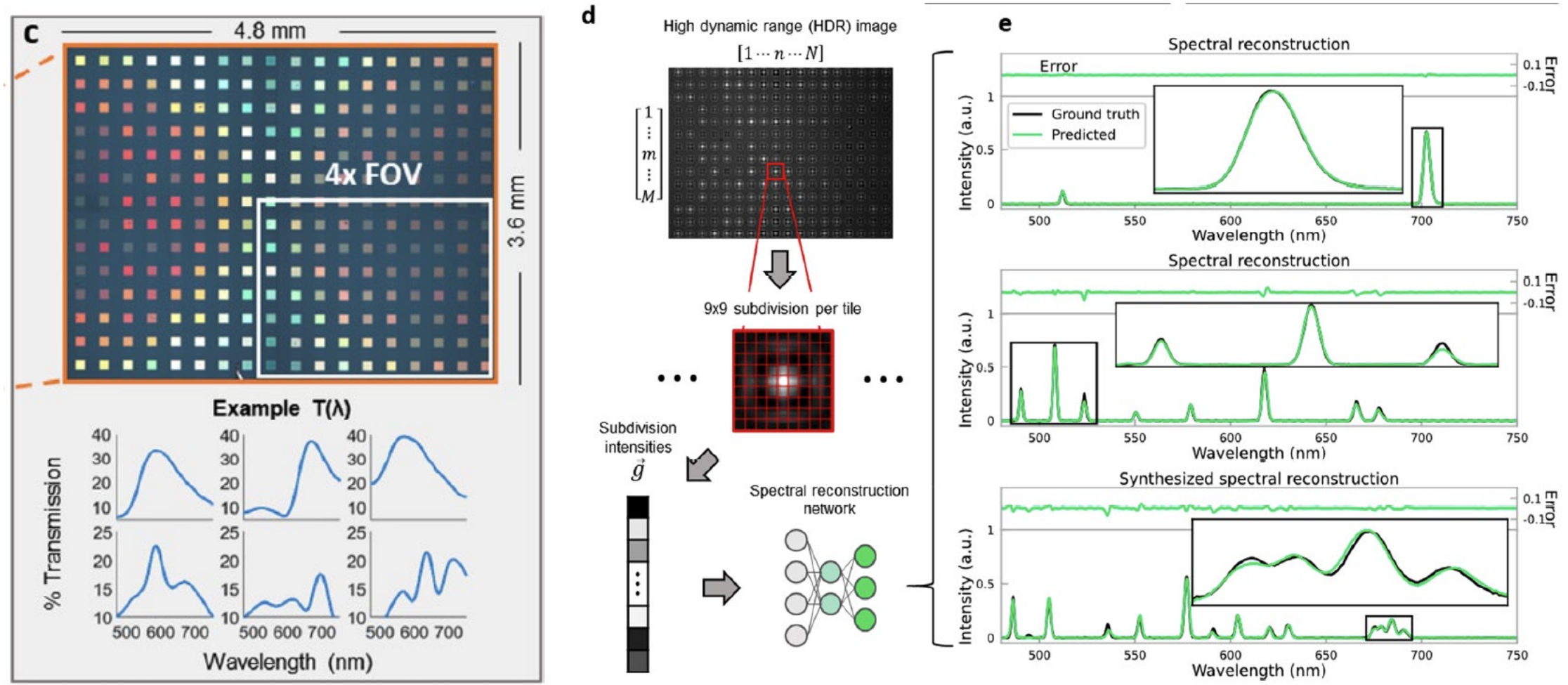
2.4 计算重构型微光谱仪-计算光谱视频



M. Yako *et al.*, ***Nature Photonics*** 17, 218 (2023)

松下公司基于随机滤光片的计算型光谱相机，可以工作在34 帧/秒，VGA分辨率

2.4 计算重构型微光谱仪-机器学习重构



C. Brown *et al.*, ***ACS Nano*** 15, 6305 (2021)

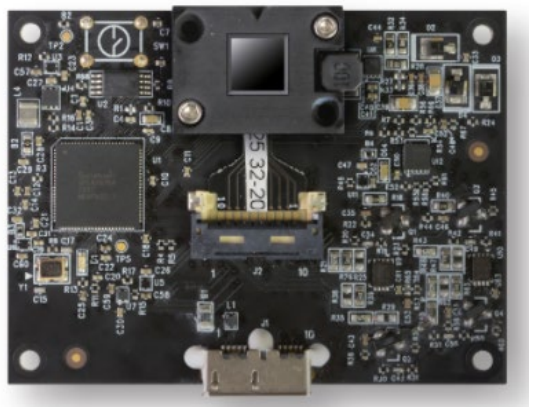
2. 计算重构型微光谱仪总结

之前提到的色散型、滤光片型和傅里叶变换型光谱仪的微型化需要克服的主要难点是色散元件和光路的缩小带来的性能退化和成本上升，而计算光谱是把这部分的技术挑战转嫁到算力和算法上，为光谱仪的微型化带来了全新的思路。计算光谱带来的好处是色散元件和光路可以不需要精细又精确，对材料和加工的要求大大降低，尺寸和成本的问题迎刃而解。因此，计算型光谱仪会是未来发展的一个热点，也是未来最有前景的光谱仪。

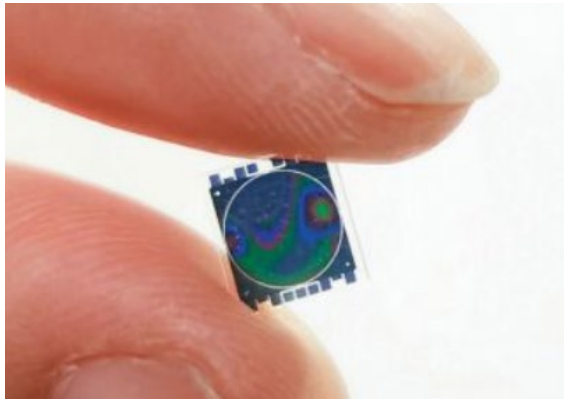
但是计算光谱也不是万能的，带来好处的同时也产生了一些新的问题，比如计算光谱的数据可靠性远不如基于传统方案的光谱仪，还有计算光谱的大批量生产一致性差，计算重构需要较多算力和时间等。这些问题也是制约计算光谱真正走上实用化的最大挑战。

3.成像光谱仪微型化技术-扫描式

几种扫描式微型光谱成像技术方案和产品



Unispectral：静电驱动
F-P腔微型成像光谱仪

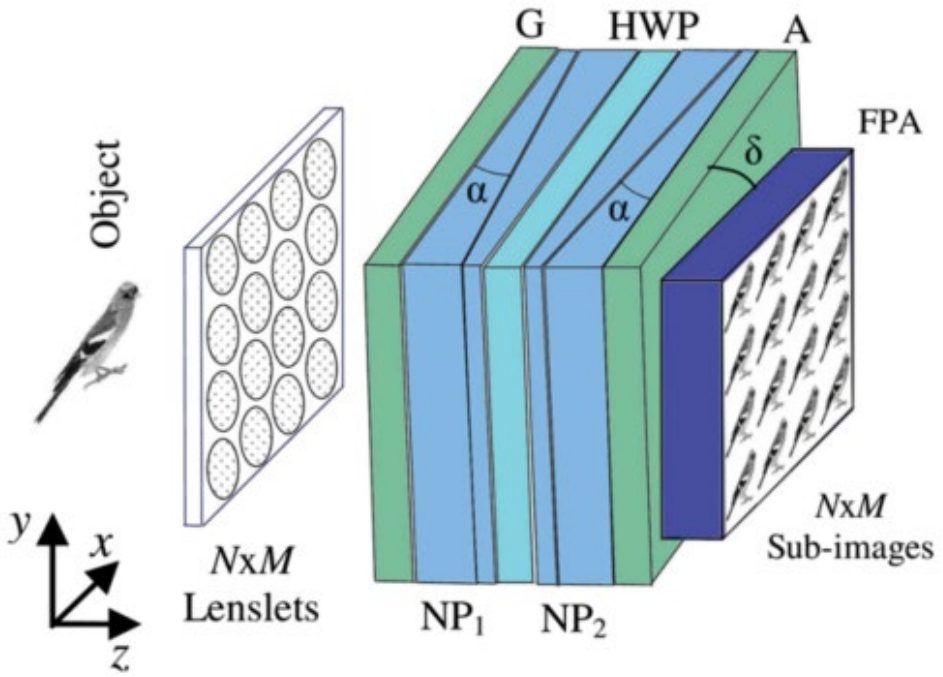


深圳海谱：压电驱动
F-P腔微型成像光谱仪

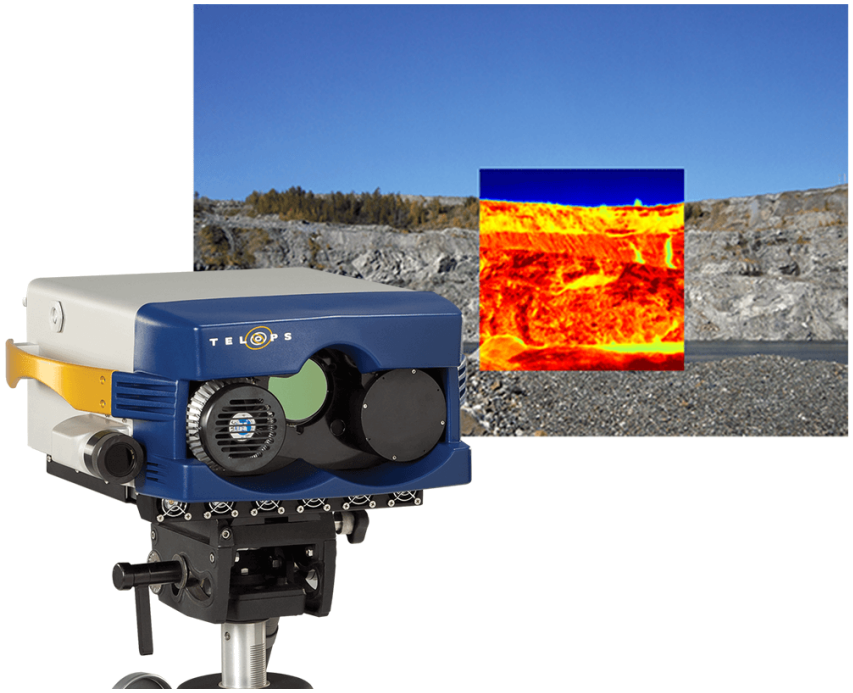
深圳中达瑞和的液晶可
调滤波器成像光谱仪

无锡谱视界线性渐变滤
光片线扫描成像光谱仪

3.成像光谱仪微型化技术-扫描式

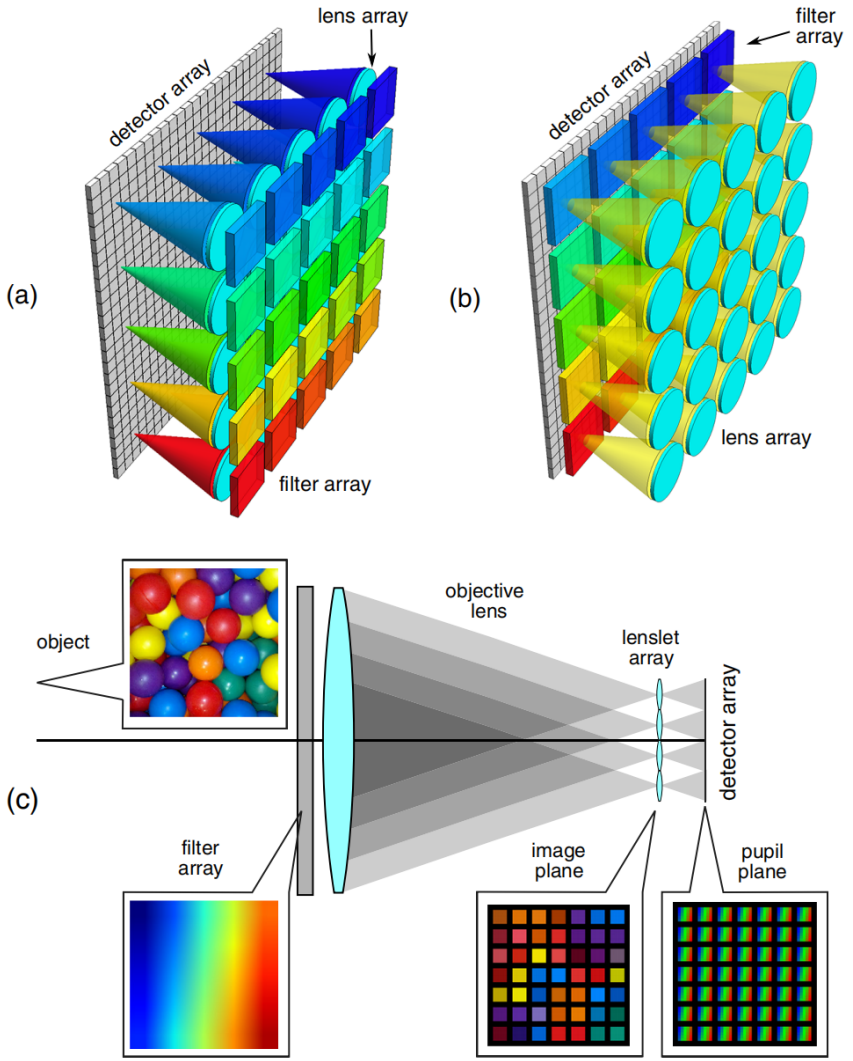


高光谱成像傅里叶变换成像光谱仪



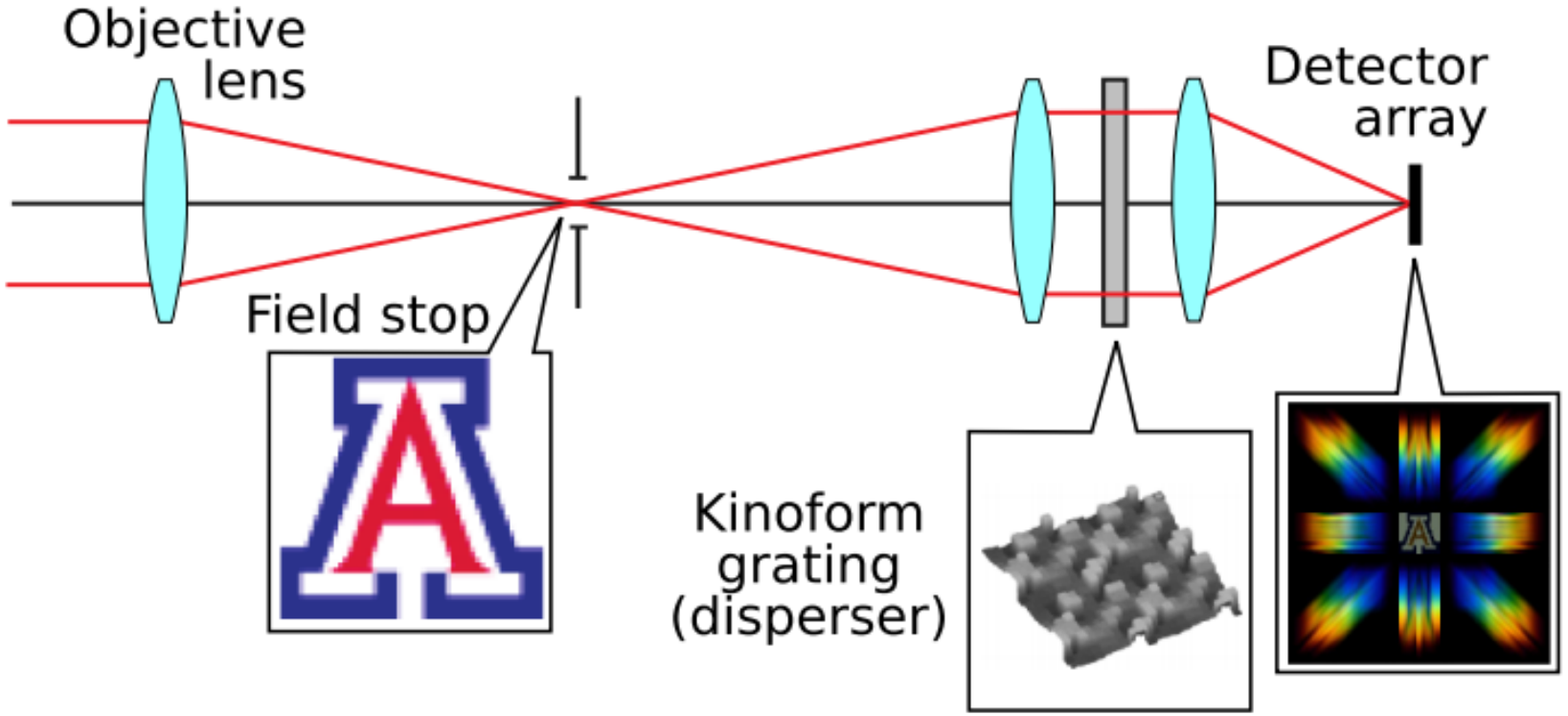
加拿大Telops高光谱成像傅里叶变换成像光谱仪

3.成像光谱仪微型化技术-快照式



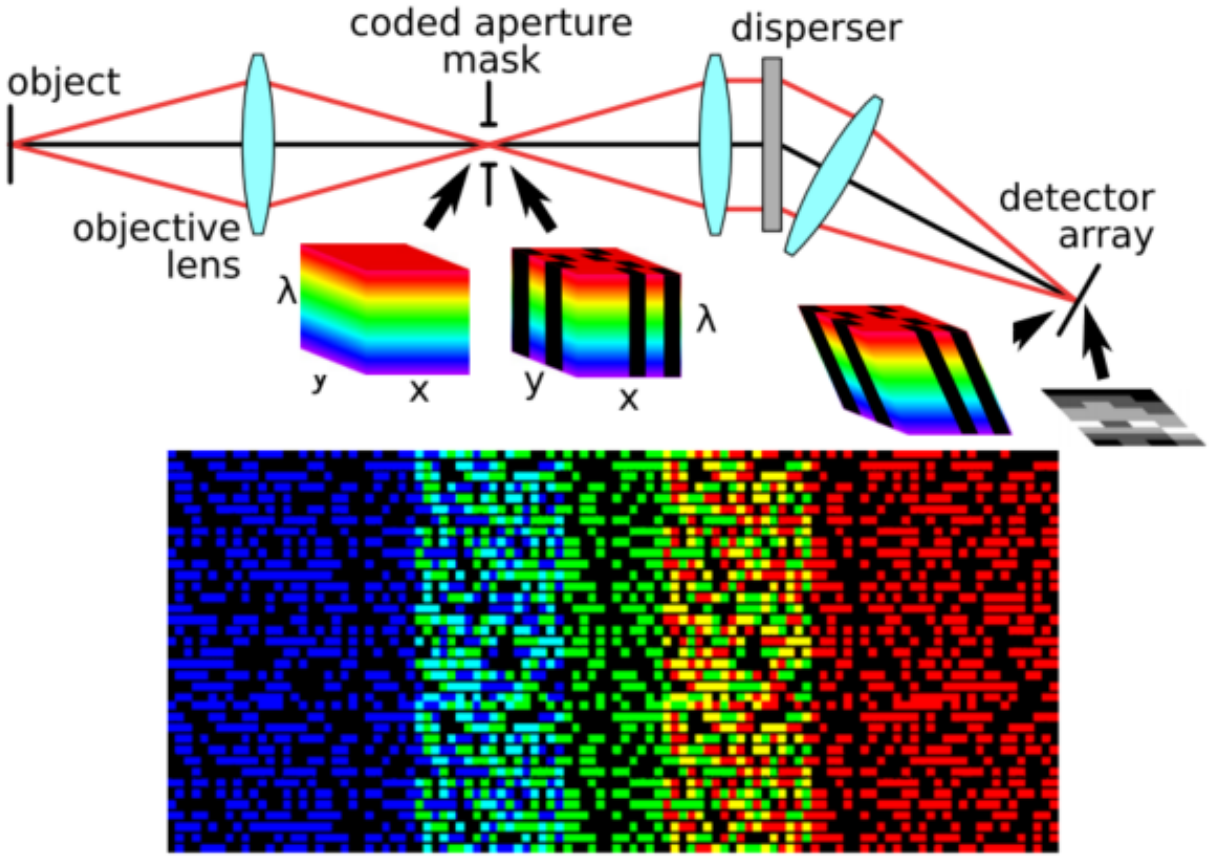
多孔径滤光片成像光谱仪

3.成像光谱仪微型化技术-快照式



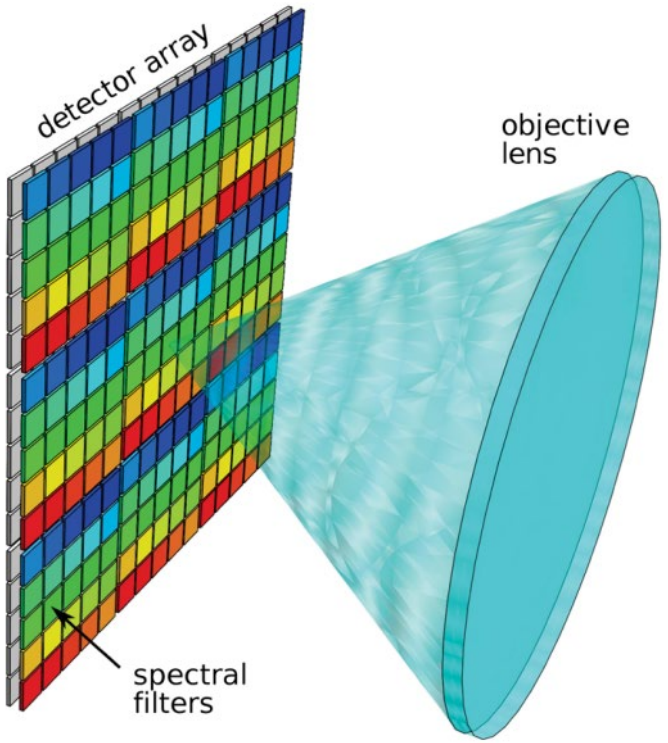
计算层析成像光谱仪

3.成像光谱仪微型化技术-快照式



编码孔径计算成像光谱仪

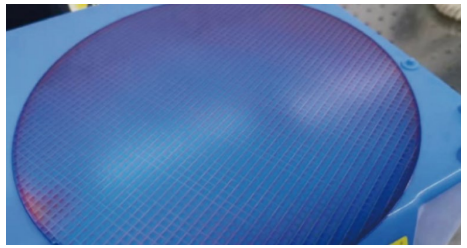
3.成像光谱仪微型化技术-快照式



马赛克滤光片阵列成像光谱仪

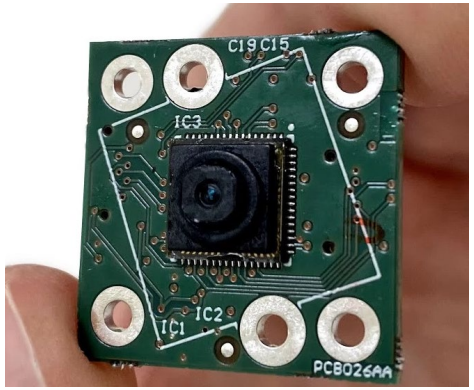
Spectral DEVICES INC.

加拿大Spectral Devices公司多光谱滤光片成像光谱仪



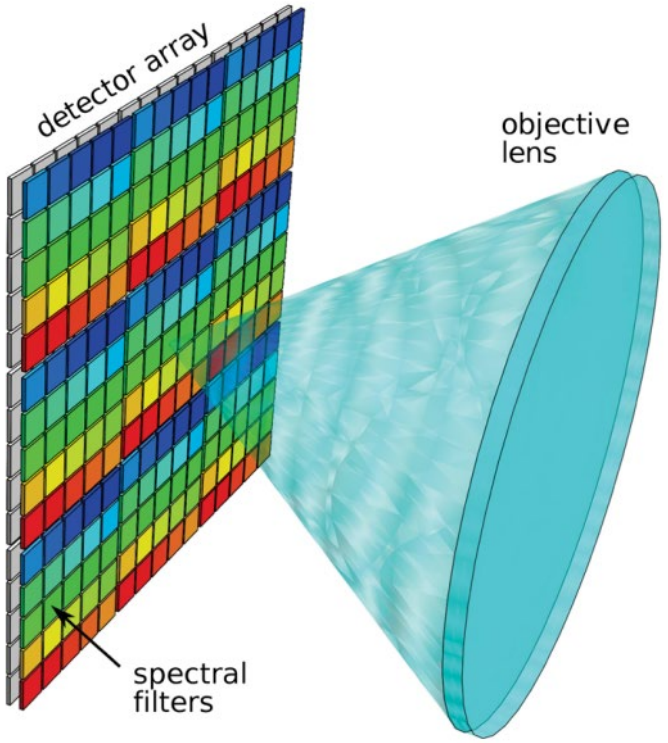
天津津航，苏州多感科技等

长春求是光谱的聚合物滤光片阵列光谱仪

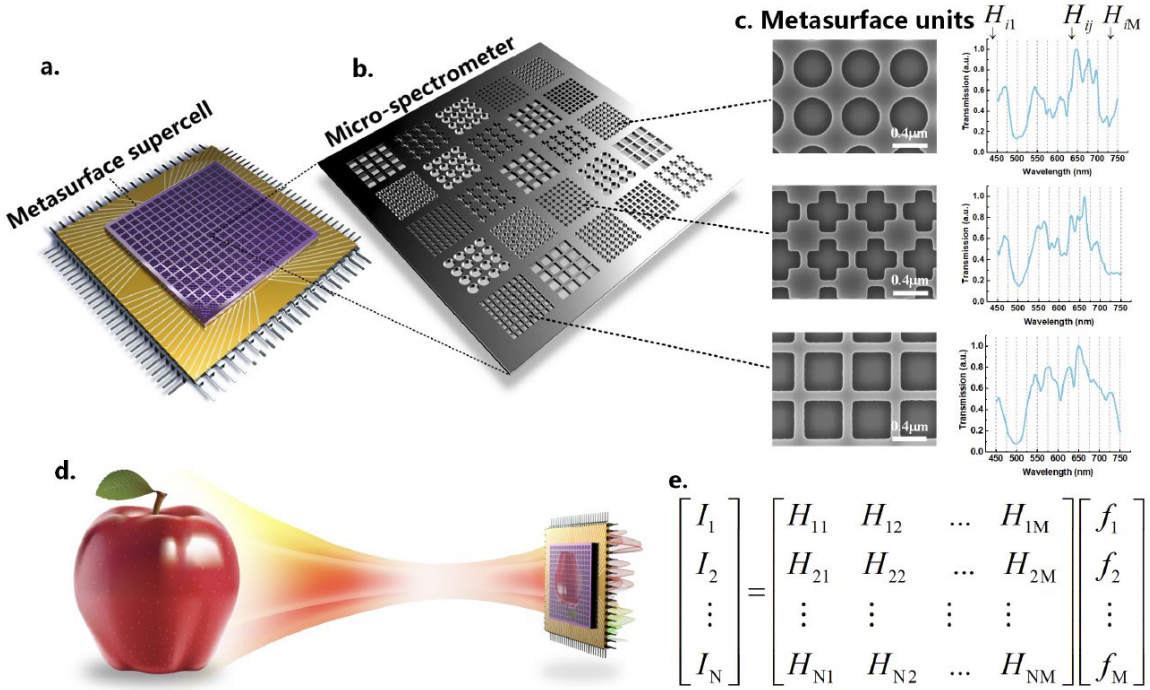


Spectricity基于F-P腔滤光片阵列成像光谱仪

3.成像光谱仪微型化技术-快照式



马赛克滤光片阵列成像光谱仪



与光科技的马赛克滤光计算成像光谱仪

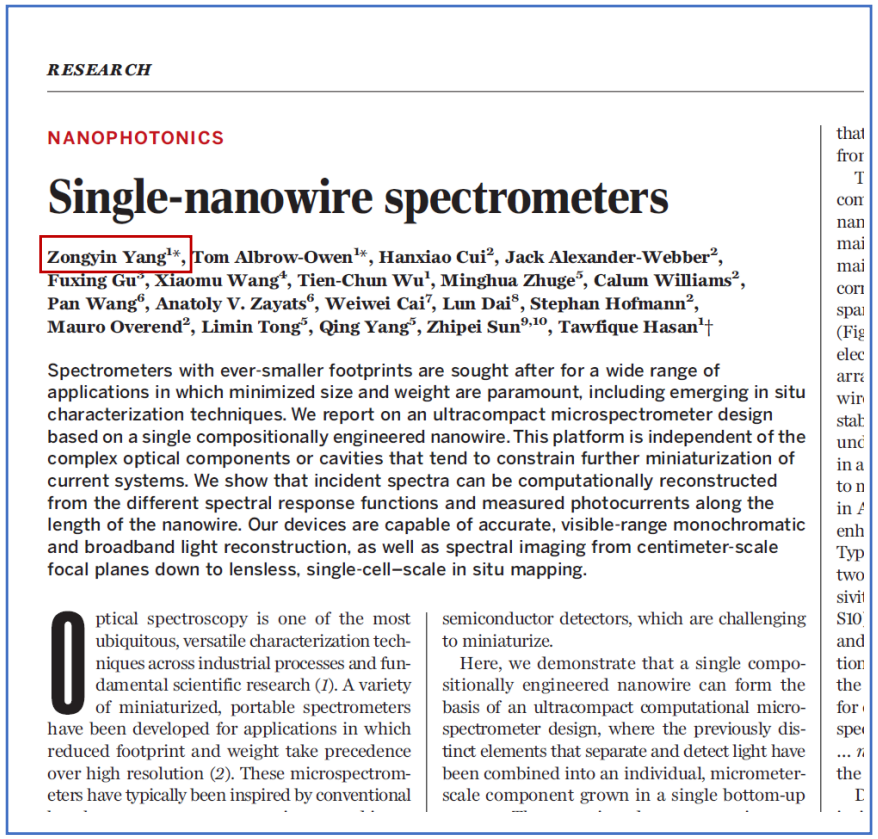
Outline

- ❑ 光谱检测基础
- ❑ 光谱仪微型化技术
- ❑ 成像光谱仪微型化技术
- ❑ **浙大光电子研究组研究工作**

浙大光电子研究组代表性论文

光谱**材料**	首次提出移动衬底法生长渐变半导体材料	Z. Yang *et al.*, ***Nano Lett.*** 11, 5085 (2011)	浙大硕士期间
	首次提出移动源法生长渐变半导体材料	F. Gu*, Z. Yang* （并一） *et al.*, ***J. Am. Chem. Soc.*** 133, 2037 (2011)	
光谱**发光**	最宽波长可调纳米激光器	Z. Yang *et al.*, ***Nano Lett.*** 14, 3153 (2014)	
	波长连续可调纳米激光器	M. Zhuge, Z. Yang *et al.*, ***ACS Nano*** 13, 9965 (2019)	剑桥博士期间
微型**光谱仪**	**世界最小**光谱仪	Z. Yang *et al.*, ***Science*** 365, 1017 (2019)	
	建立光谱仪微型化的**理论和技术体系**	Z. Yang *et al.*, ***Science*** 371, eabe0722 (2021)	浙大独立PI成果
	时间编码范德华异质结**光谱仪**	***Science*** 378, 296-299 (2022)	
	多维信息感知评论论文	Z. Y. Yang* *et al.*, ***Nature Photonics*** (2024)	
	拓扑编码解决高**分辨率与高灵敏度无法兼具的难题**	… Z. Y. Yang* *et al.*, ***Science Advances*** (2024)	
	多体问题解决高**分辨率和光谱范围无法兼具的难题**	…Z. Y. Yang* ***Light science & applications*** (2025)	
	飞秒激光加工宽光谱色散铌酸锂光栅结构	…Z. Y. Yang* *et al.*, ***Nature Commun.*** (revision)	
	基于应力调控的宽光谱范围光谱仪	… Z. Y. Yang* *et al.*, ***Science Advances*** (2025)	

基于渐变半导体纳米材料的探测器编码型光谱仪



RESEARCH

NANOPHOTONICS

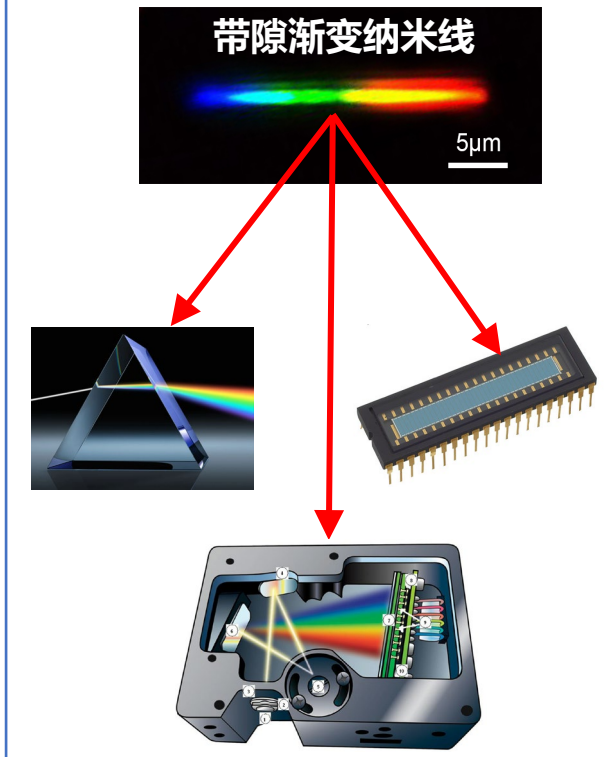
Single-nanowire spectrometers

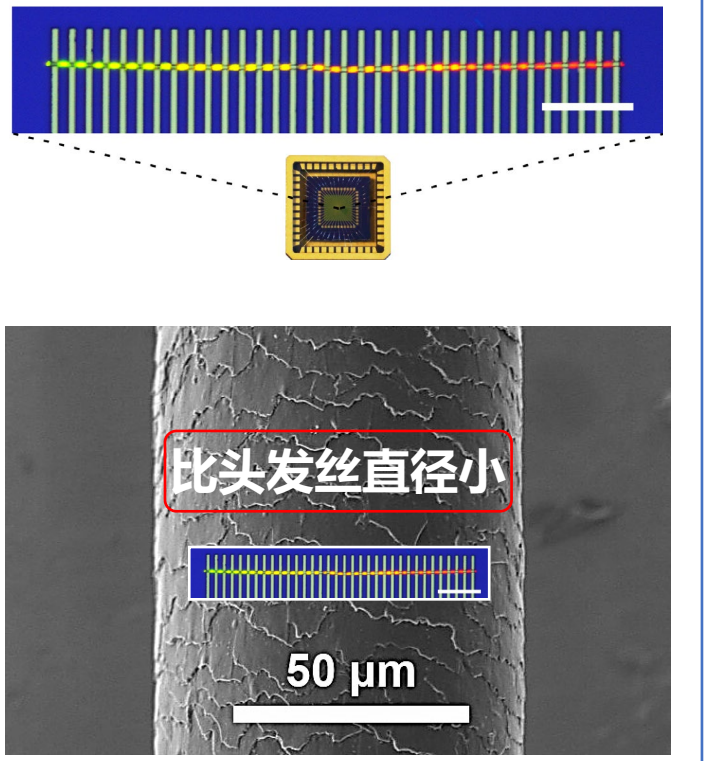
Zongyin Yang[1]*, Tom Albrow-Owen[1]*, Hanxiao Cui[2], Jack Alexander-Webber[2], Fuxing Gu[3], Xiaomu Wang[4], Tien-Chun Wu[1], Minghua Zhuge[5], Calum Williams[2], Pan Wang[6], Anatoly V. Zayats[6], Weiwei Cai[7], Lun Dai[8], Stephan Hofmann[2], Mauro Overend[2], Limin Tong[5], Qing Yang[5], Zhipei Sun[9,10], Tawfique Hasan[1]†

Spectrometers with ever-smaller footprints are sought after for a wide range of applications in which minimized size and weight are paramount, including emerging in situ characterization techniques. We report on an ultracompact microspectrometer design based on a single compositionally engineered nanowire. This platform is independent of the complex optical components or cavities that tend to constrain further miniaturization of current systems. We show that incident spectra can be computationally reconstructed from the different spectral response functions and measured photocurrents along the length of the nanowire. Our devices are capable of accurate, visible-range monochromatic and broadband light reconstruction, as well as spectral imaging from centimeter-scale focal planes down to lensless, single-cell–scale in situ mapping.

Optical spectroscopy is one of the most ubiquitous, versatile characterization techniques across industrial processes and fundamental scientific research (*1*). A variety of miniaturized, portable spectrometers have been developed for applications in which reduced footprint and weight take precedence over high resolution (*2*). These microspectrometers have typically been inspired by conventional semiconductor detectors, which are challenging to miniaturize.

Here, we demonstrate that a single compositionally engineered nanowire can form the basis of an ultracompact computational microspectrometer design, where the previously distinct elements that separate and detect light have been combined into an individual, micrometer-scale component grown in a single bottom-up



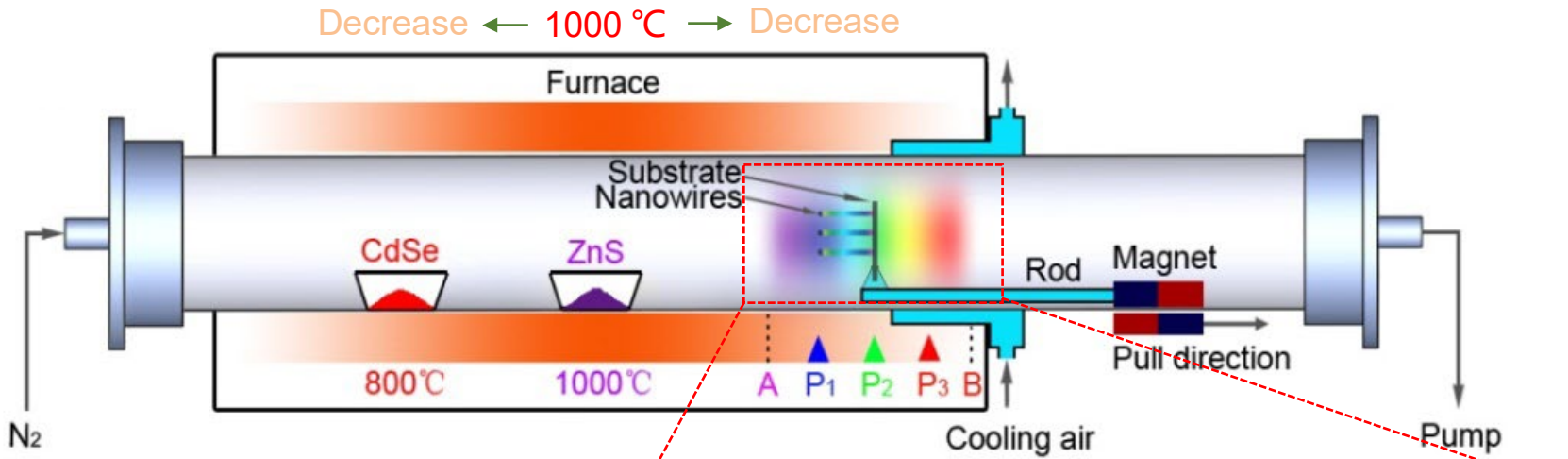


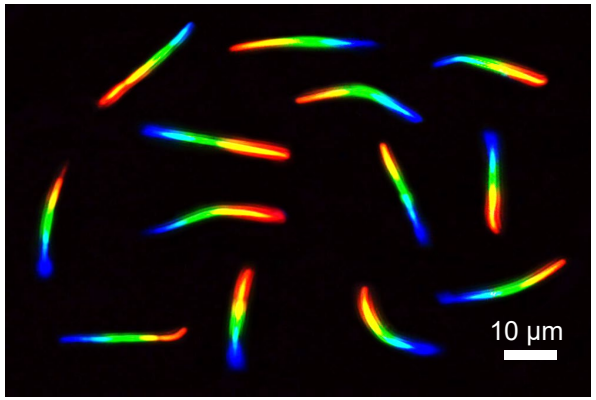
创新意义：首次提出探测器编码新机理，用一根纳米线替代了光栅、探测器和准直光路

世界最小光谱仪，有望用于解决卡脖子技术

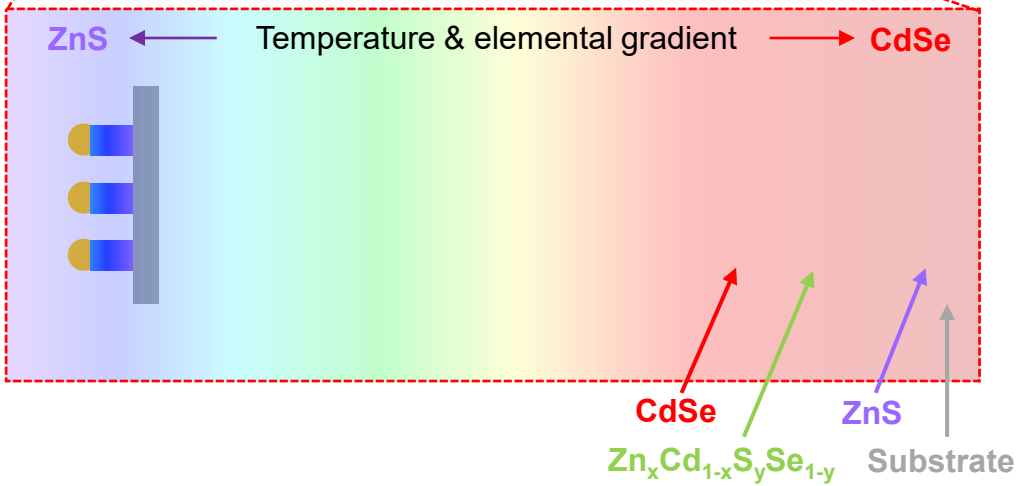
Z. Y. Yang *et al.*, ***Science*** 365, 1017 (2019)

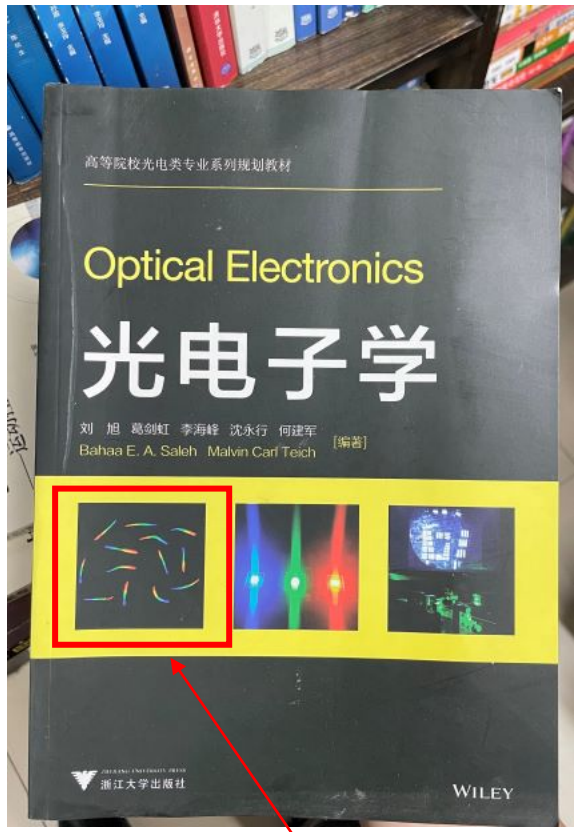
探测器编码材料合成：光谱能带渐变半导体





Real-colour micrograph of dispersed $Zn_xCd_{1-x}S_ySe_{1-y}$ nanowires under UV excitation





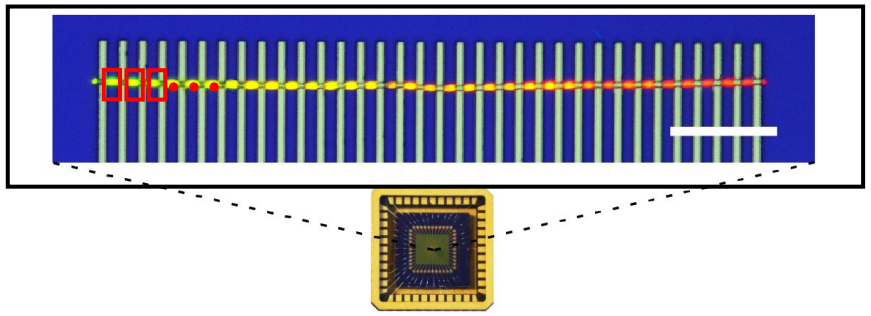
首次提出移动衬底法制备渐变材料，解决熔点不同材料合金的难题

Z. Y. Yang *et al.*, ***Nano Lett.*** 11, 5085 (2011) (10天接收)

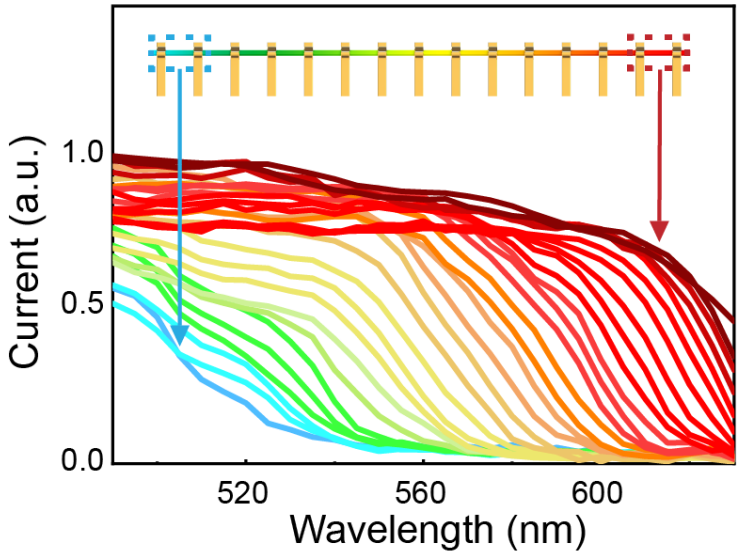
Z. Y. Yang *et al.*, ***Nano Lett.*** 14, 3153 (2014)

教材封面图片

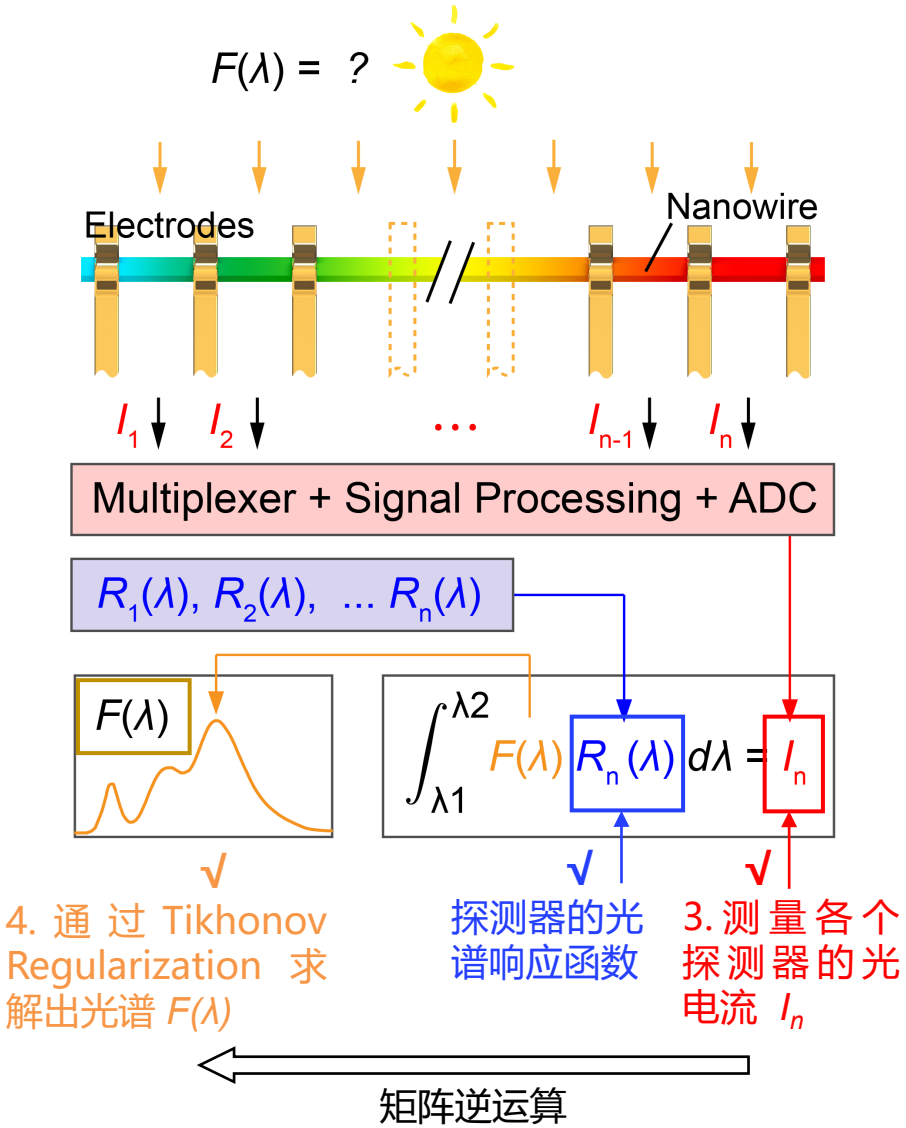
探测器编码纳米光谱仪工作原理



1.在纳米线上加工电极得到光探测器阵列

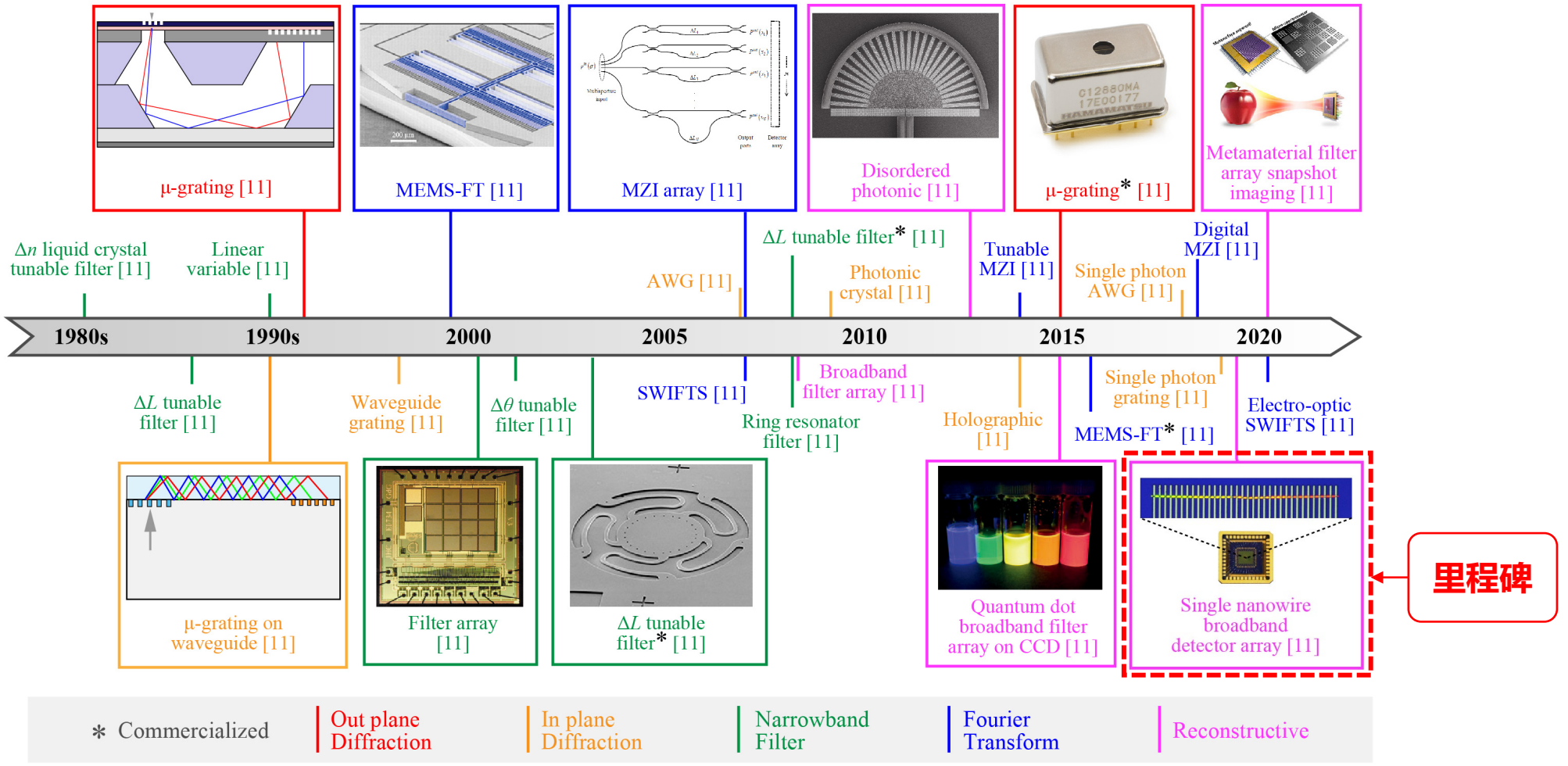


2.标定各个探测器的光谱响应函数 $R_n(\lambda)$



Z. Y. Yang *et al.*, ***Science*** 365, 1017-1020 (2019)

光谱仪发展里程碑



Science 371, eabe0722 (2021)

学术报道

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Miniaturizing spectrometers

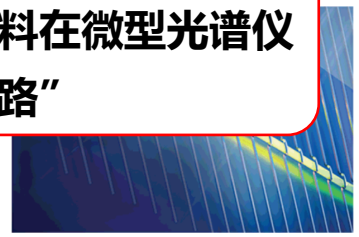
Ian S. Osborne

+ See all authors and affiliations

Science 06 Sep 2019:
Vol. 365, Issue 6457, pp. 995
DOI: 10.1126/science.365.6457.995-a

PDF

the scale to which these spectrometers can be shrunk. To address the desire for miniaturized spectrometers with a micrometer-scale (and smaller) footprint, Yang *et al.* developed such a microspectrometer based on single, compositionally engineered nanowire. This result is a practical step forward for the use of other light-sensitive nanomaterials for such ultra-miniaturized spectroscopy platforms.



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Artistic interpretation of a nanowire spectrometer

CREDIT: ELLA MARU STUDIO

Science 编辑 Ian 撰文评价：
“该研究为纳米材料在微型光谱仪中的应用铺平了道路”

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Derek Lowe's commentary on drug discovery and the pharma industry. An editorially independent blog from the publishers of *Science Translational Medicine*. All content is Derek's own, and he does not in any way speak for his employer.

By Derek Lowe

ANALYTICAL CHEMISTRY

Nanowire Spectroscopy

By Derek Lowe | 24 September, 2019

英国皇家化学会专栏作家 Lowe教授，在Science 子刊撰文高度评价：
“该工作是全新的技术路线，极大地减小光谱仪的尺寸和成本”

Katsushi Ikeuchi
Editor

Computer Vision
A Reference Guide

国家出版基金项目
NATIONAL PUBLICATION FOUNDATION

中外物理学精品书系

前沿系列 · 61

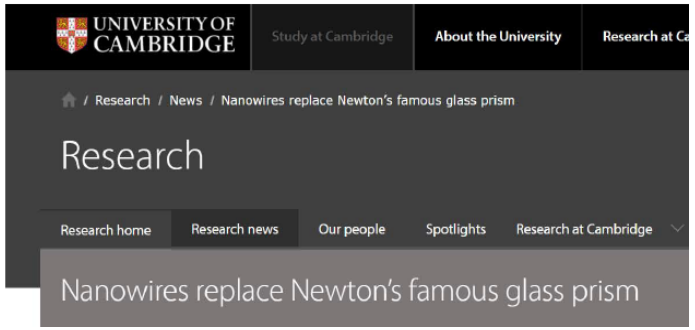
拉曼光谱仪的科技基础及其构建和应用

张树霖 著

Science审稿人评价： “该研究是集世界最先进的材料合成工艺、最高超的实验技巧、以及巧妙的算法得到的杰作（a true tour de force）”

被写进多本著作中

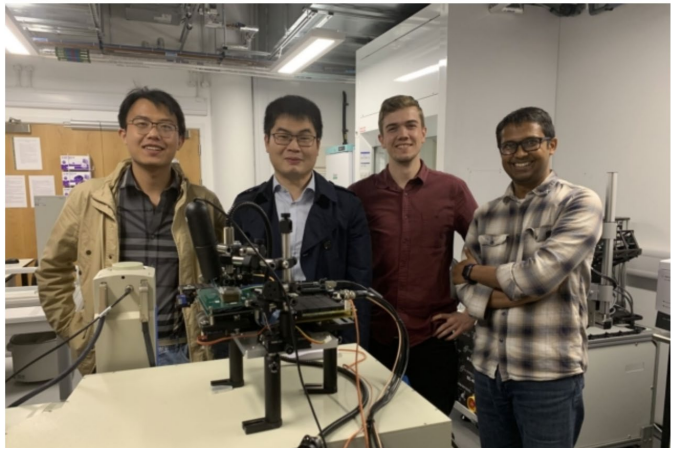
公众报道



剑桥大学以“纳米技术颠覆牛顿棱镜”为题报道了我们的工作

Single-nanowires make powerful spectrometers

24 Sep 2019 Gergő Péter Szekeres



今日Science: 世界最小光谱仪问世，纳米技术颠覆牛顿的棱镜

From:知社 知社学术圈

8年历经150次失败，浙大青年科学家用“彩虹”造出全球最小光谱仪，将成本降低数千倍 | 专访

原创 多加 DeepTech深科技 1月29日

就在刚刚，曾研制出全球最小光谱仪的浙大信息与电子工程学院研究员杨宗银，发表了其作为第一作者的第二篇 *Science* 。

失败150次后，你还会坚持吗？浙大老师用2篇《科学》回答

原创 请收藏 浙江大学 2月1日

1月29日，由浙江大学信息与电子工程学院智能传感所的百人计划研究员杨宗银，作为第一作者撰写的综述在线发表在**国际顶级期刊《科学》（Science）上。文章首次系统性总结了光谱仪微型化的技术方案和发展历程，引起国际科学界高度关注。**

剑桥大学，浙江大学公众号，DeepTech，Science Daily 等国内外上百家媒体报道，获得良好的公众影响力

技术延伸（更小）：范德华异质结微型光谱仪

Science

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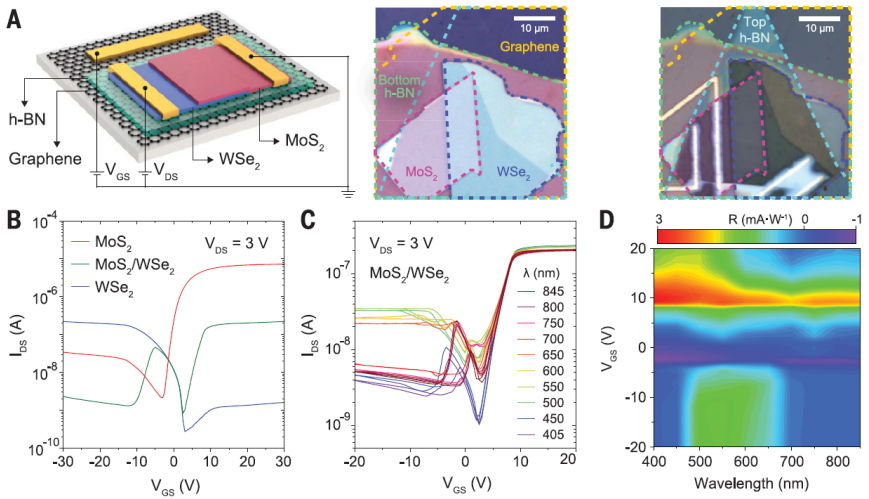
HOME > SCIENCE > VOL. 378, NO. 6617 > MINIATURIZED SPECTROMETERS WITH A TUNABLE VAN DER WAALS JUNCTION

REPORT | OPTICS

Miniaturized spectrometers with a tunable van der Waals junction

HOON HAHN YOON, HENRY A. FERNANDEZ, FEDOR NIGMATULIN, WEIWEI CAI, ZONGYIN YANG, HANXIAO CUI, FAISAL AHMED, XIAOQI CUI, MD GIUS UDDIN, [...] ZHIPEI SUN +6 authors Authors Info & Affiliations

SCIENCE · 20 Oct 2022 · Vol 378, Issue 6617 · pp. 296-299 · DOI: 10.1126/science.add8544



由几层二维材料堆叠而成，时间光谱编码，厚度和横向尺寸进一步缩小

Zhipei Sun *et al.*, ***Science*** 378, 296-299 (2022)

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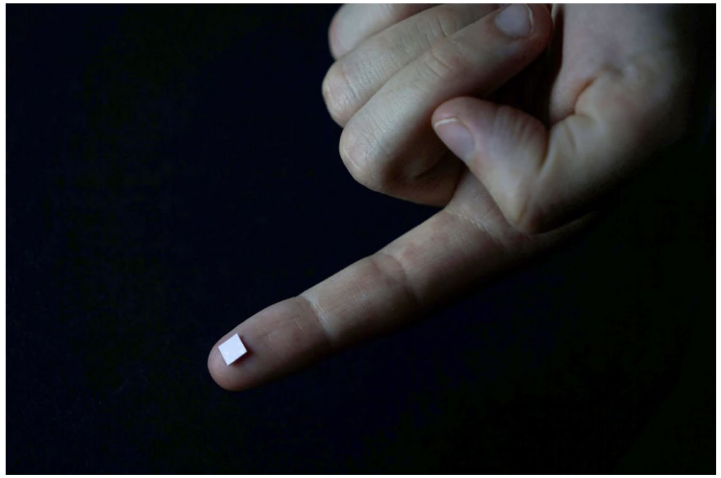
RESEARCH HIGHLIGHT | 24 October 2022

Spectrometer fits on a fingertip — but is big on precision

An instrument found on workbenches around the world has been scaled down enough to be used in a smartphone.

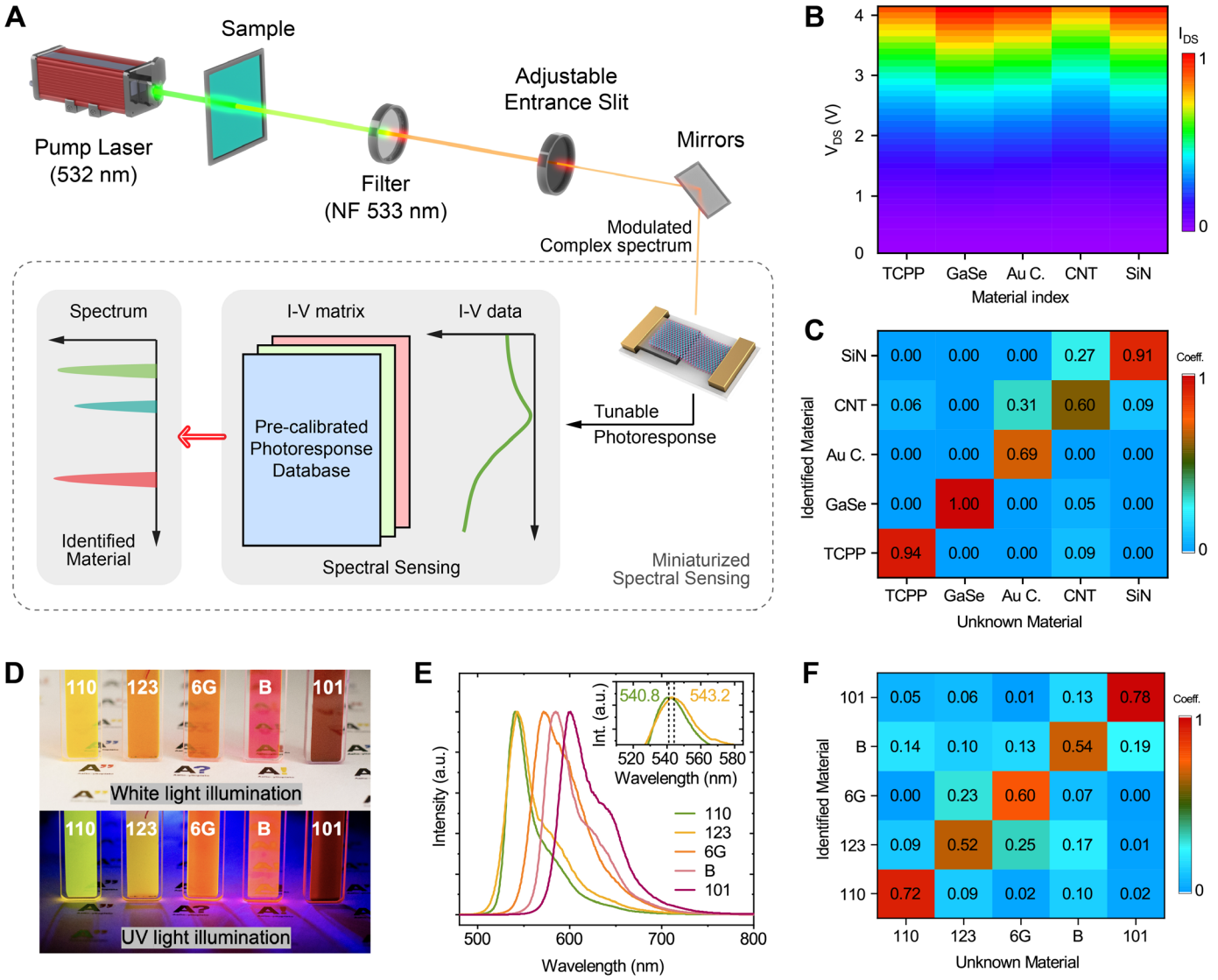


A spectrometer that uses an active area smaller than the diameter of a human hair fits on a fingertip.
Credit: Hoon Hahn Yoon, Suvi-Tuuli Akkanen

Nature 杂志highlights范德华异质结微型光谱仪的工作

https://doi.org/10.1038/d41586-022-03370-8

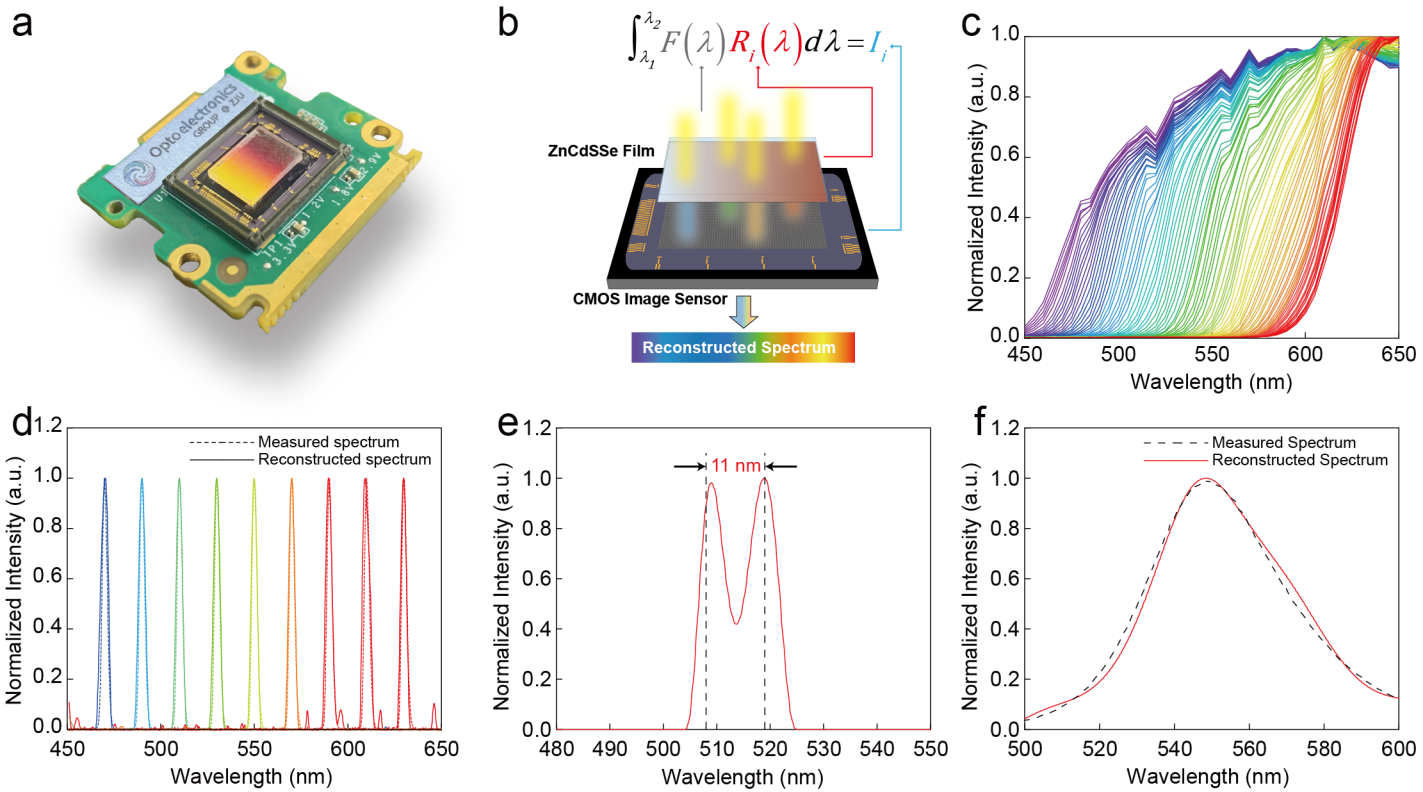
技术延伸（更小）：范德华异质结微型光谱仪物质识别

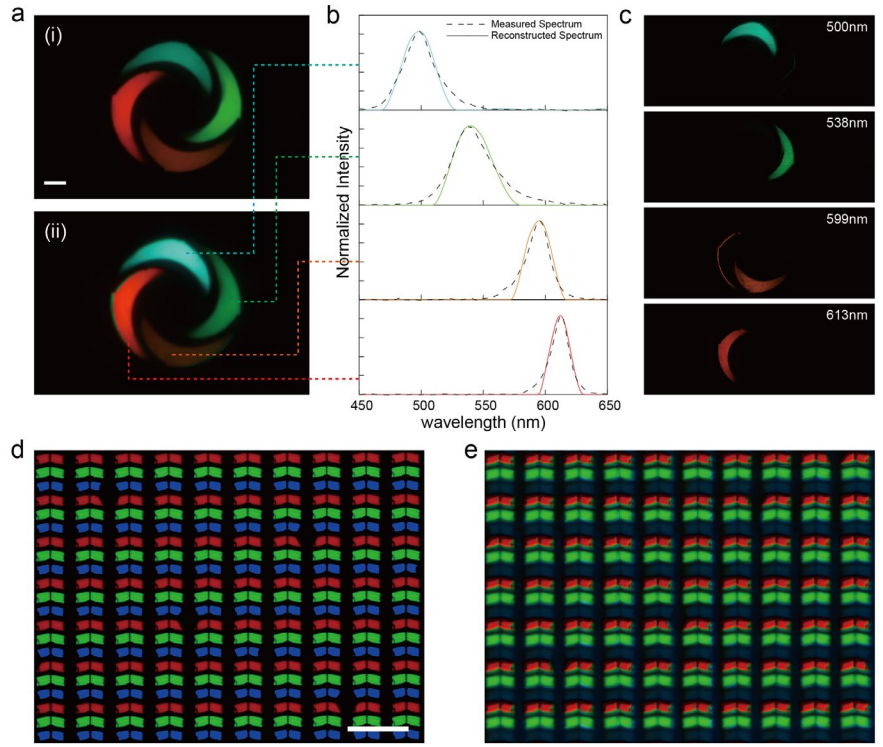


微型光谱仪用于物质识别

Z. P. Sun, **Z. Y. Yang** *et al.*, ***Science Advances*** (2024)

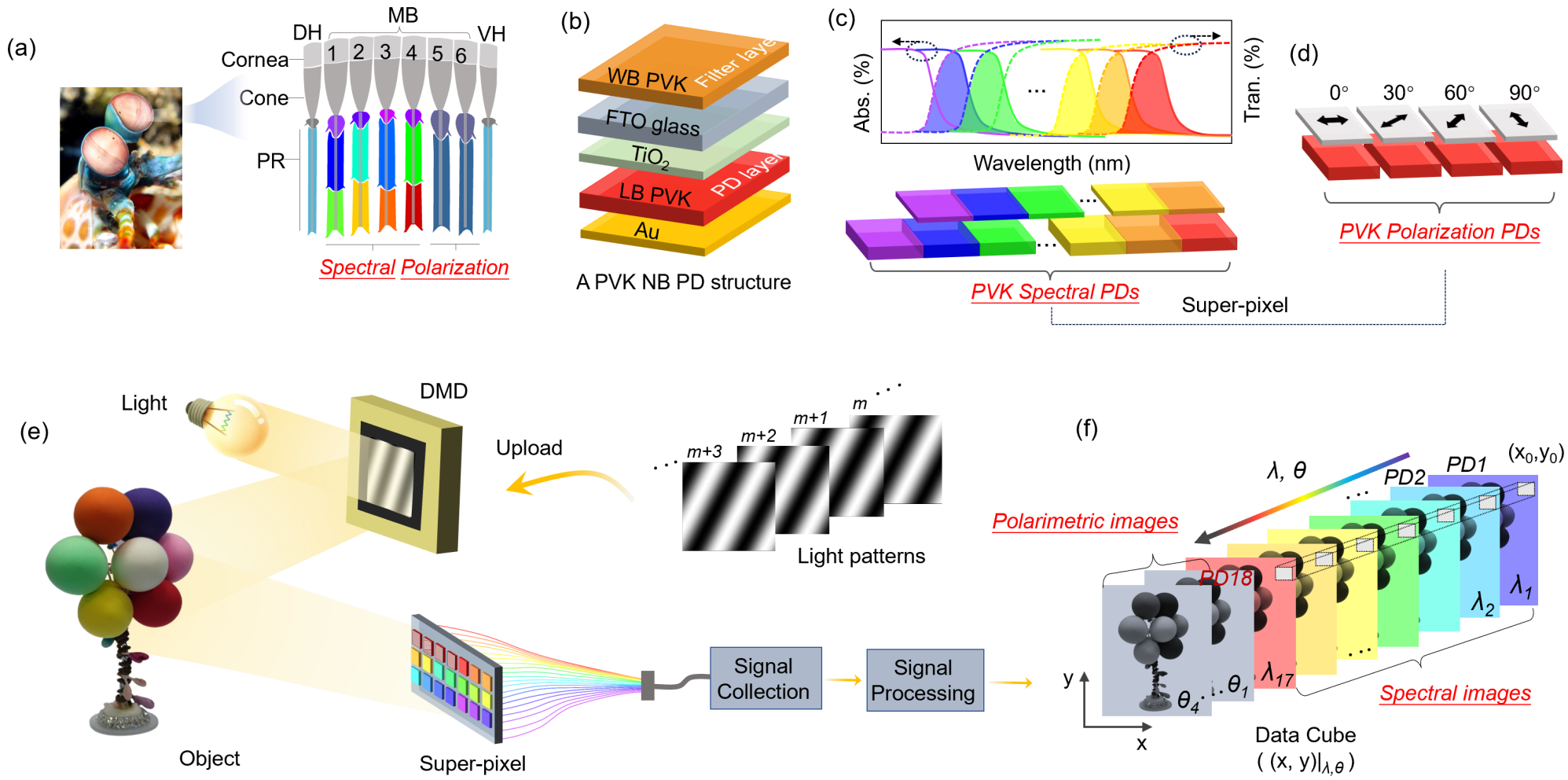
技术延伸（点→成像）：基于带隙渐变薄膜的光谱成像相机





Z. Y. Yang* *et al.*, ***Advanced Optical Materials*** (under review)

技术延伸（点→成像）：基于钙钛矿多光谱薄膜的光谱成像相机



Y.J. Liu, **Z. Y. Yang*** *et al.*, ***Science Advances*** (under review)

汇报提纲

高性能
微型光谱仪

1 研究背景

2 小型化理论与研究

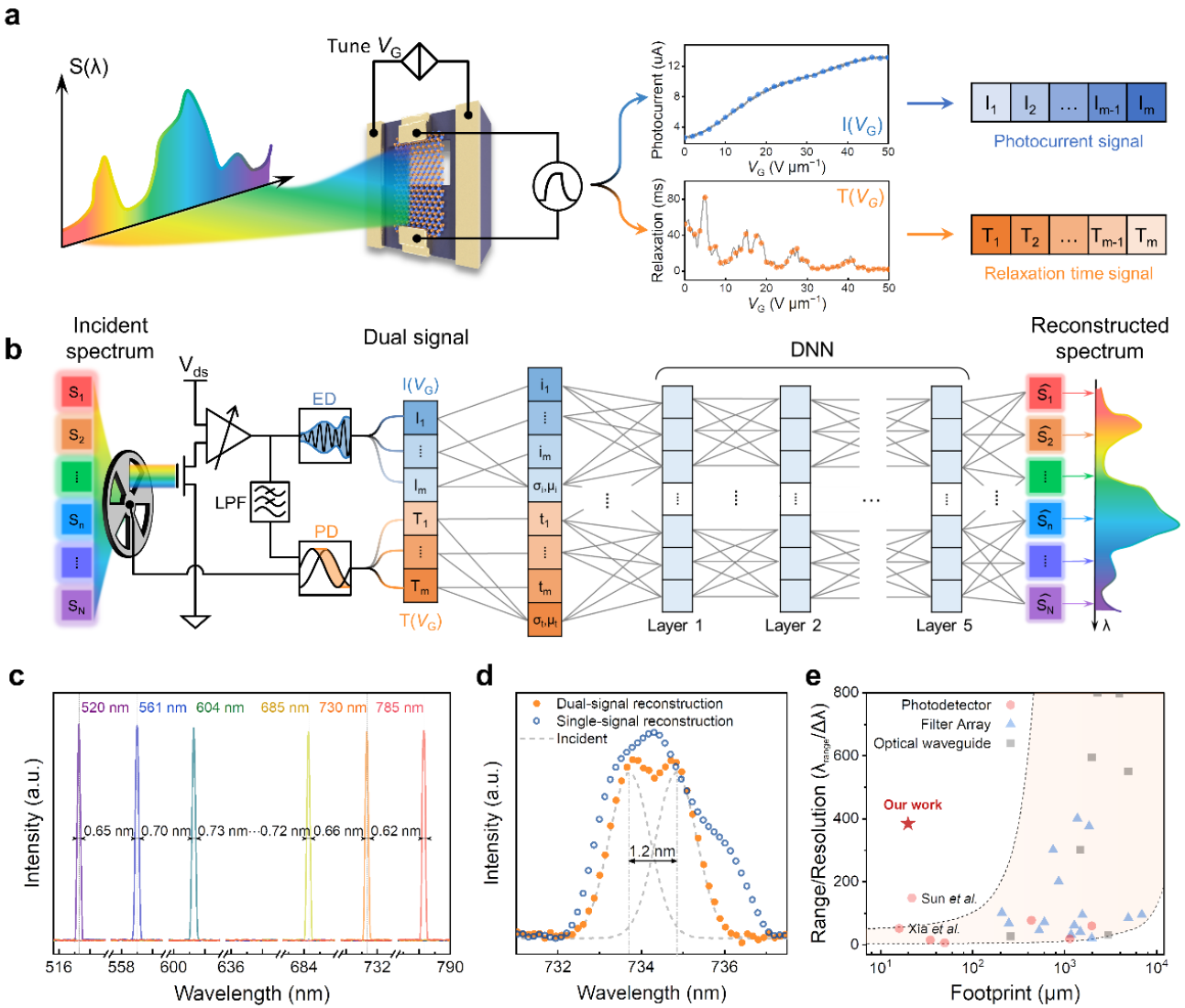
3 提高分辨率

4 拓宽光谱范围

5 提高灵敏度

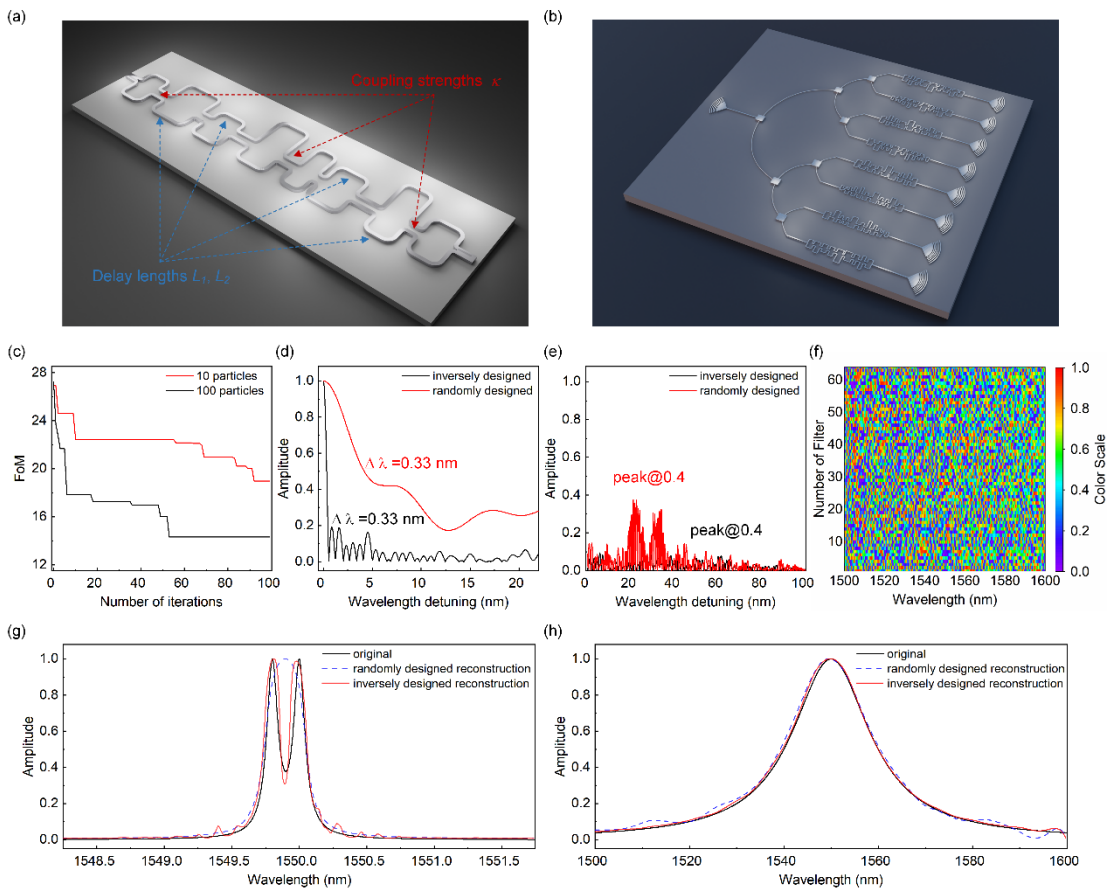
6 技术应用

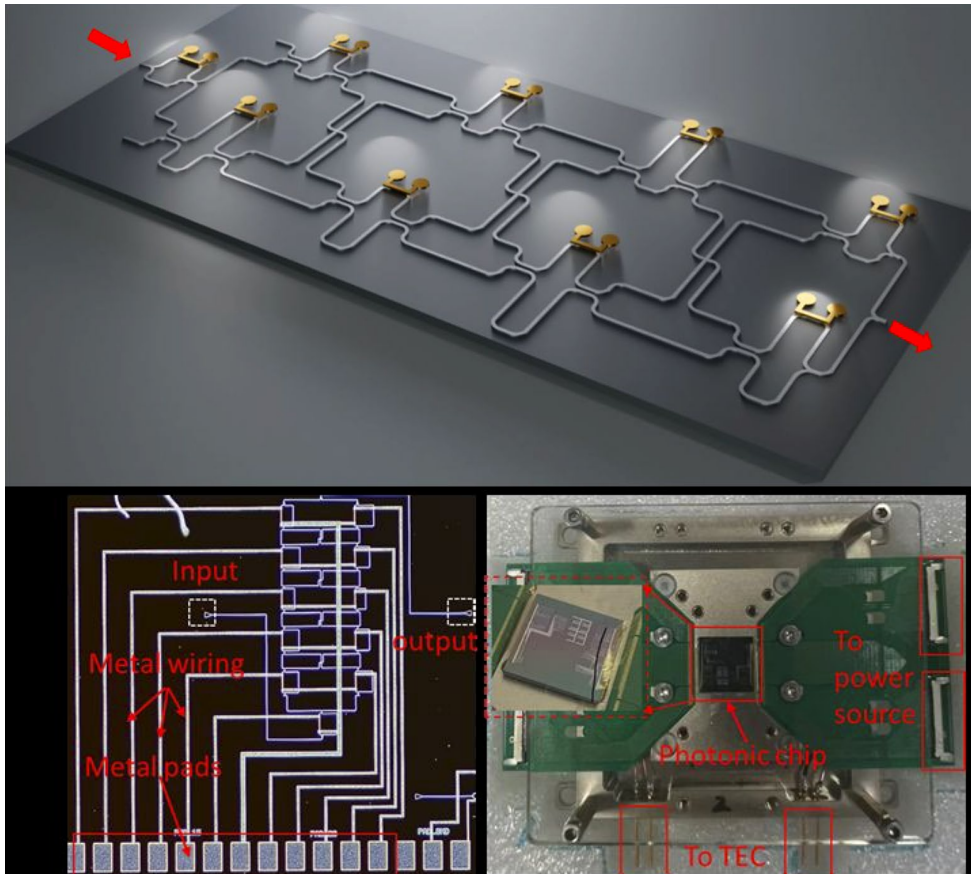
基于双重编码方案，实现更高光谱分辨率



Nature Electronics (2024)

基于逆向设计硅光芯片，解决光谱仪高分辨率与低功耗兼具的难题





A. Li, **Z. Y. Yang*** *et al.*, ***Opto-Electronic Advances*** 7, 240099 (2024)

A. Li, **Z. Y. Yang*** *et al.*, ***Engineering*** 43, 81 (2024)

汇报提纲

高性能
微型光谱仪

1 研究背景

2 小型化理论与研究

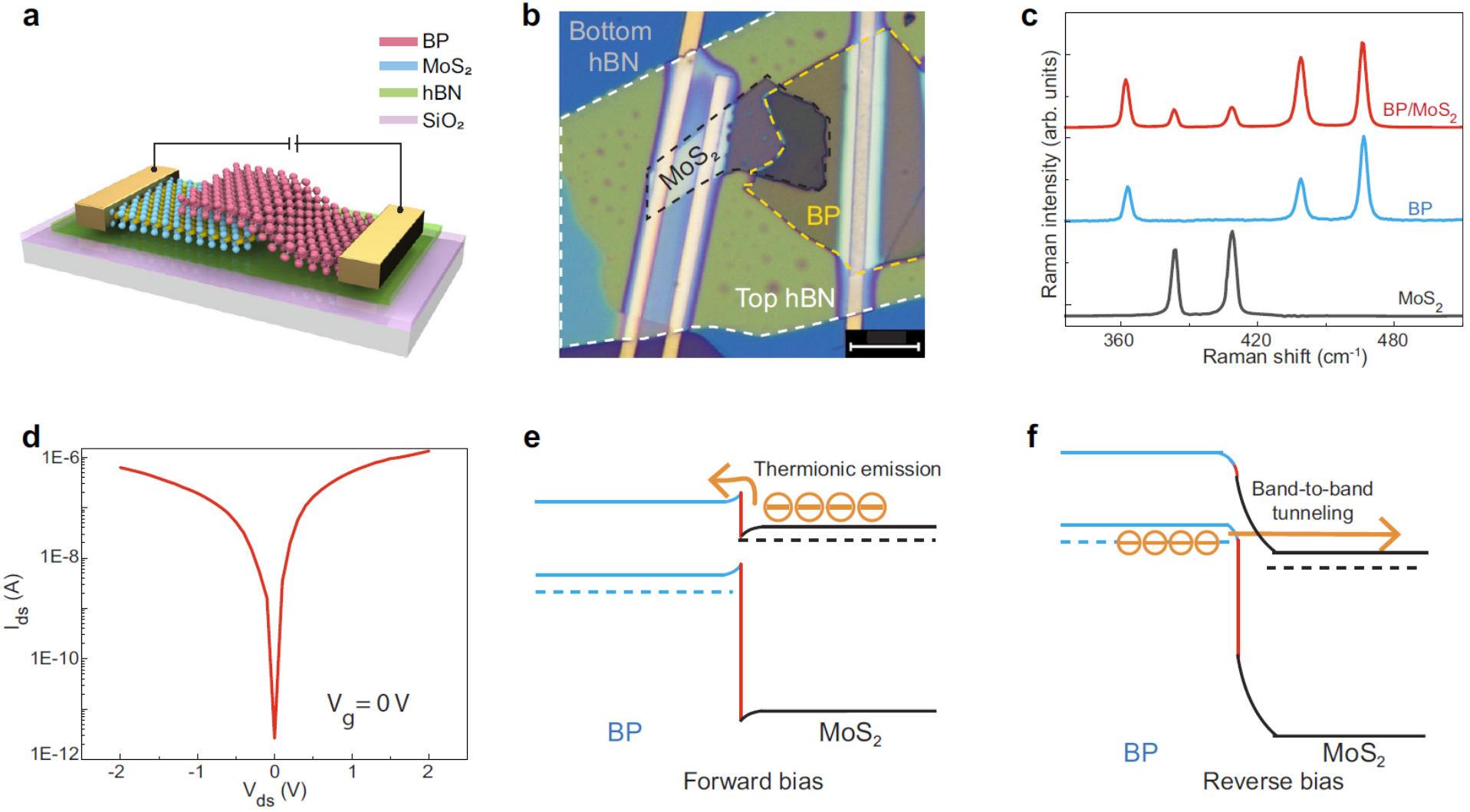
3 提高分辨率

4 **拓宽光谱范围**

5 提高灵敏度

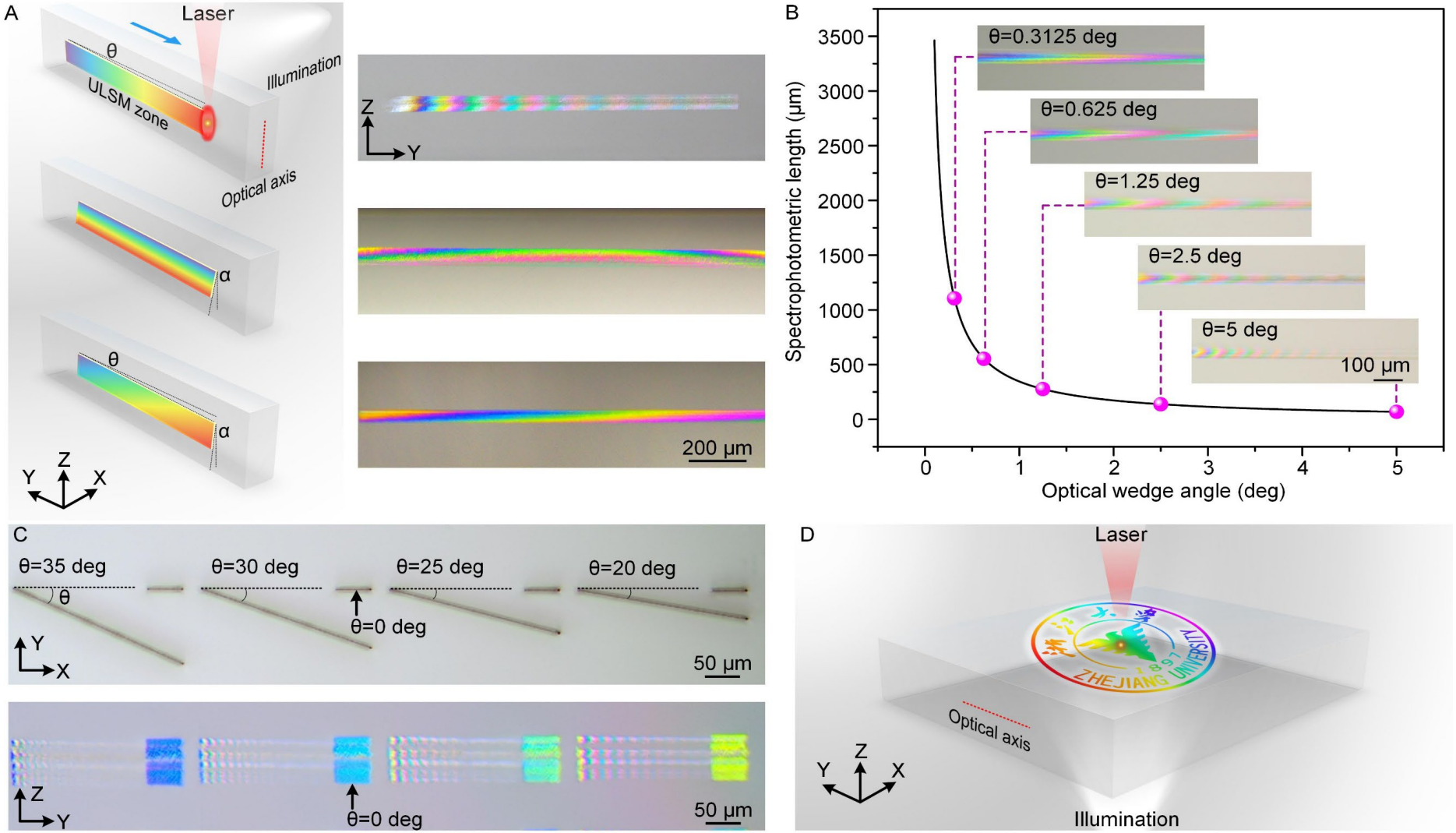
6 技术应用

更宽光谱范围：黑磷异质结光谱仪，覆盖可见到1600nm



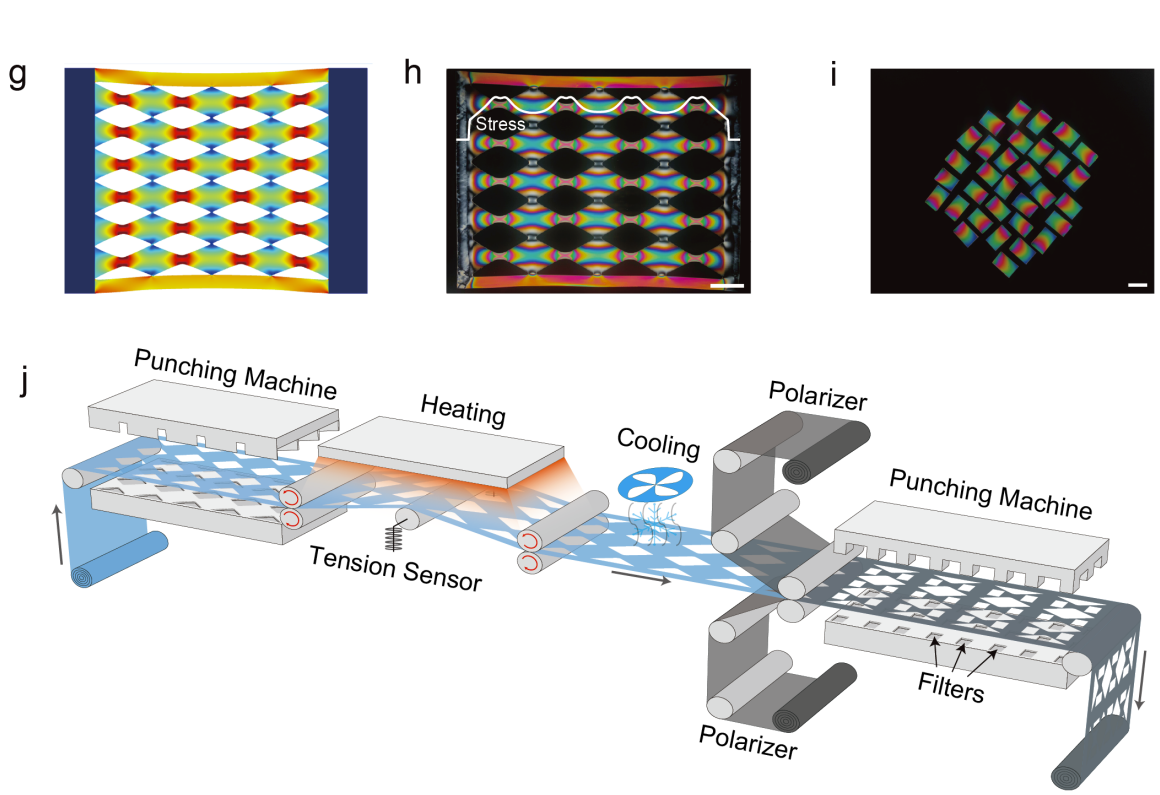
Zhipei Sun *et al.*, ***Nature Communications*** 15, 571 (2024)

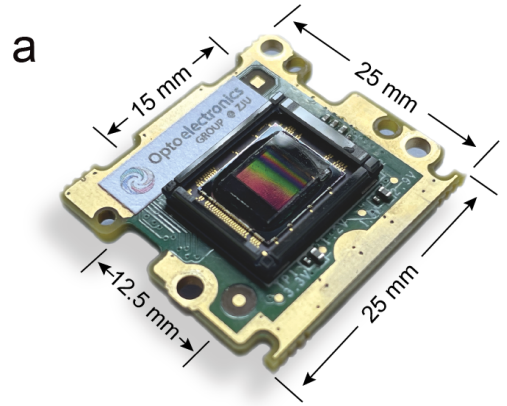
更宽光谱范围：飞秒激光诱导加工光栅，光谱分光范围覆盖可见到1600nm

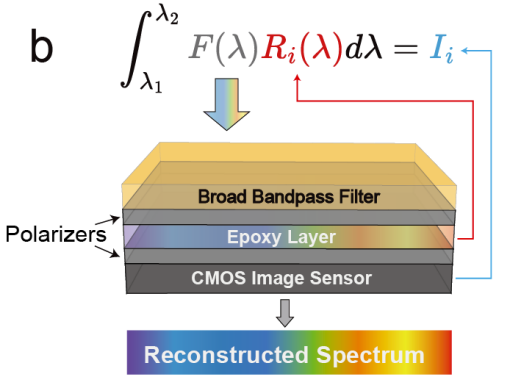


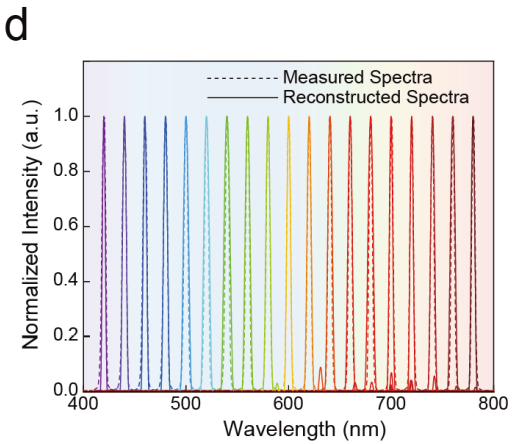
Z. Y. Yang* *et al.*, ***Nature Communications*** (under review)

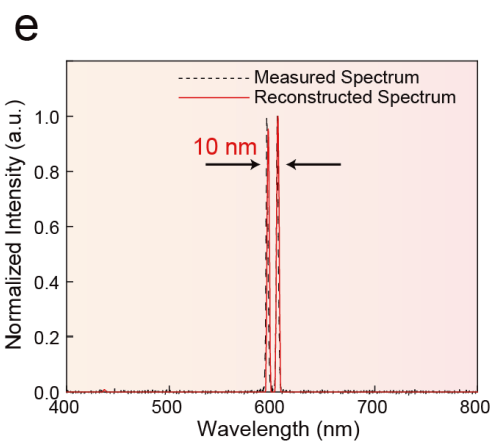
更宽光谱范围：基于应力梯度聚合物，光谱范围覆盖可见到1600nm





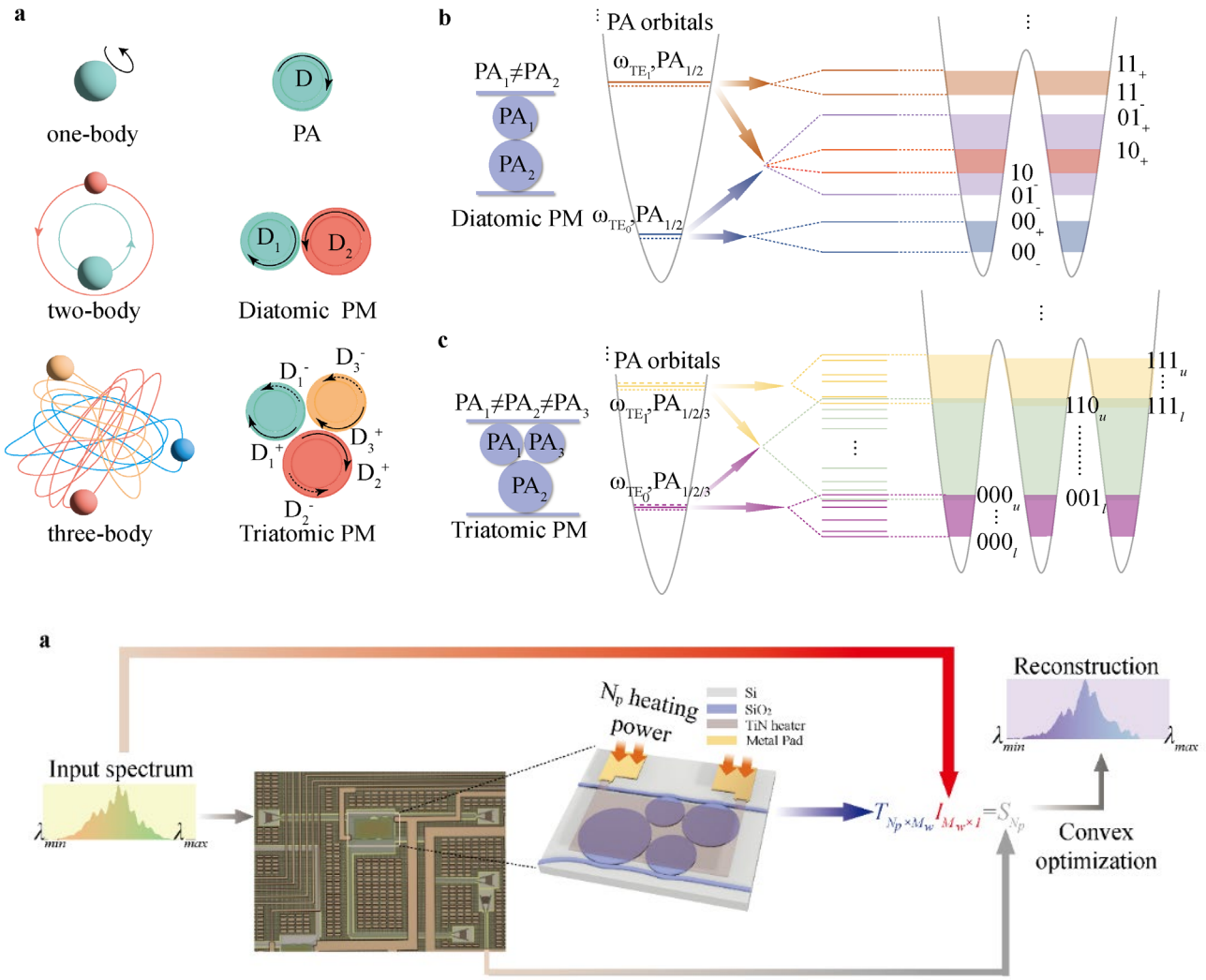






G. Y. Zhang, **Z. Y. Yang*** *et al.*, ***Science Advances*** (minor revision)

采用多体光学分子原理，解决高分辨率和宽光谱范围无法兼具的难题



X. H. Guo*, **Z. Y. Yang*** *et al.*, ***Light Science & Applications*** (2024)

汇报提纲

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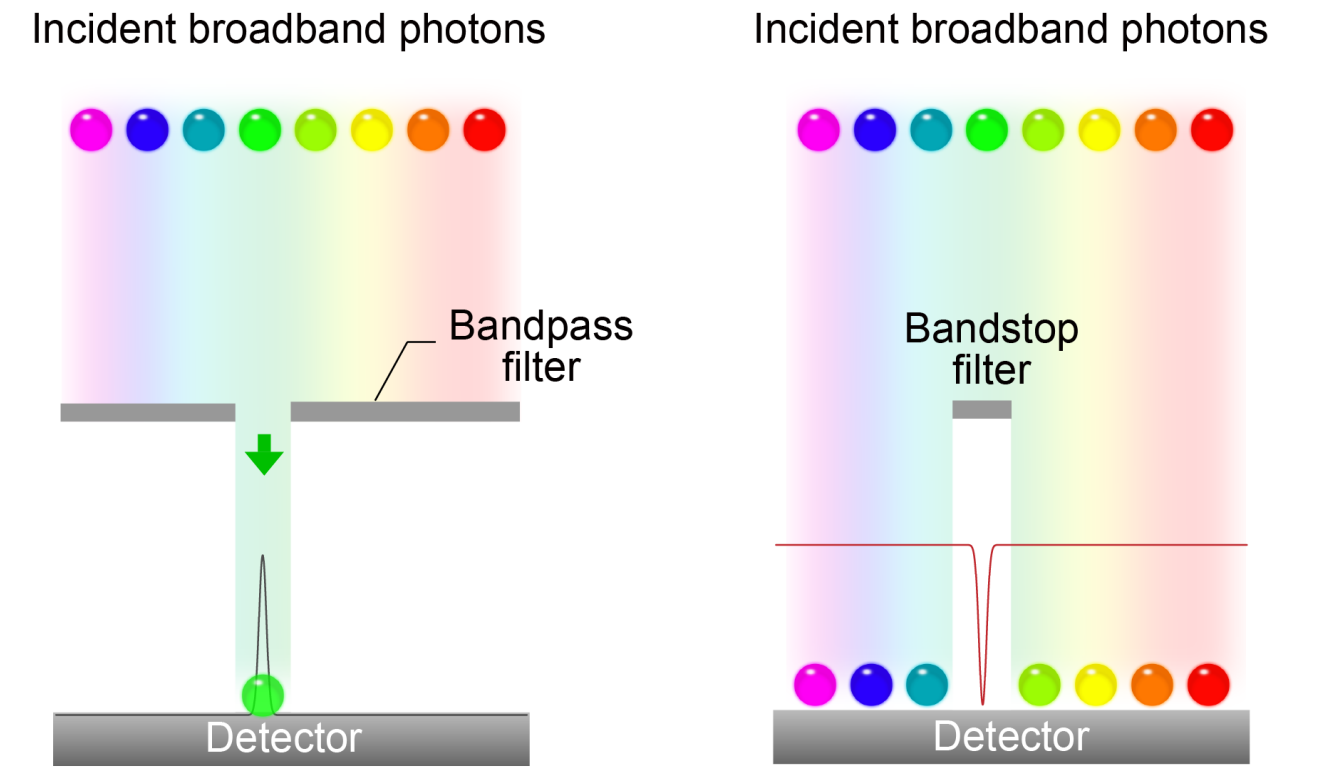
5 **提高灵敏度**

6 技术应用

基于陷波编码的高灵敏度光谱仪原理

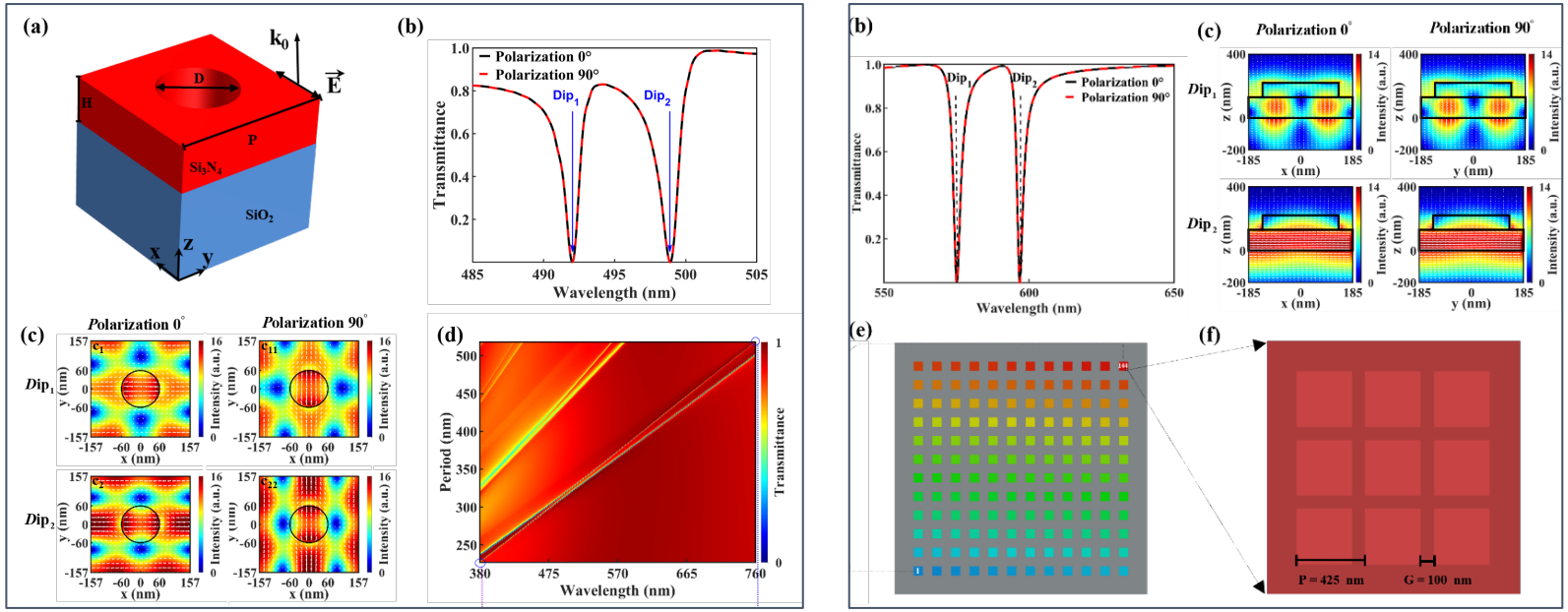


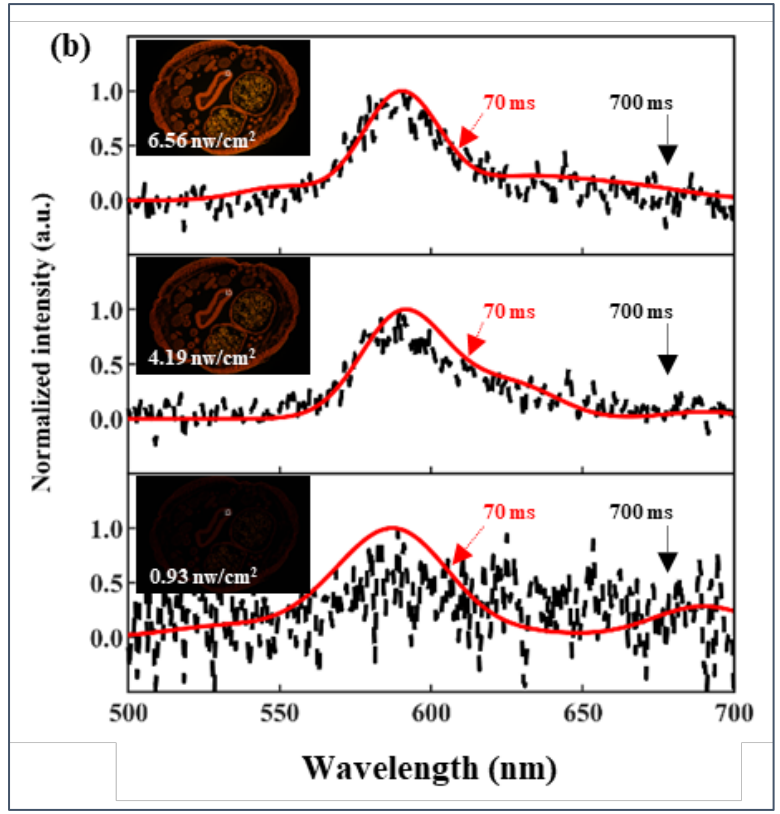
1 detected photon **<** **(n-1)** detected photons

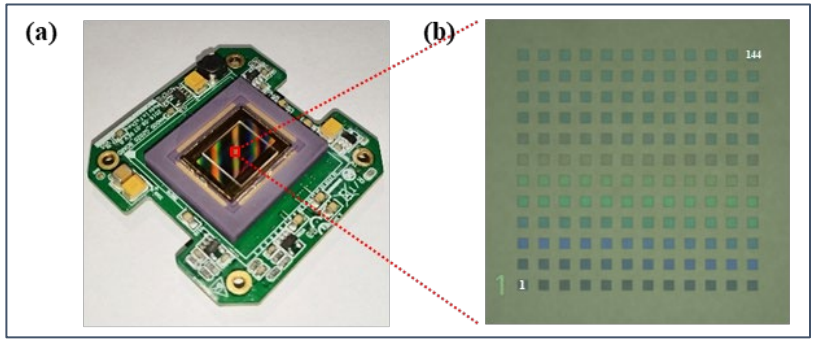
1 portion of spectral information **=** **1** portion of spectral information

F. Tang, **Z. Y. Yang*** *et al.*, ***Science Advances*** 10, eadr7155 (2024)

基于光子晶体（陷波型）阵列的高灵敏度光谱仪



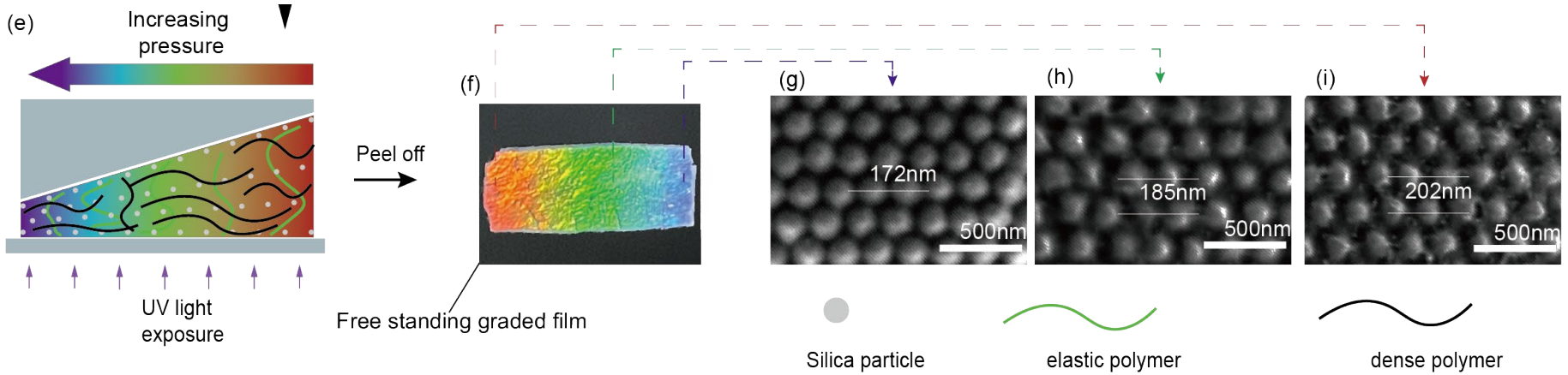


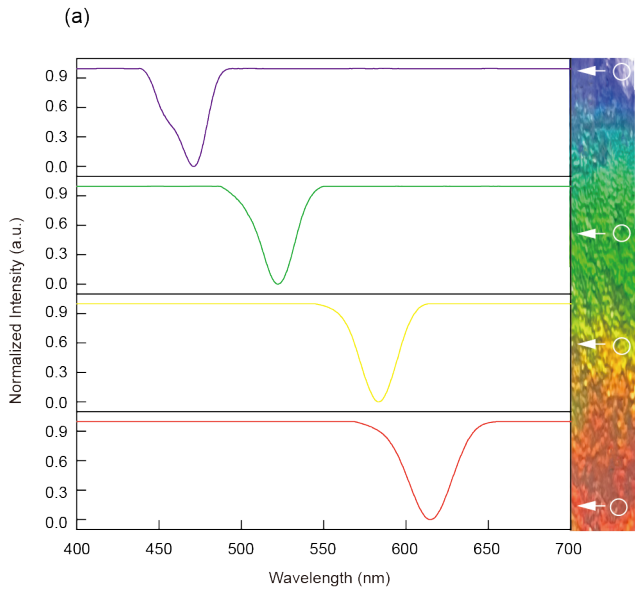


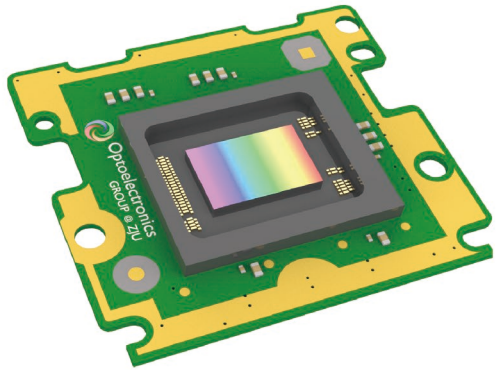
J. J. Wu, **Z. Y. Yang*** *et al.*, ***Optics Letters*** 49, 5483 (2024)

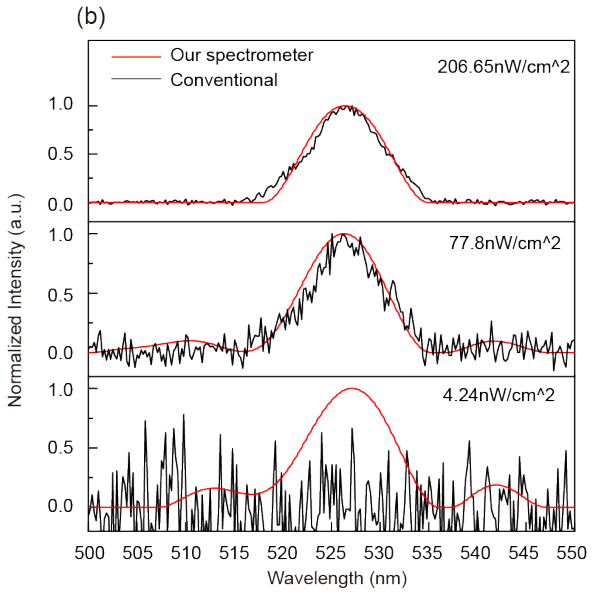
J. J. Wu, **Z. Y. Yang*** *et al.*, ***Molecules*** 29, 5580 (2024)

基于带隙渐变光子晶体薄膜（陷波型）的高灵敏度光谱仪



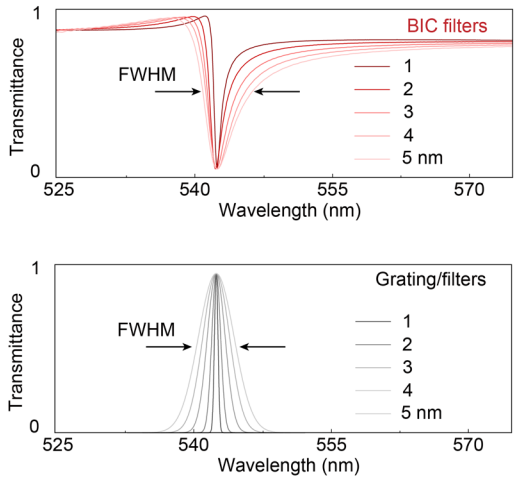


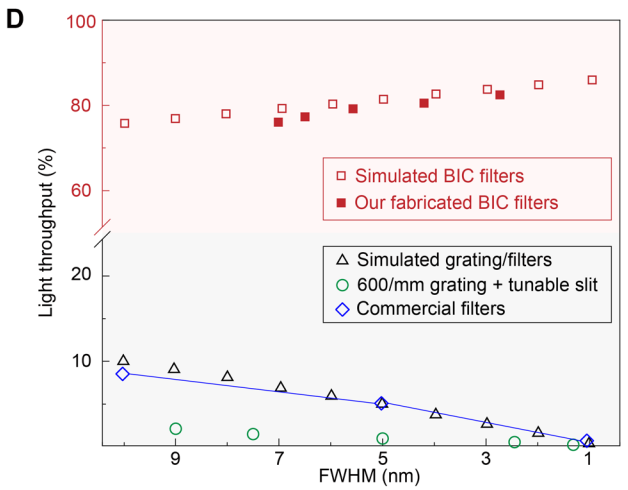


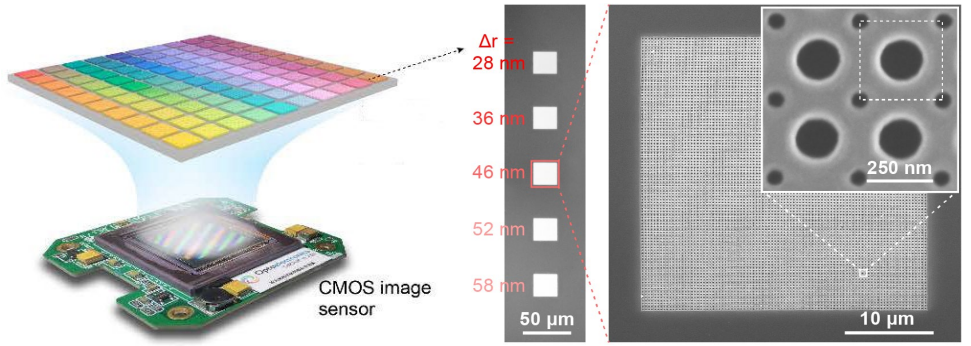


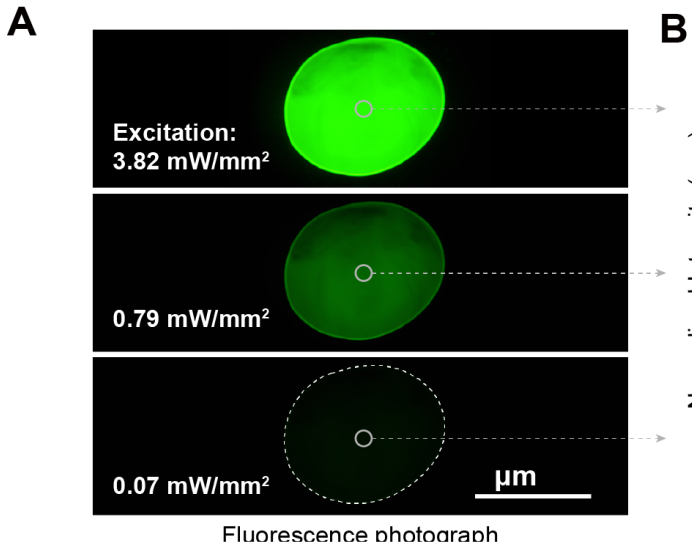
T. Yang, **Z. Y. Yang*** *et al.*, ***Optics Express*** 32, 25830 (2024)

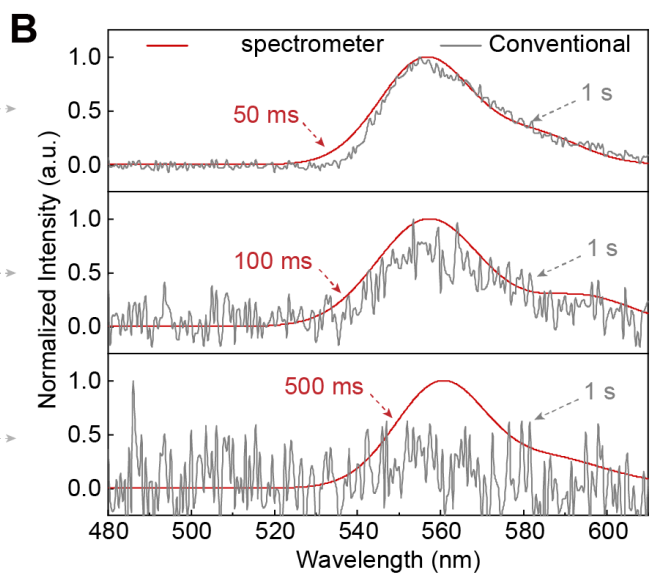
基于BIC超表面，解决光谱仪高灵敏度与高分辨率无法兼具的难题











10~100× sensitivity

F. Tang, **Z. Y. Yang*** *et al.*, ***Science Advances*** 10, eadr7155 (2024)

汇报提纲

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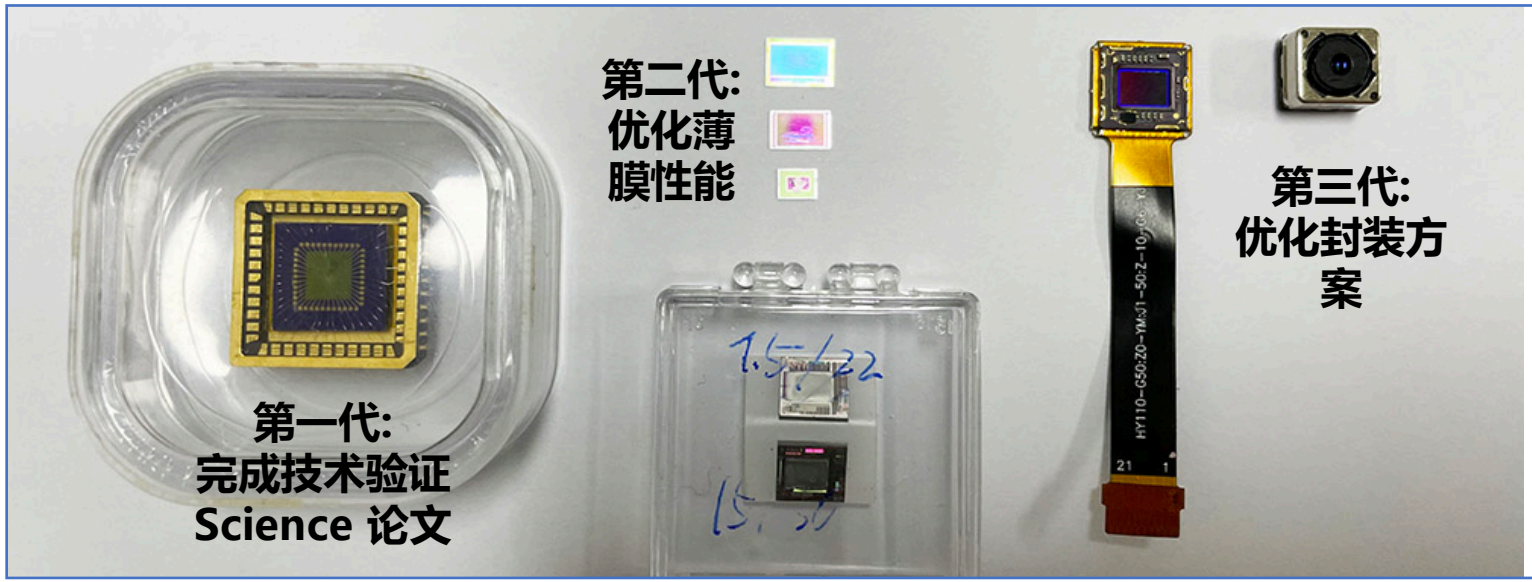
5 提高灵敏度

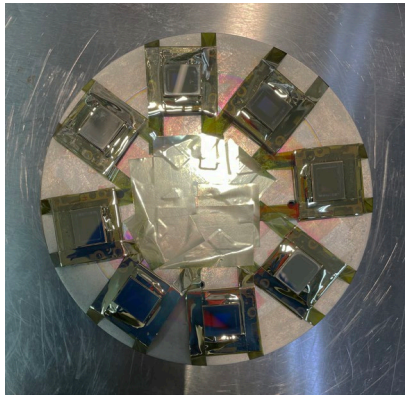
6 **技术应用**

From
Scientific research
to
Practical devices

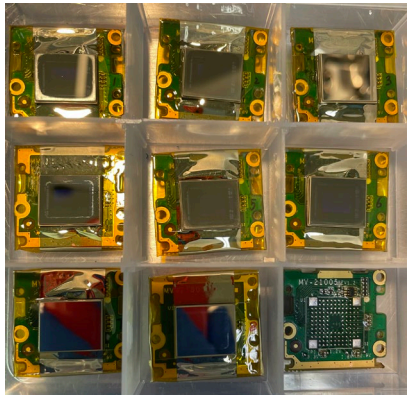
工艺积累



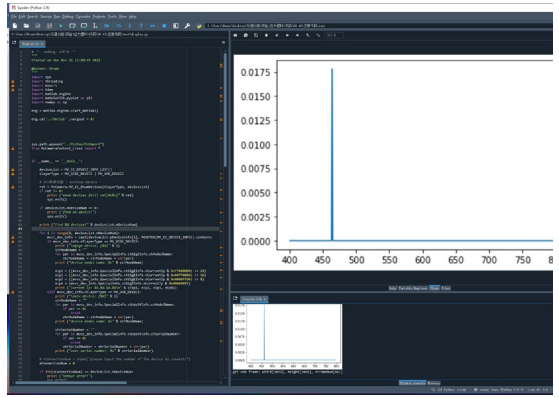
多次技术迭代，微型光谱的领跑者



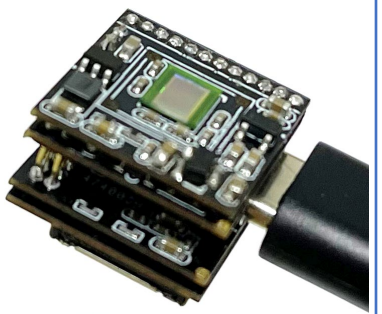
材料制备工艺



芯片封装+PCB

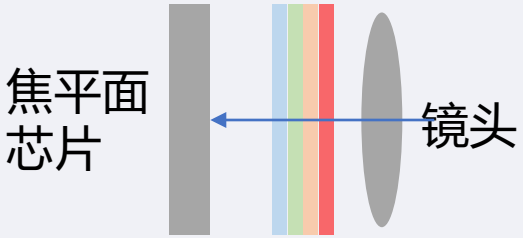


软件开发



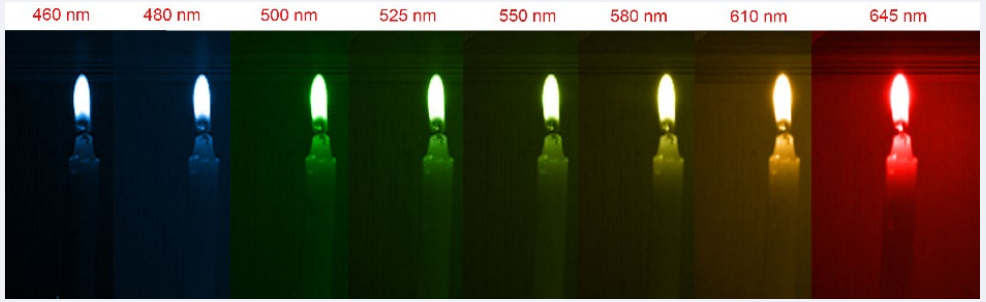
系统集成

技术应用1：小型、高速红外多光谱

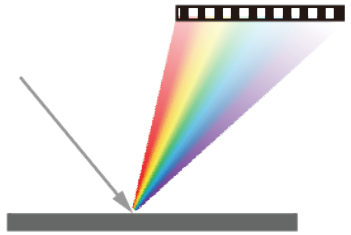
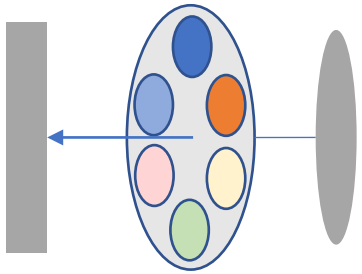
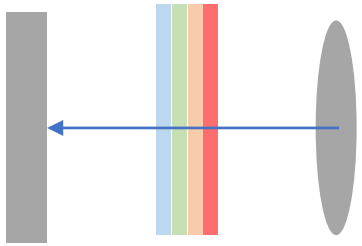


可高速切换带隙
可调膜（4波段）

原理验证样机（可见波段）

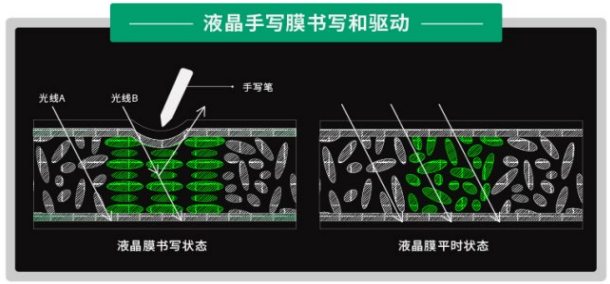


- 尺寸与微型相机接近
- 中红外光谱成像>**30 帧/秒**
- 最多可扩展到16个波段

中红外波段光谱成像		稳定性	效率	空间分辨率	成像速度
光栅、棱镜等色散元件		不稳定 机械线扫描结构复杂，易受震动影响	低(<10%)	低	慢（小于1帧/秒）
旋转滤光片轮型		不稳定 机械结构振动对成像质量有影响	高(>50%)	高	慢（小于1帧/秒）
高速带隙切换滤光膜		稳定 固态四通道快速稳定切换	高(约等于50%)	高	快（大于30 帧/秒）

应用2：从光谱相机到光擦除多色书写板

研发光谱相机过程中
开发出
电磁调色液晶膜



联想到小孩用的液晶手写板，只能**全局擦除！痛点！**

技术挑战

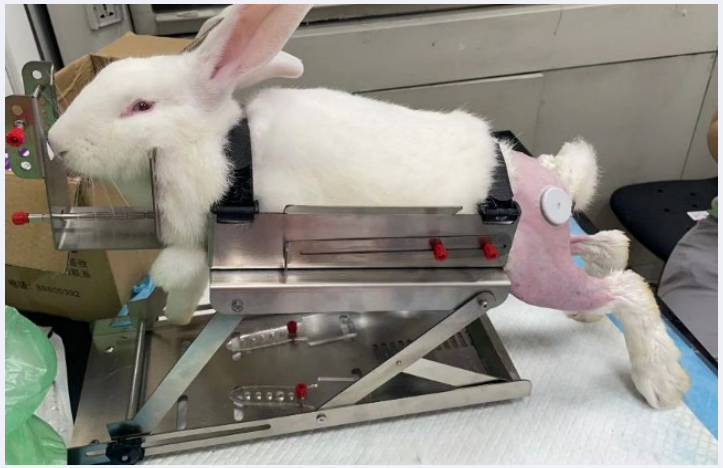
- 无法局部擦除
- 无法多色书写
- 大尺寸成品率低

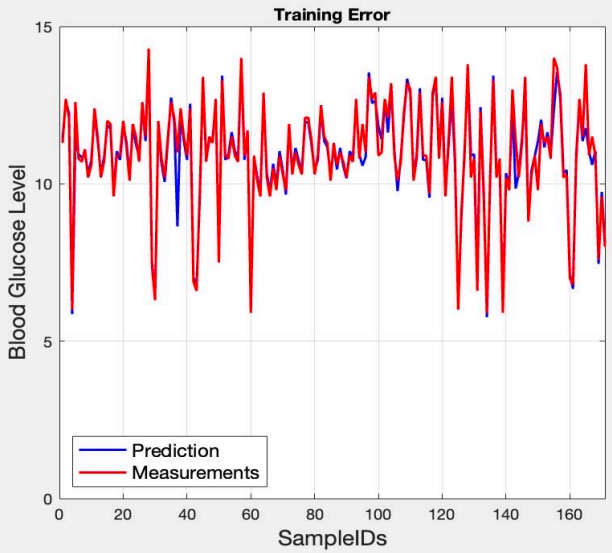
解决方案

- 首创光控液晶 局部擦除
- 首创磁控液晶 多色书写
- 表面钝化 大尺寸稳定

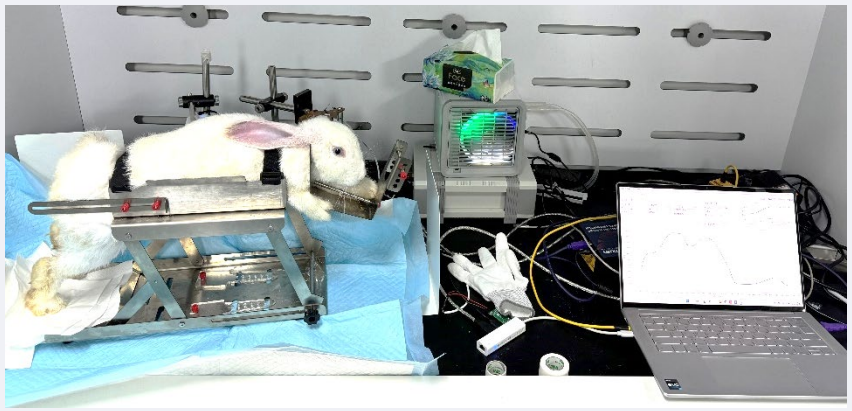
跨界解决液晶写字板行业痛点

应用3：无创血糖检测



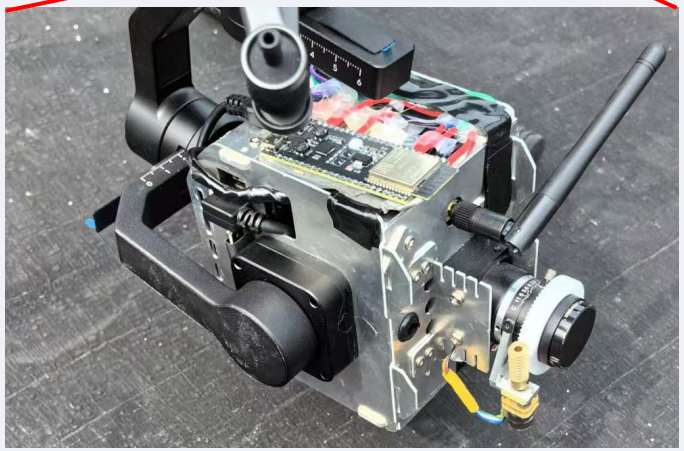


建立兔子早上2小时血糖光谱模型，可以准确预测晚上血糖值 MARD<20%



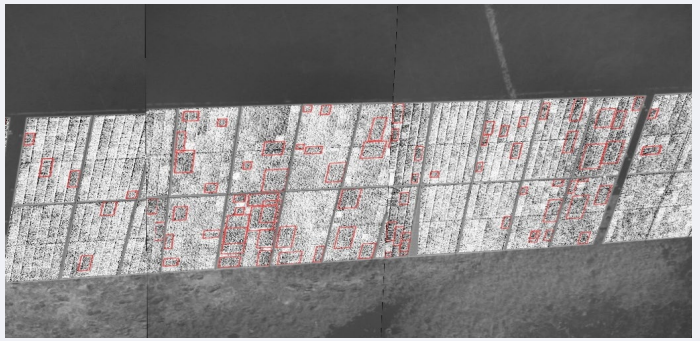
长时间测量建模

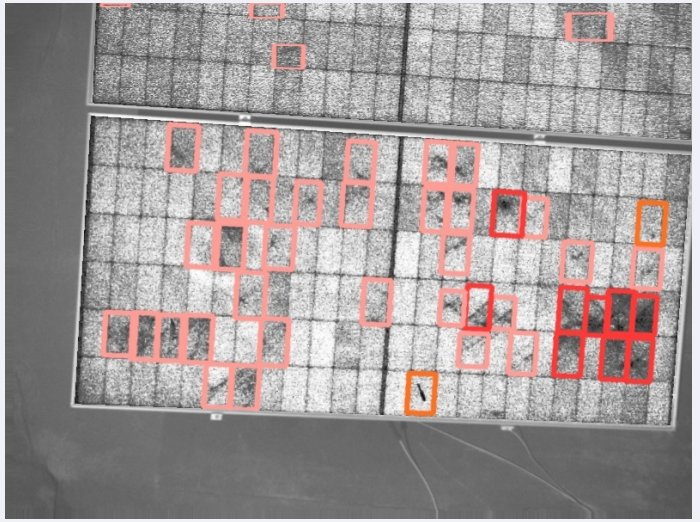
应用4：光谱相机太阳能电池板缺陷检测（华为项目）



自研近红外光谱相机+通讯系统+
机器学习图像识别算法

搭载无人机在太阳能发电站
实地检测

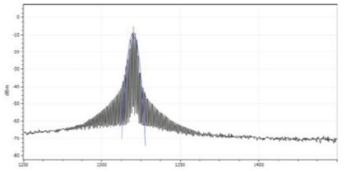




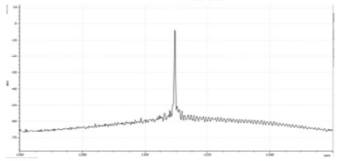
太阳能板缺陷识别准确度>95%

应用5：通讯设备中FP-LD和DFB激光器检测（华为花火价值奖）

技术背景



FP-LD光谱

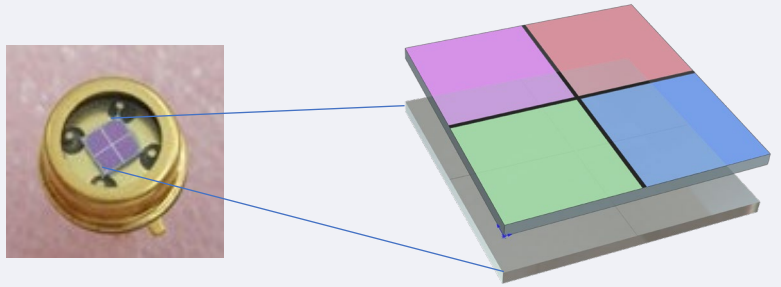


DFB光谱

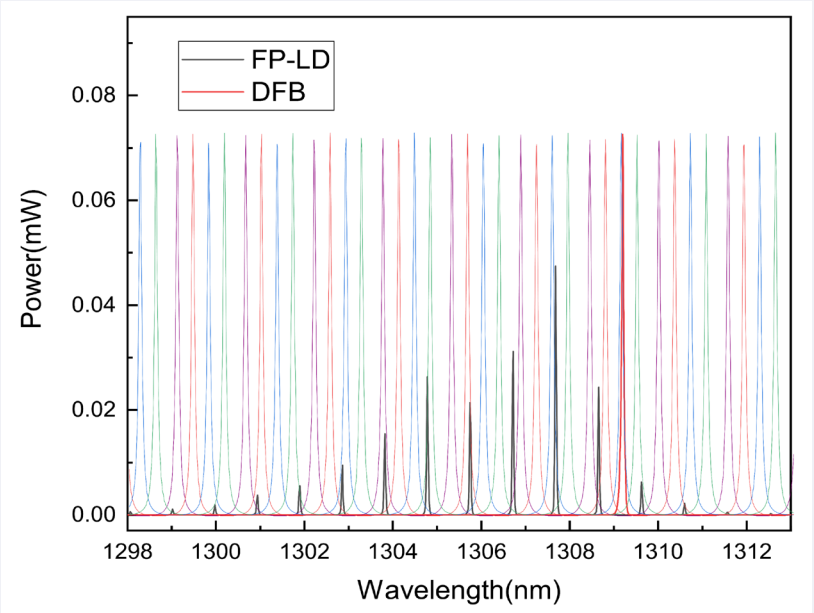
小型化低成本
FP-LD识别方式

- 通信系统中同时存在FP-LD（泵浦激光器）和DFB（半导体激光器），希望能通过有效的检测或识别手段将其区分。
- FP-LD是多纵模，呈现宽线宽光谱特征，DFB是单纵模，呈现窄线宽光谱特征，其中FP-LD与DFB在同一波长范围内。
- 当前典型手段采用光谱仪，将信号分解成光谱成分，数字化为波长的函数，并通过计算机显示出来，能较容易将FP-LD与DFB区分。但光谱仪成本且体积大，不利于工程实施和运营维护，所以需要有一种小型化低成本FP-LD识别部件/模块集成到通信设备中。

华为难题揭榜



InGaAs四象限探测器　　像元上覆盖4种FP滤光片



通过模式筛选产生的脉冲类型判断激光器类型

从技术应用到本科生培养

指导本科生获得创新创业竞赛奖

第八届中国国际“互联网+”创新创业大赛 **全国金奖**

2024中国国际大学生创新大赛 **全国金奖**

2024中国国际大学生创新大赛 **全国银奖**

第十四届“挑战杯”创业竞赛 **全国金奖**

浙江省国际大学生创新大赛 **金奖×4**

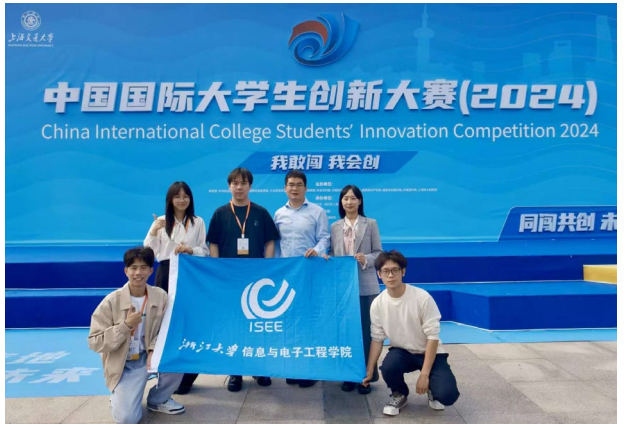
浙江省第七届“互联网+” **金奖**

浙江省第八届“互联网+” **金奖+冠军**

浙江省第九届“互联网+” **金奖**

浙江省第十三届“挑战杯” 创业计划竞赛 **金奖**

浙江省第十四届 “挑战杯” 创业计划竞赛 **金奖**



总结

光谱仪微型化

新理论、新材料，为器件研制打下基础

- 提出新的材料合成法（*Nano Lett.* 2011，*Nano Lett.* 2014；*JACS* 2011）
- 提出光谱器件新理论与技术体系（*Science 2021, Nature Photonics 2024*）

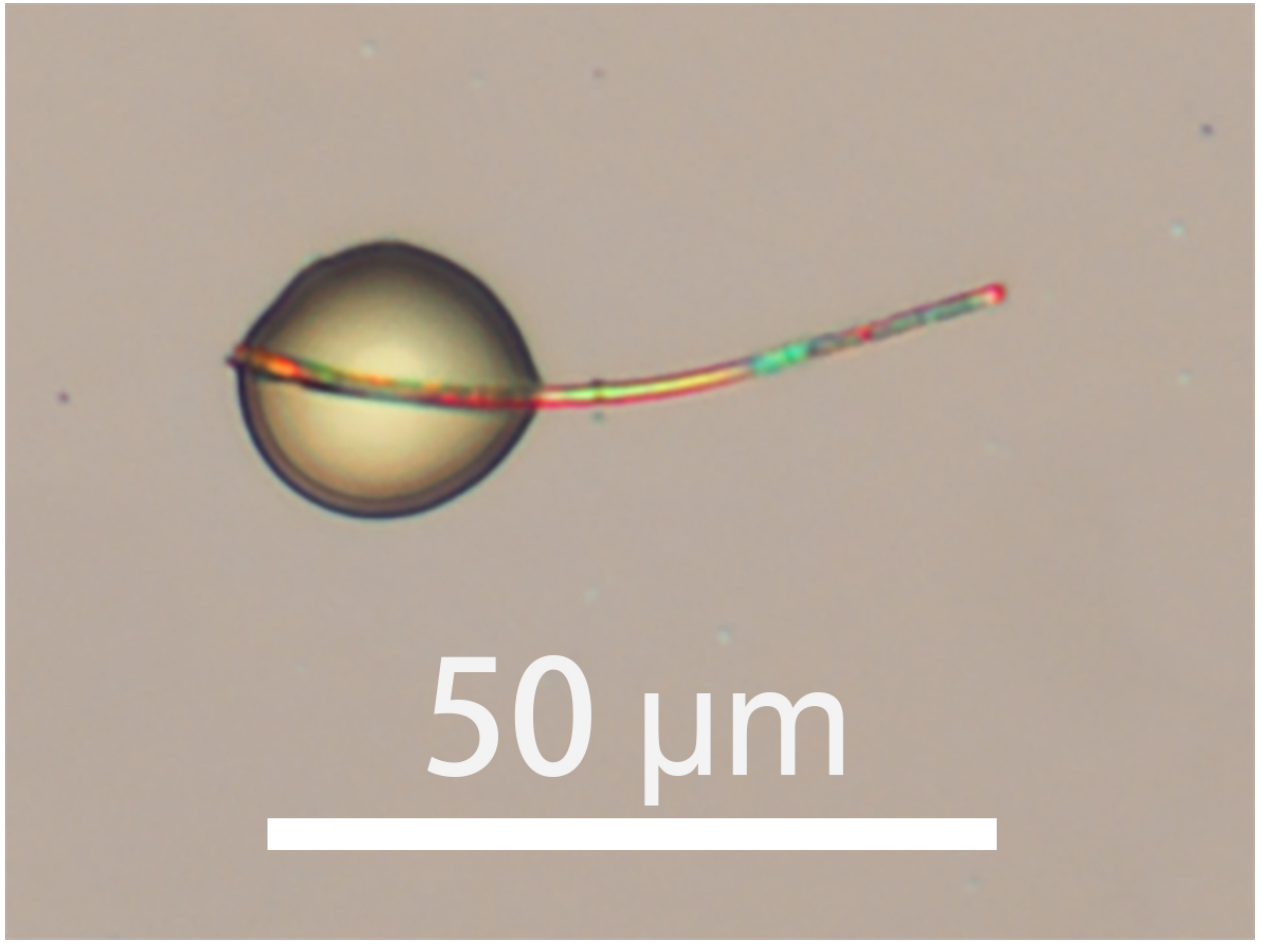
新结构、新机理，突破光谱仪尺寸性能极限

- 基于新材料突破光谱尺寸极限（*Science* 2019, *Science* 2022）
- 器件创新→解决难题→提升性能（*Nature Materials, Science Adv., Light…*）

技术应用

- 高速红外多光谱成像
- 无创血糖 等...

基础理论研究1：光制冷

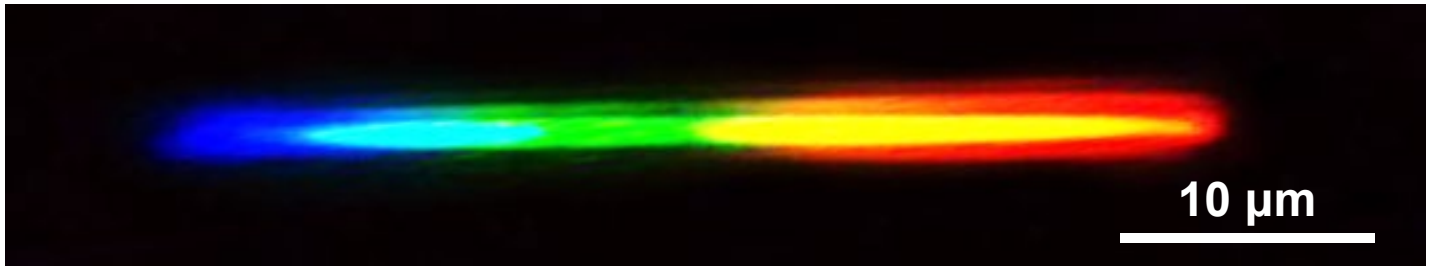


Optical refrigeration in lattice engineered materials

最小的冰箱

基础理论研究2：非互易光传输

级联激发，传输时间百纳秒
(信息传输调控，传感 等...)



光波导传输时间小于1纳秒